RSP

About the Conference

The Third International Conference on Futuristic Trends in Science, Engineering, and Management (ICFTSEM-III) 2026 promises to be a milestone event in the field of research. This Collaborative endeavor aims to bring together distinguished Scholars, Researchers, Practitioners, and Students from around the globe to delve into the latest Advancements, Trends, and Challenges. The Conference will feature insightful Keynote Speeches, Panel Discussions, Paper presentations, and Collaborations that drive innovation and progress in our field.

RSP Research Hub
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XAVIER UNIVERSITY

Abstract Book Proceedings of Third International Conference on Futuristic Trends in Science, Engineering, and Management (ICFTSEM-III) 2026 - Organized by the School of Management, Xavier University, Patna, India & Global Conference Hub, Coimbatore, Tamil Nadu, India

Date of the Conference: 27.02.2026 & 28.02.2026

Editor (s)
Dr. Piyush Ranjan Sahay
Dr. Martin Poras SJ
Ms. Shilpa Sharma
Dr. C. Somu

Abstract Book Proceedings of Third International Conference on Futuristic Trends in Science, Engineering, and Management (ICFTSEM-III) 2026 - Organized by School of Management, Xavier University, Patna, India & Global Conference Hub, Coimbatore, Tamil Nadu, India.

Editors

Dr. Piyush Ranjan Sahay,

Dean,

School of Management, Xavier University, Patna.

Dr. Martin Poras SJ,

Vice Chancellor,

Xavier University, Patna.

Ms. Shilpa Sharma,

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Department of Business Administration, Xavier University, Patna.

Dr.C. Somu

Chairman & Managing Director, ECLearnix EdTech Private Limited,

KCT Tech Park, Coimbatore, Tamil Nadu, India.

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Preface

Third International Conference on Futuristic Trends in Science, Engineering, and Management (ICFTSEM-III) 2026 - Organized by School of Management, Xavier University, Patna, India & **Event Partner:** Global Conference Hub, Coimbatore, Tamil Nadu, India, held successfully on February 27th & 28st, 2026. 223 teams from 05 countries, including Australia, Egypt, Iraq & Malaysia, Japan, Nigeria and PAN India, including 18 Indian States from 34 reputed universities and 70 colleges, participated in this conference. The objective of this conference is to bring together entrepreneurs, academicians, research scholars, and postgraduate students from around the world to encourage, acknowledge, and support research in all these areas by providing opportunities for them to exchange and share their experiences, fresh concepts, and research findings and discuss the pragmatic challenges encountered and solutions adopted in the aforementioned interdisciplinary areas through a wide range of research activities and publications.

During the conference, keynote speaker was **Dr. Monica Bhutani (Ph.D. IIT Delhi)**, Postdoctoral Researcher and Research Associate at Lincoln University College, Malaysia., Associate Professor, Department of Electronics and Communications, Bharati Vidyapeeth's College of Engineering, New Delhi, India., **Topic:** Importance of AI in Research Excellence.

Conference Chairperson of each session are as **Prof. M. Firdos Sheikh,** Assistant Professor & Deputy HoD, Department of Computer Science & Engineering, Poornima University, Jaipur, Rajasthan, India., **Dr. Ajay N. Upadhyaya,** Professor & I/C Principal, Department of Computer Engineering, SAL Education, Ahmedabad, Gujarat, India., **Dr. S. Mythili,** Professor, Department of Biomedical Engineering, PSNA College Of Engineering And Technology, Dindigul, Tamilnadu, India., **Dr. Hemavathi P,** Professor, Department of Computer Science & Engineering, Bangalore Institute of Technology, Bangalore, India., **Dr. Avni Suresh Thakkar,** Assistant Professor, School of Management, Shri Ramdeobaba University, Nagpur, Maharashtra., **Dr. Pawanjeet Kaur,** Assistant Professor (Chemistry), School of Mechatronics Engineering, Symbiosis Skills and Professional University, Pune, Maharashtra, India, **Dr. K. K. Baseer**, Associate Professor, Department of Computer Science and Engineering, GITAM School of Technology, GITAM (Deemed to be University), Bengaluru, Karnataka, India., **Dr. Sambathkumar M,** Assistant Professor, Department of Mechanical Engineering, Kongu Engineering College, Perundurai, Erode, Tamilnadu, India., **Dr. B. Senthil Rathi,** Assistant Professor (Grade II), Department of Chemical Engineering, SSN College of Engineering, Chennai, Tamilnadu, India., **Dr.Vishal Rathi,** Assistant Professor, Department of Electronics and Telecommunication Engineering, Sipna College of Engineering and Technology, Amravati, Maharashtra, India., **Dr. Ashish Tiwari,** Associate

Professor, Department of Computer Science and Engineering (HoD), Babu Banarasi Das Institute of Technology & Management, Lucknow, Uttar Pradesh, India, **Dr. Reshma S,** Associate Professor, Department of Artificial Intelligence & Machine Learning, Dayananda Sagar College of Engineering, Bangalore, Karnataka, India., **Dr. Manvi Mishra,** Associate Professor, Head of the Department, Department of Computer Science & Engineering, Shri Ram Murti Smarak College of Engineering Technology and Research, Bareilly, Uttar Pradesh, India., and **Dr. Sarika Ghanshyam Jadhav,** Assistant Professor, School of Computer Science, Engineering, and Applications, D Y Patil International University, Pune, Maharashtra, India.

15 Oral Presentation sessions are planned and successfully held under the joint efforts of conference chair members, presenters, and conference members through hybrid mode. Many recent trend topics are discussed. The best presentations were selected under the UG, PG, Research Scholar, and Faculty categories, which were evaluated by conference chair members and the conference team as per the given rubric sheet. 223 abstracts are included in these proceedings and have been classified into six focus research areas for corresponding sessions held at the conference. We would like to express sincere gratitude to all the authors for their dedicated contributions to the proceedings. We would like to extend our thanks to all the technical committee members and reviewers for their constructive comments on all papers also, to the organizing committee for the sincere and dedicated work. Finally, we would like to thank the Global Conference Hub for producing this volume. We strongly believe that the participants of (ICFTSEM-III) 2026 have enjoyed a wonderful and fruitful time during the conference.

PROCEEDINGS OF

(ICFTSEM-III) 2026

Third International Conference on Futuristic Trends in Science, Engineering, and Management

Organized by

School of Management, Xavier University, Patna, India

&

Global Conference Hub, Coimbatore, Tamil Nadu, India

Abstract Proceedings

(Special Edition)

About the College

Xavier University, Patna is a private university established under the Bihar Private Universities Act, 2013, and officially notified to function as a centre of higher learning in Bihar. The campus is spread across a lush, green 36-acre area in Digha-Aashiyana, Patna, offering students a vibrant academic and cultural environment. At its core, XUP follows the Jesuit tradition of holistic education, focusing on intellectual, emotional and spiritual development to prepare students as future leaders and change makers. The university offers a broad spectrum of programs including undergraduate and postgraduate courses in arts, commerce, management, communication, science, and technology, designed to align with modern industry needs. Students can pursue specialized degrees such as BBA with multiple majors and a strong Bachelor of Commerce program that blends theory with practical exposure and internships. XUP's academic system incorporates semester-based learning and continuous internal assessment to ensure outcome-based education. The faculty comprises experienced professionals committed to quality teaching, research, and mentorship, encouraging students to think critically and engage deeply with diverse subjects. A vibrant campus life includes cultural fests, sports tournaments, student council activities, and community service through NSS, promoting teamwork, leadership, and social responsibility. In addition to academics, the university emphasizes extracurricular engagement such as seminars, workshops, and industry visits to enhance practical learning. Students also benefit from supportive cells like the Student Grievance Redressal Committee, ensuring their voices are heard and issues resolved promptly. The Internal Quality Assurance Cell works to maintain high standards across programs and administrative processes. XUP encourages a culture of inclusivity, ethical awareness, and global outlook through its vision of creating a just, humane and sustainable society. The campus infrastructure is designed to be safe and eco-friendly, with facilities for sports, hostels, outreach activities, and student clubs. With strong industry linkages and focus on placements, the university strives to connect students with real-world opportunities and career pathways. XUP's mission is to develop well-rounded professionals who are equipped to make a meaningful impact in their fields and in society.

About the Department

The Third International Conference on Futuristic Trends in Science, Engineering, and Management (ICFTSEM-III 2026) is jointly organized by the School of Management, Xavier

University, Patna, and Global Conference Hub, Coimbatore, and will be conducted in online mode on 27th and 28th February 2026. The conference aims to bring together researchers, academicians, industry professionals, and students to share innovative ideas, research findings, and emerging trends across multidisciplinary domains. It provides a global platform for knowledge exchange in areas including science, engineering, technology, management, and social sciences, encouraging collaborative research and real-world problem solving. All accepted papers will be published in Google Scholar indexed journals with DOI, and selected high-quality papers will be published as Scopus-indexed book chapters at an additional cost, making ICFTSEM-III 2026 a strong opportunity for academic visibility and professional networking.

About the Global Conference Hub

Global Conference Hub is organizing an international peer-reviewed conference dedicated to Advancements in Sciences, Technology and Management. It promotes collaborative excellence between academicians and professionals from academics. The objective of the Global Conference Hub is to provide an opportunity for academicians and industrialists from various fields with cross disciplinary interests to bridge the knowledge gap and promote research esteem and the evolution of pedagogy. This conference is an amalgamation of industrialists, and academia where they can gear up knowledge. Our gratitude towards people who are concerned about advancements in the hub of research, and we cordially invite them to gear up and make the congress an unforgettable successful event.

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Chairman & Managing Director, ECLearnix
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Dr. Monica Bhutani (Ph.D. IIT Delhi)

Postdoctoral Researcher and Research Associate at
Lincoln University College, Malaysia
Associate Professor, Department of Electronics and Communications, Bharati Vidyapeeth's College of Engineering, New Delhi, India.

Topic: Importance of AI in Research Excellence

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Dr. Manvi Mishra

Associate Professor, Head of the Department, Department of Computer Science & Engineering, Shri Ram Murti Smarak College of Engineering Technology and Research, Bareilly, Uttar Pradesh, India.

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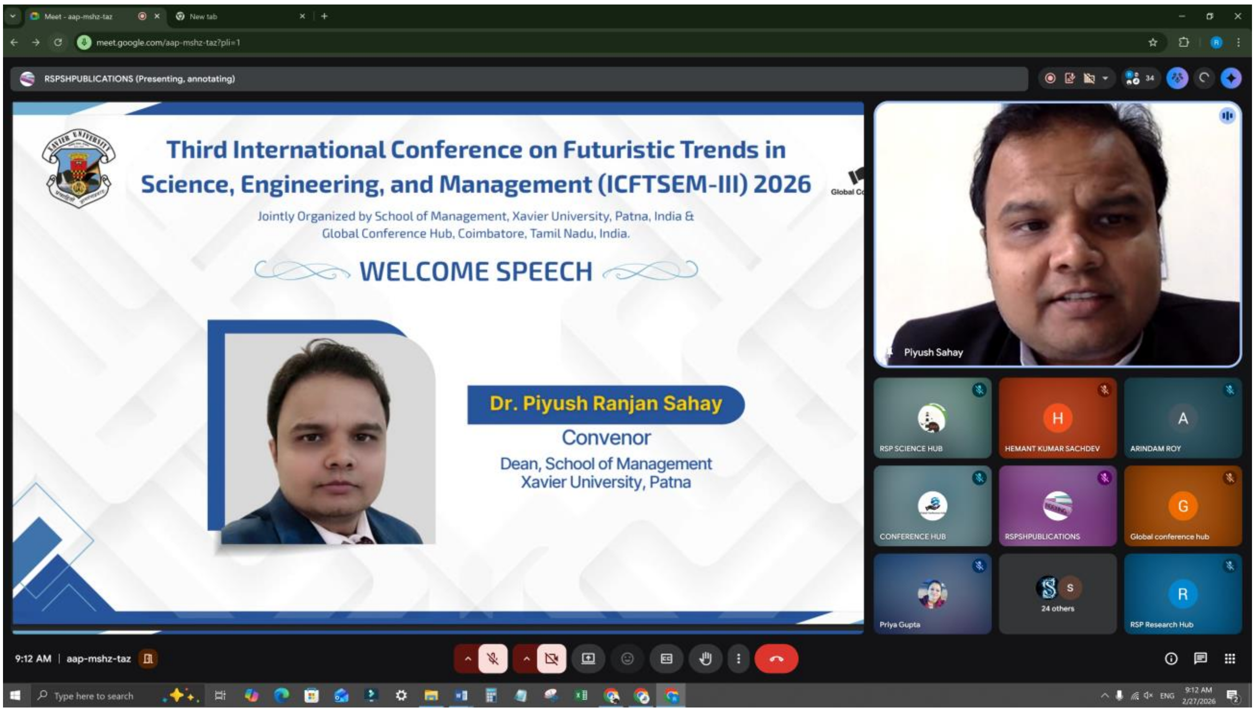

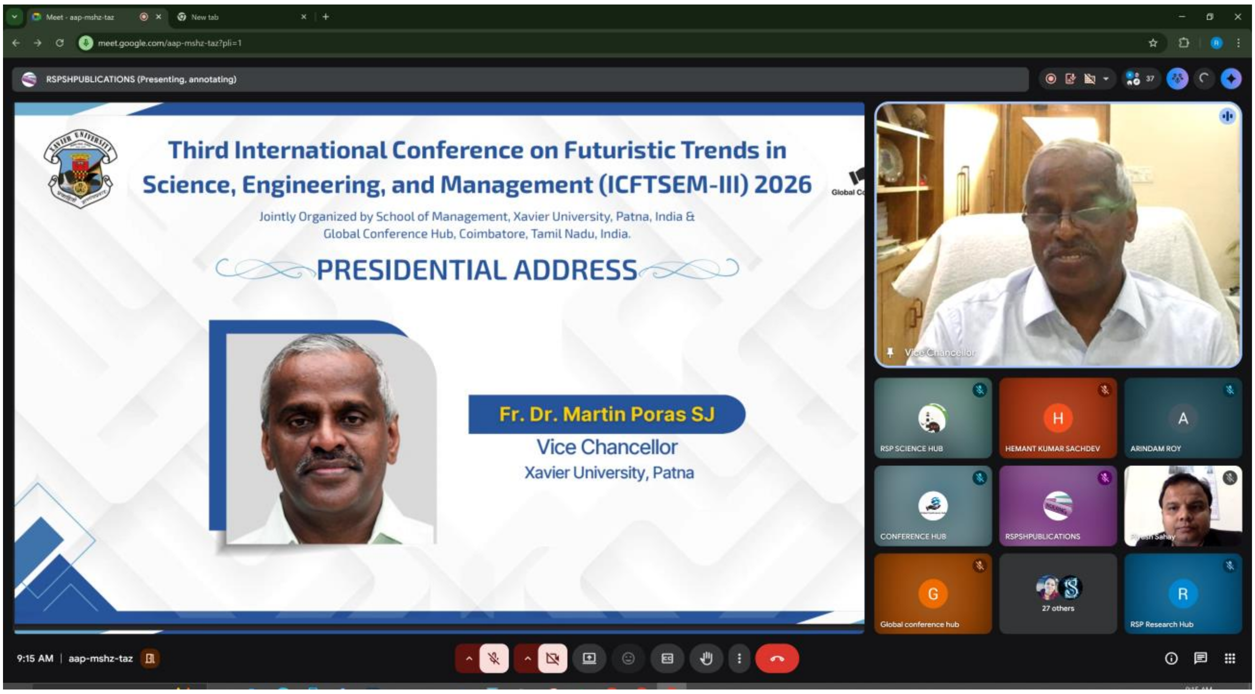

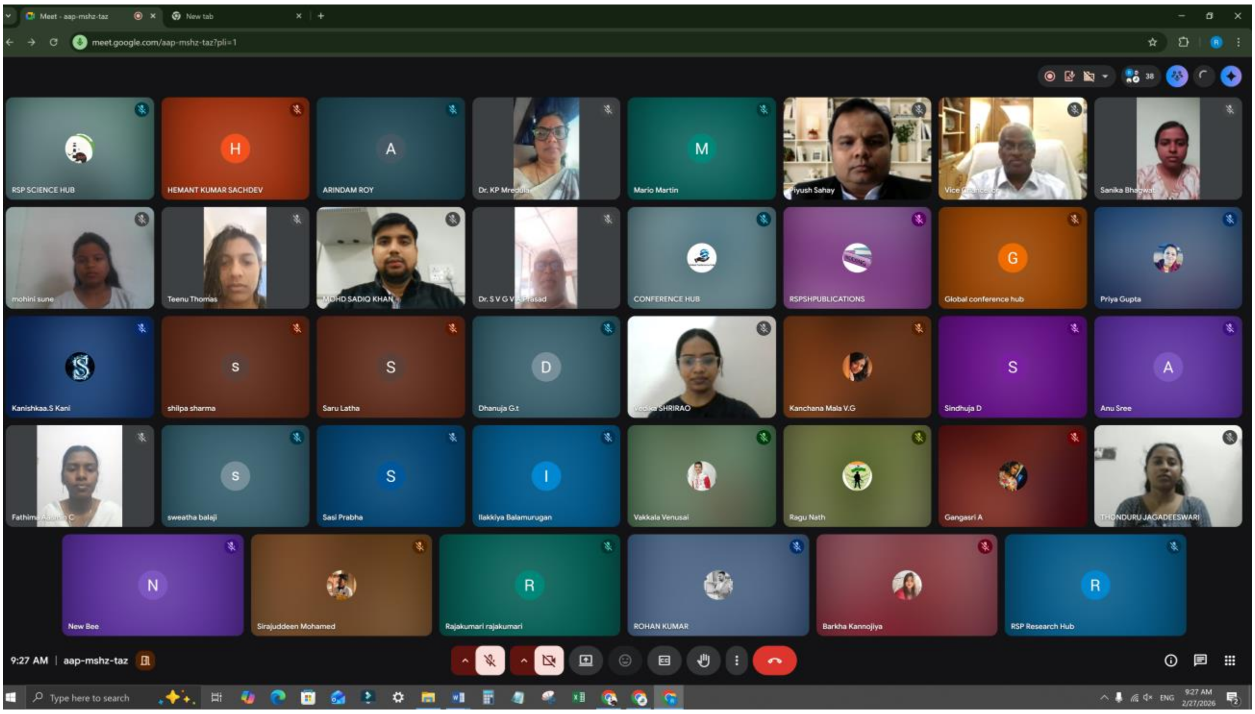

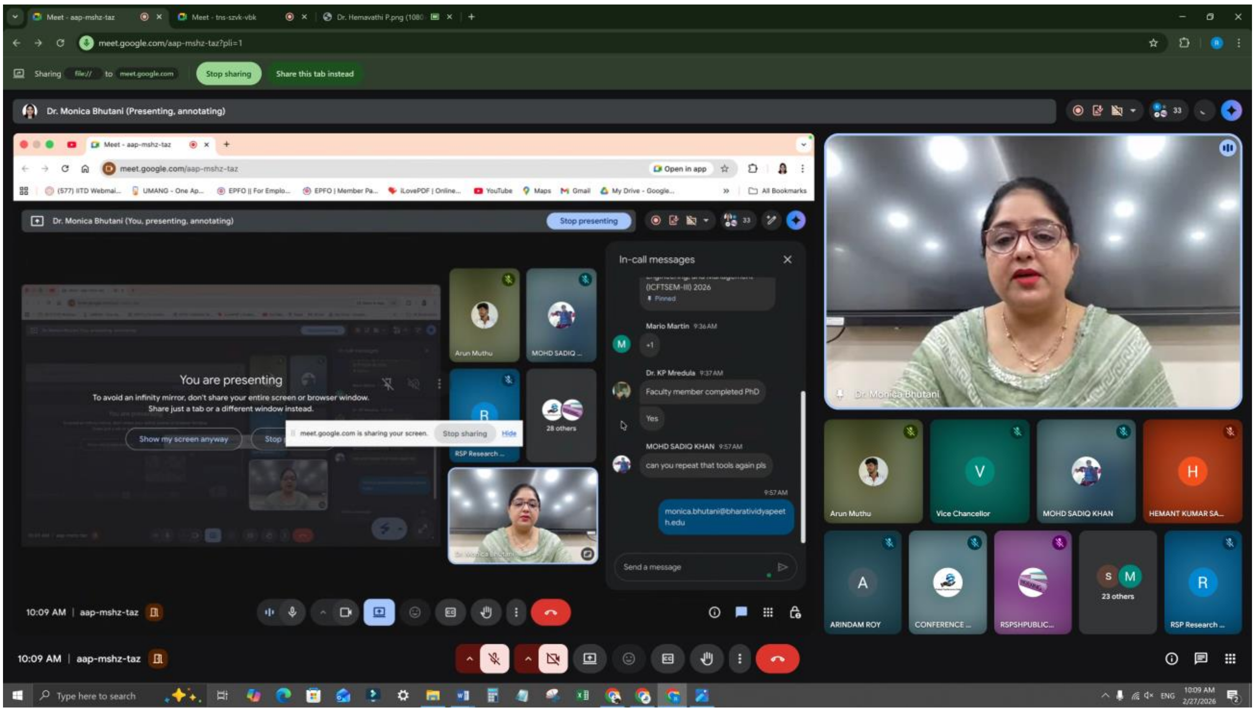

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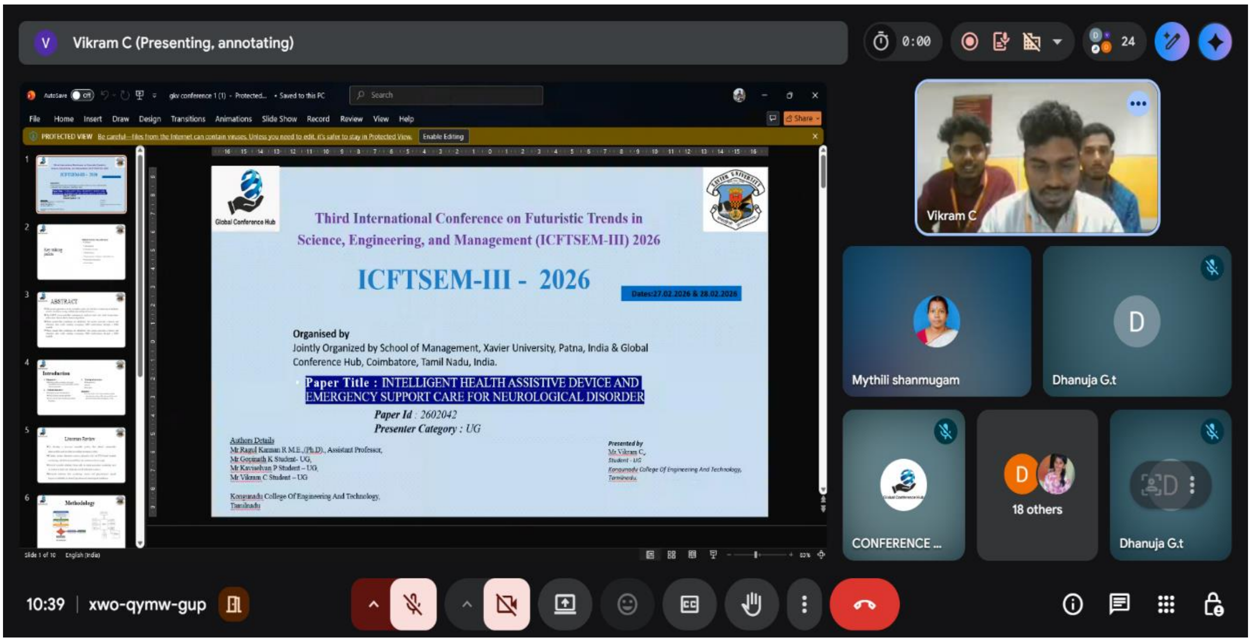


Sivasubramanian J (Presenting, annotating)
Global Conference Hub
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Paper Title : Deep Learning-Based EEG Communication Aid For Paralyzed Patients
Paper Id : 2602120
Presenter Category : UG
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Roney Varughese Joseph,
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Puducherry
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Puducherry
Roney Varughese Joseph
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Vangara Bhavana
Jeyani
CONFERENCE H...
9 others
RSP Research H...
12:35 PM | tns-szvk-vbk


Jayshree Devikar (Presenting, annotating)
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Dates:27.02.2026 & 28.02.2026
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Jointly Organized by School of Management, Xavier University, Patna, India & Global Conference Hub, Coimbatore, Tamil Nadu, India.
Paper Title : intelligent Prompt Optimization Tool for Generative AI Models
Authors Details
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Ms. Sakshi Phadatare
Ms. Samruddhi Gorde
Ms. Akanksha Kapadnis
Prof. Supriya Bhujbal
Student – UG
JSPM's BSIOTR,
Pune, Maharastra.
Paper Id : 2602267
Presenter Category : UG
Presented by
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Ms. Sakshi Phadatare
Student - UG
JSPM's BSIOTR
Pune, Maharastra
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Saptha Sanil
Dr. Pawanjeet Kaur
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Rohit Patil
15 others
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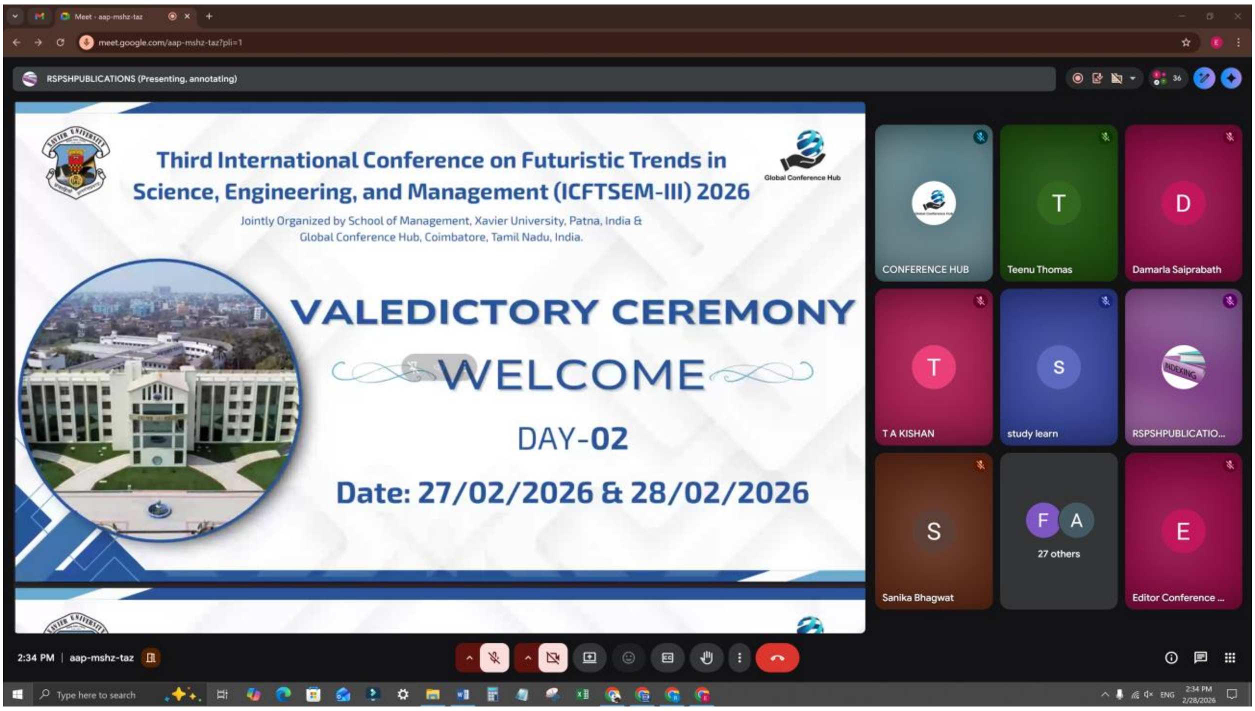
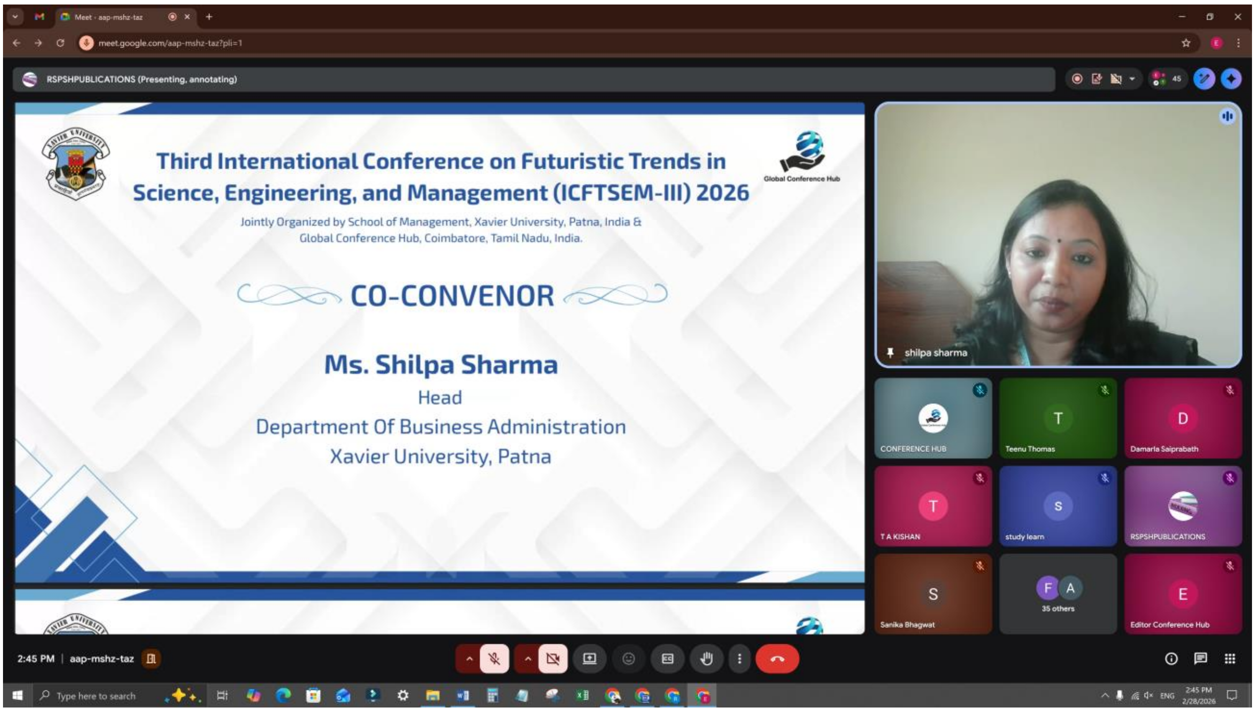
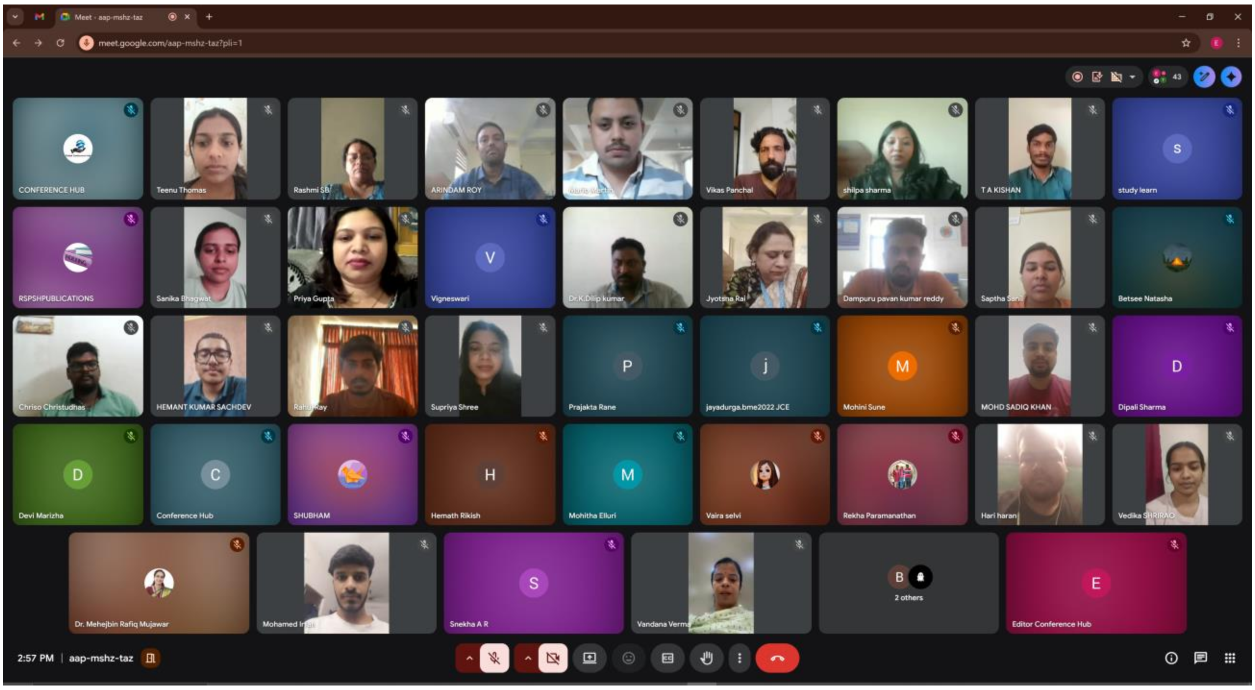
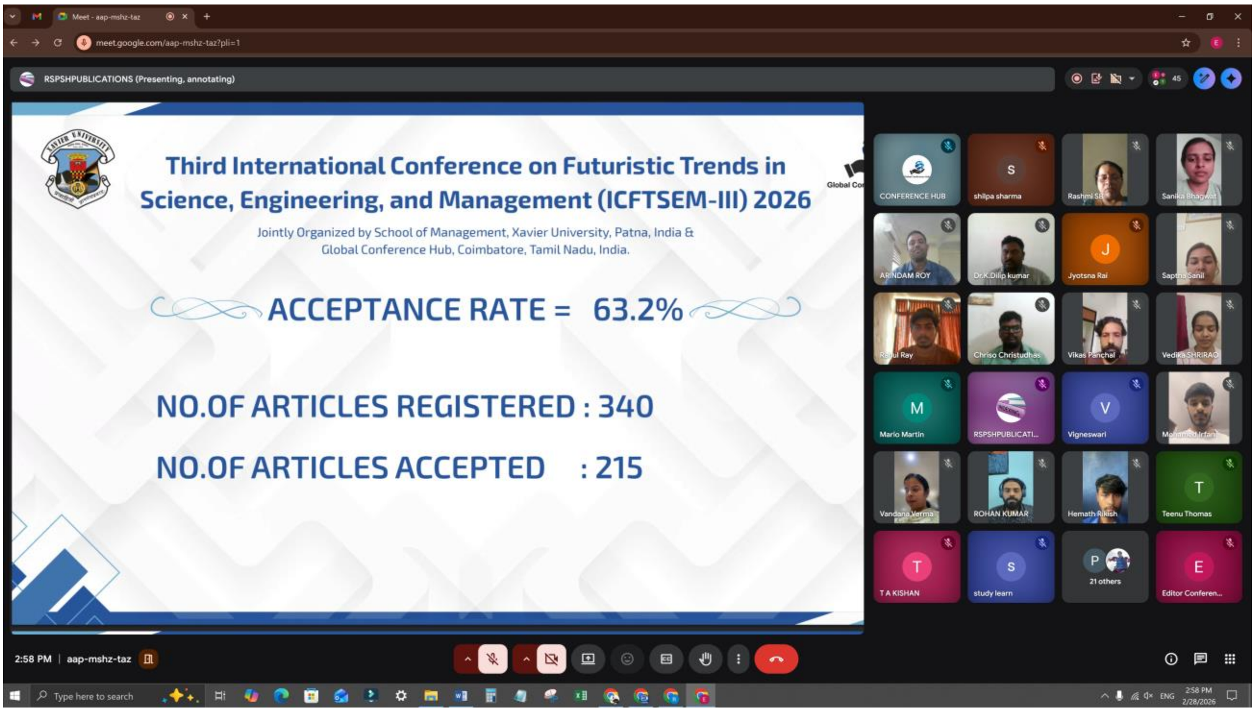
Valedictory – ICFTSEM - 2026





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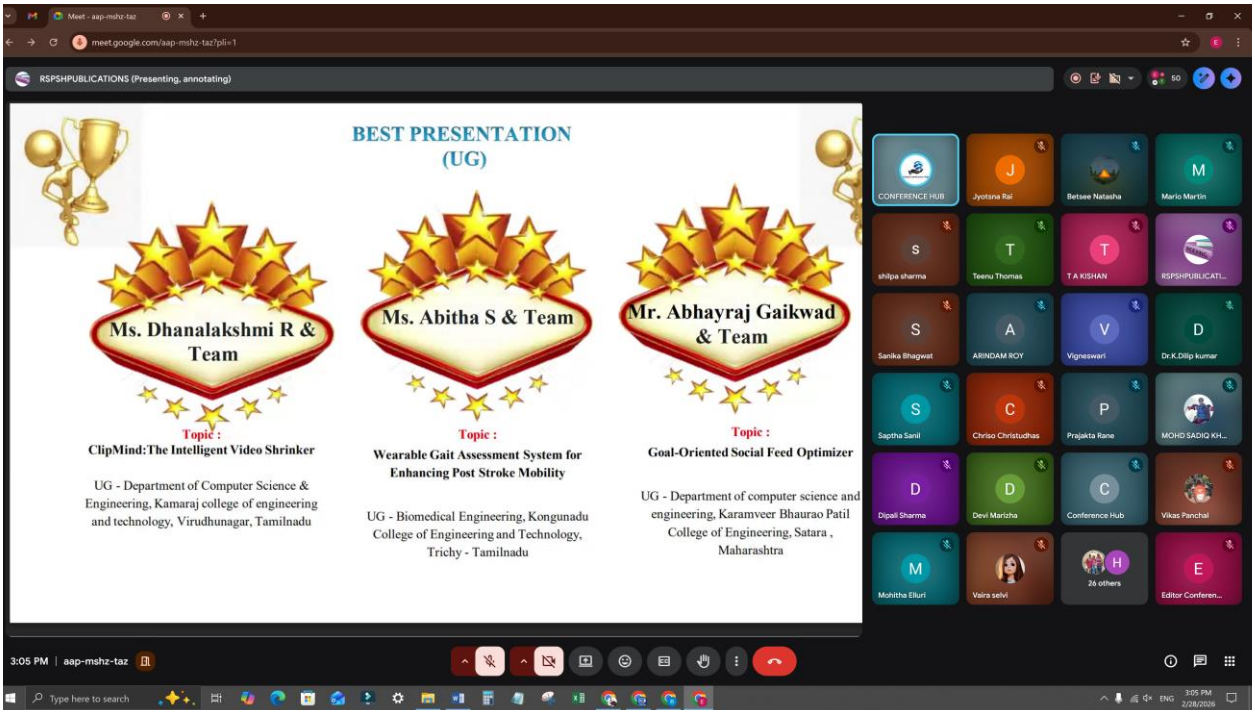
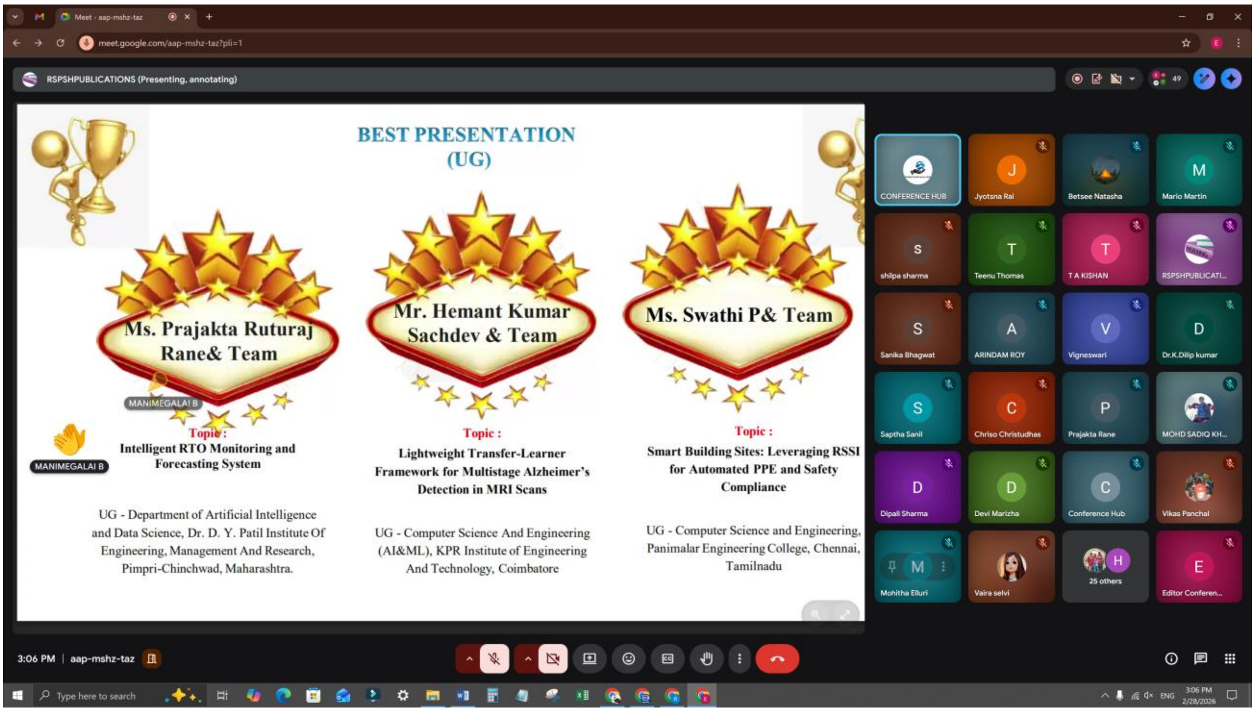
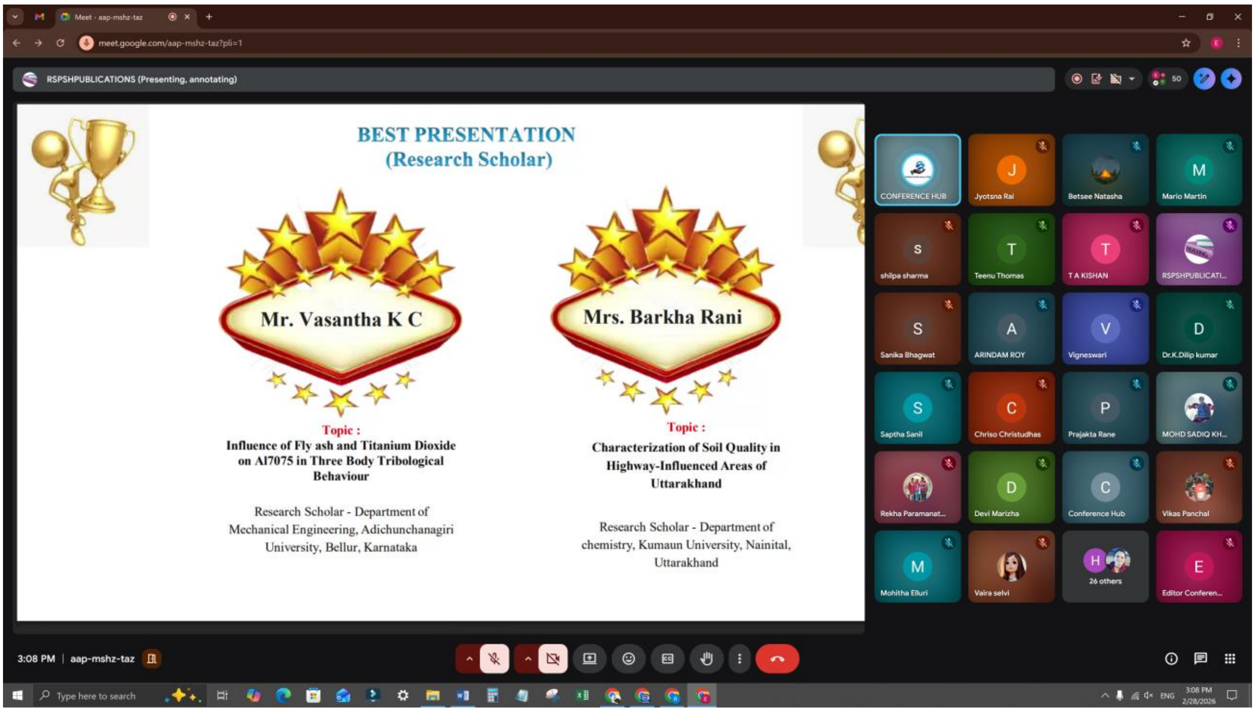
ACCEPTANCE RATE = 63.2%
NO.OF ARTICLES REGISTERED : 340
NO.OF ARTICLES ACCEPTED : 215

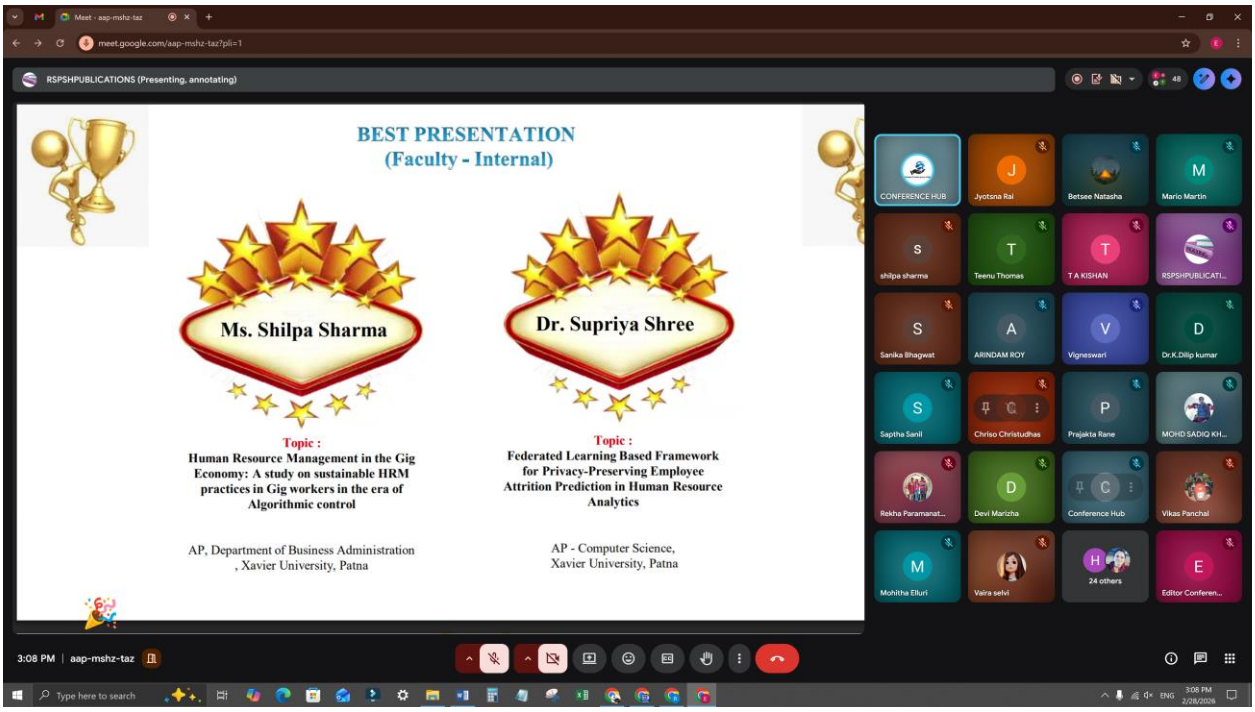
Best Presentation of ICFTSEM - 2026






BEST PRESENTATION
(PG)
Mrs. Rajakumari R
Topic :
IOT Based real time road hazard detection and alert system using smart streetlight and smart road studs
PG - Electronics And Communication Engg , Thiruvalluvar College Of Engg&tech Vandavasi, Tamilnadu
Ms. M Saranya
Topic :
Freelancer Skill Gap Analyzer and Upskilling Pathfinder
PG - Department of Computer Application, Dayananda Sagar College of arts science and commerce ,Bangalore University, Karnataka


BEST PRESENTATION
(Research Scholar)
Mr. Vasantha K C
Topic :
Influence of Fly ash and Titanium Dioxide on Al7075 in Three Body Tribological Behaviour
Research Scholar - Department of Mechanical Engineering, Adichunchanagiri University, Bellur, Karnataka
Mrs. Barkha Rani
Topic :
Characterization of Soil Quality in Highway-Influenced Areas of Uttarakhand
Research Scholar - Department of chemistry, Kumaun University, Nainital, Uttarakhand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BEST PRESENTATION
(Faculty - Internal)
Ms. Shilpa Sharma
Topic :
Human Resource Management in the Gig Economy: A study on sustainable HRM practices in Gig workers in the era of Algorithmic control
AP, Department of Business Administration , Xavier University, Patna
Dr. Supriya Shree
Topic :
Federated Learning Based Framework for Privacy-Preserving Employee Attrition Prediction in Human Resource Analytics
AP - Computer Science, Xavier University, Patna
3:08 PM | aap-mshz-taz


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BEST PRESENTATION
(Faculty - External)
Mrs. Shruthi DV
Topic :
AuraVote: A Secure, Decentralised Voting System with AI-Powered Liveness and Identity Verification Using ArcFace and Blockchain
AP - Department of Information science and Engineering , Malnad College of engineering, Hassan,
Dr. KP Mredula Pyarelal
Topic :
Artificial Intelligence based mathematical education in sustainable development for academics
AP - Applied Science and Humanities Department, Sardar Vallabhbhai Patel Institute of Technology, Vasad
Mr. Dheeraj Sharma
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AP - Electrical Engineering Department, Government Engineering College Baran, Rajasthan
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Understanding Digital Transformation: Building Futures

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ABSTRACT

Digital Transformation is becoming instrumental for optimizing use of emerging technologies to improve performance & integration of business processes and accomplish seamless operations, sustainability and growth. It includes but not limited technologies linked to process automation, analytics, data management & processing. While there is a common perception of digital transformation involving primarily translation of pre-existing business processes into digital format, organisations need to have sufficient foresight to prioritise developing & deploying capabilities for building futures (as opposed to the mere process digitalization). It involves coming to terms and understanding dynamic scenarios & trends within the respective industries and marketplaces, establishing digital environment-centred operational protocols and last but not least – focusing on the ‘’future’’ rather than events of the past! Furthermore, the emerging digital transformation drivers are likely to require refocusing from cut-throat competition within the respective industries but establishing and implementing collaborative approaches towards developing a prosperous benefit-all operational eco systems.

Keywords: Digital Transformation, Building Futures, Business Process Re-engineering.

ICFTSEMAP002

A Study of the Histories of People for Gerontology through Steps of Artificial Intelligence in Text Book

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ABSTRACT

Artificial intelligence (AI) has a big potential to help tackle educational system, especially during pandemic it has been issued for worldwide people's interests in many reasons. Particularly, education has very important issue for all people in the world. Nowadays, Al has the power to improve teaching and learning methods throughout all age from birth through death even before death and life after death. Consciousness is the most important to improve ability than memorization. In order to improve consciousness, there are among four cycling to grow such as Layered knowledge 層(Sou)知(Chi), round 環(Wa), practice for a lifetime 一生稽古(Isshoukeiko), and unnatural wonder 不自然(Fushigen)の(no)妙(Myou). These are AI goals to seek wisdom through experience. We can understand gradually meaning of Master 名人(Meijin), Expert 達人(Tatsujin), Ironman 鉄人(Tetsujin): Creation and evolution of intelligence through time and eternity.

Keywords: Gerontology, Artificial Intelligence, education, Consciousness.

ICFTSEMAP003

Collaboration between Teaching English and Renewable Energy: Fostering Sustainable Development and Language Proficiency

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ABSTRACT

In today's interconnected world, the intersection of education and sustainable development has gained significant attention. This abstract explores the potential collaboration between teaching English as a second language (ESL) and renewable energy education, with the goal of fostering sustainable development goals and enhancing language proficiency simultaneously. Renewable energy has emerged as a crucial solution to address the challenges of climate change and transition to a sustainable future. Education plays a vital role in equipping individuals with the knowledge and skills necessary to embrace renewable energy technologies and practices. Meanwhile, English language proficiency has become a global necessity due to its role as a lingua franca in academia, diplomacy, and the professional world. This abstract proposes that a collaboration between ESL and renewable energy education can lead to numerous benefits. By integrating renewable energy topics into ESL curricula, students can gain an interdisciplinary perspective on sustainable development, developing an understanding of the environmental, social, and economic implications of renewable energy technologies. This can foster a sense of global citizenship and responsibility among students. Incorporating renewable energy concepts into ESL education can also enhance language proficiency by providing students with contextualized learning experiences that require critical thinking, problem-solving, and communication skills. This can help students develop a deeper understanding of the language and its applications in real-world scenarios. Overall, the collaboration between ESL and renewable energy education has the potential to promote sustainable development and language proficiency, equipping students with the knowledge and skills necessary to address the challenges of the 21st century. Incorporating renewable energy concepts into ESL instruction can significantly improve language acquisition and proficiency. By integrating relevant terminology, debates, and discussions into the classroom, students are provided with engaging and meaningful content, which in turn enhances their motivation and participation. Through expressing opinions, debating

renewable energy policies, and collaborating on group projects related to sustainable energy solutions, students can develop their language skills in a practical and meaningful way. Moreover, collaboration between ESL and renewable energy education can create opportunities for authentic language use. Field trips to renewable energy installations, lectures from industry experts, and interactive projects allow students to practice English in real-life contexts, thereby enhancing their language fluency and cultural understanding. This experiential learning approach fosters a deeper appreciation of the language and its practical applications, while promoting cross-cultural understanding and exchange.

ICFTSEMAP004

A Study on Implementation of eLearning and Students Learning Behavioral Changes

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ABSTRACT

It is a well-known fact that education is essential in this modern age, in this modern era, society is undergoing many changes, but education is the main reason for it, although there are many factors involved. The demand for education is increasing day by day and the development and evolution of education is growing very fast. In this research article, this ariticle have based researcher experience on the opinions of many scholars about how to focus on the mindset of students in Information and Communication Technology Education or eLearning in the modern era. Many books and already written articles have been qualitatively researched to write this research paper.

Keywords: ELearning, ICT, Education, Developemt, Students.

Contribution Title Exploring Algorithmic Paradigms in Message Classification: Insights from the Enron Email Dataset

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ABSTRACT

This research focuses on message classification, specifically distinguishing between legitimate and spam messages. The paper emphasizes the importance of preprocessing textual data using vectorizers, introducing Count Vectorizer and TFIDF Vectorizer for this purpose. These vectorizers convert text into numerical representations. The dataset is split into training and testing data to facilitate model development and evaluation. Python, along with libraries such as scikit-learn and nltk, is used for model implementation, providing machine learning and natural language processing capabilities. Various algorithms, including decision trees, random forests, support vector machines, logistic regression, and neural networks, are employed, each initialized with specific parameters for optimization. Data is sourced from the Enron email dataset on Kaggle, comprising around 500,000 emails linked to Enron's investigation by the Federal Energy Regulatory Commission. The research objectives include training models with selected algorithms to accurately categorize messages and evaluating their performance using metrics like accuracy, precision, recall, and F1 score. Findings reveal weak positive correlations between message characteristics and the target variable. The developed models show promising performance, emphasizing the need to consider diverse factors and techniques in message classification. The study contributes insights into the relationships between message characteristics and classification accuracy, aiding the development of effective models across various domains.

Keywords: Message Classification, Spam Detection, Algorithms, Decision Trees, Random Forests, Support Vector Machines, Logistic Regression, Neural.

Green Fabrication of Candle Soot–Based Superhydrophobic Surfaces for Selective Oil–Water Separation

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ABSTRACT

Nowadays, the application prospects of superhydrophobic surfaces in oil-water separation have attracted more & more attention. Therefore, preparing a simple, low-cost, eco-friendly superhydrophobic surface remains challenging. The dip-coating approach successfully created the superhydrophobic sponge after candle soot nanoparticles, silica nanoparticles, and polyethylene were combined in acetone as a solvent to create a mixed solution. The findings demonstrate the superhydrophobic sponge's superior oil-water separation efficiency and high water contact angle (168°). This paper suggests a straightforward approach for creating a superhydrophobic oil-water separation material. The material is readily available, the preparation process is environmentally friendly, and it is expected to be widely used in practical manufacturing. The superhydrophobic sponge exhibits excellent durability and reusability, maintaining its performance after multiple cycles of oil-water separation. Its porous structure contributes to rapid absorption and effective separation of various oils from water. This method offers a promising solution for environmental remediation and industrial wastewater treatment applications. The sponge's fabrication process is scalable and adaptable to different substrates, enhancing its versatility. Additionally, its resistance to harsh chemical environments ensures long-term functionality in diverse conditions. Future research could focus on optimizing the composite ratios to further improve separation efficiency and mechanical strength.

Keywords: Superhydrophobic Surface, Candle Soot, Polymer, Oil-Water Separation, Water Contact Angle.

Students' Holistic Development in Higher Education: A Multidimensional Conceptual Perspective

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ABSTRACT

Higher education is more likely to foster rounded growth of students other than standard educational results. The modern discourse of education stresses the fact that the success of students is conditioned by the comprehensive development of cognitive, emotional, social, ethical, physical, career, and digital aspects. Nevertheless, the available literature tends to discuss these areas separately and this has led to the piecemeal picture of student development. To fill this gap, the current research adopts a conceptual research design to integrate the proven theories, empirical literature and the world education frameworks in the context of the holistic development of students in higher education. The paper conceptualizes the holistic development as a dynamic and interconnected process drawing on the perspectives of cognitive, humanistic, developmental, and policy oriented perspectives. The paper presents the seven fundamental dimensions of whole person development in a systematic way and the connections among them, their theoretical basis and conceptual relationships. The paper provides a complex structure combining old educational theories and the recent studies and the international agenda, which contributes to the richness of the conceptualization of the holistic educational research. The suggested viewpoint highlights the topicality of holistic growth in solving the existing higher education issues, such as student well-being, employability, ethical responsibility, and digital transformation. The paper adds to the existing knowledge in higher education by having an all-inclusive conceptual base that can be used in future empirical studies, development of frameworks and teaching practice to produce well-rounded, flexible and socially responsible students.

Keywords: Holistic Student Development; Higher Education; Multidimensional Learning; Student Well-Being; Employability.

ICFTSEMAP008

Multi AI Agent Planner

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ABSTRACT

This paper presents a **Multi-AI Agent Planner**, a scalable and modular framework designed to coordinate multiple autonomous agents across diverse domains such as travel planning, project management, education, event organization, and startup strategy development. Unlike traditional single-agent or domain-specific systems, the proposed planner enables real-time collaboration, synchronized task execution, and intelligent resource allocation among heterogeneous agents. Each agent operates independently using domain-specific reasoning models, while a centralized orchestration layer ensures effective communication, conflict resolution, and fault tolerance. The architecture supports adaptive task prioritization, dynamic reassignment of responsibilities, and continuous learning through feedback mechanisms. By leveraging distributed planning, modular system integration, and interoperable communication protocols, the planner enhances efficiency, flexibility, and scalability in complex multi-domain environments. The framework is designed for extensibility, allowing new agents and functionalities to be seamlessly integrated without disrupting existing operations. Experimental evaluations and practical use-case analyses demonstrate the system's effectiveness in managing complex workflows with minimal human intervention, positioning it as a robust solution for next-generation intelligent planning systems.

Keywords: Multi-AI Agent Planner, Multi-Agent Systems, Autonomous Agents, Distributed Planning, Task Orchestration, Intelligent Resource Allocation, Modular Architecture, Cross-Domain Applications, Agent Coordination, Intelligent Systems.

ICFTSEMAP009

Career Bot: A Speech-Driven, Skill-Based Career Guidance System with Smart Scheduling and ATS-Friendly Resume Generation

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ABSTRACT

Career planning and getting ready for a job are very important for students and people who already have jobs. The old ways of getting career advice are usually done by hand. They take a lot of time. They do not consider the individual and what they want. Many students and people looking for jobs have a time finding careers that are a good fit for them. They want to do something they're good at and something they like to do. They also want to use what they studied in school. Career planning and job preparation are necessary for career development and, for students and people who already work career planning is important. A lot of students and people looking for jobs have a time finding the right career. They want a career that's a good fit for the things they are good, at and the things they like to do. Careers should match their skills and interests. Finding a job and getting ready for a career is really tough these days. Many companies use computers to look at resumes before a person even sees them. So resumes have to be set up in a way and include certain words to get through the computer check. Career planning is a part of career development and job preparation and career planning is very important, for career development and job preparation. Career planning and job preparation go hand in hand. The proposed AI-based Career Guidance and ATS Resume Generation System integrates skill analysis, career recommendation, and resume generation into a single platform. The system assists users in selecting suitable career paths and generates ATS-compliant resumes based on their skills and preferences. This approach aims to enhance employability, reduce career-related confusion, and provide an efficient solution for career planning and job preparation.

Keywords: Career Guidance System, Artificial Intelligence, ATS Resume Generator, Skill Analysis, Career Recommendation, Learning Plan.

AI - Based Food Freshness Detection and Recipe Recommendation System

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ABSTRACT

Food spoilage and wastage remain significant challenges in household and small-scale food management due to the lack of simple and reliable freshness assessment methods. This paper presents an AI-based food freshness detection and recipe recommendation system that leverages computer vision techniques to classify food items based on visual appearance. A Convolutional Neural Network (CNN) is trained on a labeled image dataset to categorize food items into fresh, medium, and spoiled classes. Based on the predicted freshness category, the system provides suitable recipe recommendations to promote timely consumption of food items nearing spoilage. The proposed solution operates entirely in a software environment, ensuring low cost, ease of deployment, and accessibility for end users. Experimental evaluation demonstrates satisfactory classification accuracy, highlighting the effectiveness of deep learning for practical food quality assessment. The system supports sustainable food practices by reducing wastage and enhancing informed decision-making in food utilization.

Keywords: Food Freshness Detection, Convolutional Neural Network, Computer Vision, Image Classification, Food Waste Reduction.

Real-Time Fake News Detection using DeBERTa-V3 and Transformer

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ABSTRACT

Fake news dissemination through digital and social media has grown into a real threat, as it distorts public opinion, decision-making, and confidence in institutions. Manual fact-checking is time-consuming and slow, hence inappropriate for a real-time environment where the misinformation spreads within minutes. This study proposes a real-time fake news detection framework that incorporates the pre-trained DeBERTa V3 transformer model for text classification, fact-checking based on Wikipedia, and sentiment analysis to provide explainable and transparent outputs. The system is trained and tested by the LIAR factchecking dataset of short political statements labeled, and its deployment is performed via an intuitive interface using Gradio so that users can easily input the news text and get classifications, sentiments, and evidence that support those sentiments in real-time. The experimental results indicated that the proposed system can effectively classify fake from real news, while providing interpretable justifications, hence improving users' trust and promoting responsible consumption of online information. The proposed approach is appropriate for embedding into media monitoring tools, browser extensions, and learning environments to mitigate the effects of misinformation.

Keywords: Fake News Detection, DeBERTa-V3, Transformer, LIAR Dataset, Fact Verification, Sentiment Analysis, Misinformation.

ICFTSEMAP012

Development of a Soft-Switched Four-Port Converter for Hybrid PV-Battery Systems in Frequency Regulation Applications

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ABSTRACT

The growing need for sustainable energy solutions is driving the adoption of standalone photovoltaic (PV) systems, which generate electricity independently of the utility grid by harnessing solar power. These grid systems are particularly valuable in remote or rural regions where extending grid infrastructure is economically or logistically unfeasible. Traditional converter ensures a switching across a wide operating range is difficult, leading to suboptimal efficiency and higher ripple under variable PV and storage voltages and power levels . Therefore this project proposes a high gain three port DC-DC converter for standalone photovoltaic (PV) applications, featuring low ripple current at the PV input. The system aims to integrate multiple DC input sources such as a PV system, battery, and a DC supply to provide a stable AC output to the load. The core of the system is the three port DC-DC converter, which takes inputs from these diverse sources and delivers a regulated DC voltage to a three phase voltage source inverter. Soft-switching techniques reduce switching losses, enhancing efficiency and reliability under varying power demands. A dual layer control scheme employs a DSPIC30F4011 controller generating PWM signals for both port level regulation through a PI controller managing the DC-link voltage and grid side control utilizing DQ0 transformation, PI controllers for real and reactive current regulation, and precision power calculation blocks to ensure accurate active power injection. The VSI outputs are filtered and synchronized with the grid. This architecture optimizes energy sharing among PV and battery units ensuring rapid response to frequency deviations while maintaining DC link stability. This project is implemented by using MATLAB SIMULINK 2024a and DSPIC30F4011 controller.

Keywords: Soft Switching, Multiport Converter, Hybrid Grid, Frequency Support, Photovoltaic Integration, Battery Management, Control Algorithms, DC-DC Topology.

Piezoelectric Electricity Harvesting Method Using Road Bumps

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ABSTRACT

This paper presents a piezoelectric energy harvesting system integrated into road bumps, targeting the capture of mechanical energy generated from vehicular motion and its conversion into usable electrical energy for sustainable transportation infrastructure. The proposed system incorporates high-force piezoelectric stacks embedded within a mechanically amplified road bump structure to enhance stress concentration and electrical output. The generated alternating voltage is processed through an efficient power conditioning circuit consisting of rectification, filtering, and voltage regulation stages to produce stable DC power. The harvested energy is stored using a hybrid supercapacitor-battery configuration to ensure reliable power delivery for auxiliary applications such as LED street lighting, IoT sensor networks, and smart traffic monitoring systems. Experimental analysis under simulated and controlled traffic conditions validates the feasibility of the proposed system for micro-energy generation. Performance results indicate that while individual vehicle energy yield remains modest, cumulative energy generation under moderate traffic volumes can significantly contribute to localized infrastructure power requirements. The study highlights both the practical advantages and operational limitations of piezoelectric road bumps as a supplementary renewable energy solution for intelligent transportation systems.

Keywords: Piezoelectric Energy Harvesting, Road Bumps, Smart Transportation, Electric Vehicles (EVs), Renewable Energy, Power Conditioning, Energy Storage, Sustainable Infrastructure, Micro-Energy Applications.

ICFTSEMAP014

Enhace Text to SQL-Parser

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ABSTRACT

Enhancing Text-to-SQL parsing is an important development in the field of natural language processing, allowing users to access and interact with databases using simple textual queries. A Text-to-SQL parser converts natural language input into structured SQL commands, making database systems more accessible to non-technical users. Despite significant progress, challenges such as understanding complex database schemas, handling ambiguous user intentions, and generating accurate multi-table SQL queries remain. This project focuses on improving the efficiency and accuracy of Text-to-SQL parsing by incorporating better semantic analysis, schema linking strategies, and contextual learning models. The proposed enhancement aims to reduce query errors, improve robustness across different database domains, and support real-time query generation. By strengthening the mapping between user language and SQL structure, the enhanced parser can provide reliable data retrieval for applications in business intelligence, healthcare, education, and other data-driven sectors, bridging the gap between humans and database systems.

Keywords: Text-to-SQL, Parser, Natural Language Processing, Database Query, Schema Linking.

ICFTSEMAP015

Colon Cancer Classification from Histopathological Images Using Convolutional Neural Networks

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ABSTRACT

Colon cancer is one of the major causes of cancer-related deaths, and early diagnosis is essential for effective treatment. Traditional histopathological image analysis relies on expert pathologists and is often time-consuming and subjective. This paper proposes an automated colon cancer classification system using Convolutional Neural Networks (CNNs) to improve diagnostic efficiency and accuracy. Histopathological images from the LS25000 colon dataset are pre-processed using resizing, normalization, and data augmentation techniques such as rotation and flipping. The CNN model automatically extracts relevant features and classifies images into normal, benign, and malignant categories. Model performance is evaluated using standard metrics including accuracy, precision, recall, F1-score, and confusion matrix. In addition, Grad-CAM visualization is employed to highlight important image regions influencing the model's predictions, enhancing interpretability. Experimental results demonstrate that the proposed approach provides accurate and consistent classification, making it a reliable decision-support tool for assisting pathologists in early colon cancer detection.

Keywords: Colon Cancer; Convolutional Neural Networks; Deep Learning; Histopathological Images.

ML – Based Personalized Learning Path Generator

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ABSTRACT

Personalized learning remains a major challenge in traditional e-learning systems due to the lack of adaptive mechanisms that address individual learner differences in knowledge level, learning pace, and goals. This paper presents a machine learning–based personalized learning path generator that leverages learner data analysis to recommend customized learning sequences. Machine learning algorithms are applied to learner profiles and performance metrics to classify learners into appropriate proficiency levels and identify skill gaps. Based on the predicted learner category, the system generates an optimized learning path consisting of prerequisite, intermediate, and advanced learning modules. The proposed solution operates entirely in a software-based environment, ensuring scalability, low deployment cost, and ease of integration with existing e-learning platforms. Experimental evaluation demonstrates improved learning efficiency and engagement, highlighting the effectiveness of machine learning techniques in adaptive education. The system supports learner-centric education by enabling informed, data-driven learning progression and continuous personalization.

Keywords: Personalized Learning, Machine Learning, Adaptive E-Learning, Learning Path Generation, Intelligent Education Systems.

Smart College Event Tracker

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ABSTRACT

The Smart College Event Tracker is a web and cloud-based application designed to manage and track college events such as seminars, workshops, cultural programs, and technical fests. It provides a centralized digital platform to overcome issues like missed updates and low participation caused by traditional notice board communication. Faculty members can create event details including name, date, time, venue, and description, while students receive instant notifications, view upcoming events, and register online with a unique QR code generated for attendance tracking. Faculty scan the QR codes during events to ensure secure and accurate attendance recording. After participation, the system automatically generates digital certificates using predefined templates, where faculty upload only the authorized in-charge’s signature, and students can download their certificates directly. Additionally, faculty can generate reports showing total registrations, attendance count, and complete event history for effective record keeping and analysis. Overall, the system improves communication, reduces paperwork, saves time, enhances transparency, and streamlines college event management.

Keywords: Event Management System, QR Code Attendance, Student Notification System, Digital Certificates, Attendance Tracking, Report Generation.

Food Waste Management system

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ABSTRACT

The Food Waste Management System is a Flutter-based cross-platform mobile application designed to address inefficiencies in the food donation and waste management ecosystem. By connecting restaurants, NGOs, delivery agents, and orphanages in a unified digital platform, the system facilitates real-time listing, assignment, and tracking of surplus food donations, ensuring efficient redistribution to those in need. A key innovation is the integration of a spoiled food detection and rerouting mechanism, which identifies unsafe food during storage or transit and directs it to biodegradable waste converters for environmentally responsible disposal. Developed using Firebase Cloud Firestore, Node.js, and Google Maps API, the system ensures transparency, accountability, and end-to-end traceability across the donation process. With modules covering user management, donation handling, delivery coordination, and administrative oversight, the solution not only prevents foodborne illnesses and reduces landfill waste but also promotes community collaboration and supports sustainable development goals. This project offers a scalable, impactful, and technologically robust approach to transforming food surplus into a resource for social good while minimizing environmental harm.

Keywords: Food Waste Management, Flutter, Real-Time Tracking, Spoiled Food Detection, Sustainability.

ICFTSEMAP019

A Survey on Federated Learning for Label-Efficient and Robust Networks

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ABSTRACT

Machine learning has significantly changed how medical images are analyzed by allowing automatic diagnosis and helping doctors make better decisions. However, creating accurate models needs a lot of labeled data, which is hard to get because of privacy issues, the high cost of labeling, and legal limits on sharing data across institutions. Federated learning is a privacy-friendly approach that lets different organizations work together to train models without sharing their sensitive data. A current label-efficient federated learning system uses self-supervised learning and methods that account for data differences to enhance accuracy, generalization, and privacy using data that hasn't been labeled. Despite these advances, the system still struggles with data diversity and limited labeled examples, which lower its effectiveness. To address these problems, the Label-Efficient Self-Supervised Federated Learning (LE-SSFL) framework uses self-supervised contrastive learning to make better use of unlabeled data and an adaptive method to handle different types of client data, which helps improve the accuracy, scalability, and privacy of medical image analysis models.

Keywords: Federated Learning, Self-Supervised Learning, Medical Imaging, Data Heterogeneity, Label Efficiency.

ICFTSEMAP020

MRI Based Early Detection and Multi Stage Classification of Alzheimer's Disease using a Hybrid CNN-LSTM Deep Learning Model

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ABSTRACT

The most prevalent cause of dementia, Alzheimer's disease (AD), is linked to a persistent deterioration in mental and cognitive capacities. Dementia refers to the gradual deterioration of neuropsychiatric functions, which affects memory, reasoning, and the ability to perform daily activities. As they show structural alterations in various brain regions, brain medical images especially Magnetic Resonance Imaging (MRI) are frequently used to assess the formative stages of Alzheimer's disease. Deep learning algorithms and sophisticated computer based methods have been employed more frequently in recent years to identify and categorize Alzheimer's disease. In order to diagnose Alzheimer's disease early and classify it into multiple classes using MRI scans, this study uses a hybrid deep learning architecture that blends Deep Convolutional Neural Networks (DCNN) and Long Short-Term Memory (LSTM) networks. High level spatial characteristics are extracted from brain pictures by the DCNN component, and to improve classification accuracy, the LSTM layer records sequential relationships within these features. To improve prediction on unseen data, class-weighted learning and regularization techniques are applied to address data imbalance and prevent overfitting. Standard performance criteria including accuracy, precision, recall, F1-score, and validation loss are used to assess the suggested CNN–LSTM model. According to experimental results, the suggested strategy outperforms traditional CNN based techniques in properly identifying various stages of Alzheimer's disease, with a classification accuracy of 97%.

Keywords: Convolutional Neural Network (CNN), Alzheimer's Disease (AD), Deep Learning(DL), Dementia Detection, ADNI Dataset, Magnetic Resonance Imaging (MRI), Long Short-Term Memory (LSTM).

ICFTSEMAP021

Wearable Knee Biomechanics and Plantar Force Analysis for Functional Screening of Knee Osteoarthritis

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ABSTRACT

Knee osteoarthritis is a common degenerative joint condition that affects walking ability due to pain, stiffness, and restricted joint movement. In many cases, early gait-related functional changes are not easily identified using routine clinical assessment methods, which are often costly and limited to hospital environments. In this study, a wearable and non-invasive system is developed for the functional screening of knee osteoarthritis by jointly analyzing knee biomechanics and plantar force during walking. Knee joint motion is measured using an inertial measurement unit, while plantar force variations are captured using a load cell placed beneath the foot. The sensor data are processed using an ESP32 microcontroller and transmitted wirelessly for real-time observation. The combined assessment helps identify gait deviations such as reduced knee movement and uneven weight distribution, supporting early-stage screening and rehabilitation monitoring.

Keywords: Knee Osteoarthritis, Gait Analysis, Wearable Sensors, Knee Biomechanics, Plantar Force.

ICFTSEMAP022

Encephalon Controlled Thermo-Care Management Helmet

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ABSTRACT

Maintaining an optimal brain temperature is essential for normal neurological activity and for improving patient outcomes during critical medical conditions such as cardiac surgery, brain trauma, and postoperative recovery. Any deviation from the normal cranial temperature range can negatively influence cerebral metabolism, neural signaling, and overall cognitive function. Existing brain temperature management systems are often bulky, expensive, hospital-dependent, and require complex microcontroller-based control, limiting their accessibility and portability. This project presents the design and development of an Encephalon Controlled Thermo-Care Management Helmet, a non-invasive and hardware-based solution for effective scalp temperature regulation. The system utilizes a thermoelectric Peltier module to provide controlled cooling or heating, regulated by a W1209 digital thermostat to maintain the desired temperature range accurately. To overcome heat transfer issues caused by hair density, mechanical thermal probes are incorporated to ensure consistent thermal contact with the scalp. An independent safety mechanism using an LM393 comparator is implemented to automatically disconnect the power supply when unsafe temperature limits are exceeded, ensuring patient safety without relying on software control. The proposed helmet is compact, cost-effective, and easy to operate, making it suitable for clinical as well as supervised home-care environments. This system offers a practical alternative to conventional cranial temperature management devices by combining simplicity, safety, and effective thermal control.

Keywords: Brain Temperature Regulation, Thermo-Care Helmet, Peltier Module, W1209 Thermostat, Non-Invasive Thermal Therapy.

Intelligent Health Assistive Device and Emergency Support Care for Neurological Disorder

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ABSTRACT

Epilepsy is a neurological condition marked by sudden episodes that may occur without warning, making continuous supervision difficult and increasing the risk of injury when immediate help is unavailable. To address this challenge, this paper presents a wearable health assist system designed to support epilepsy patients through real-time monitoring and automatic emergency response. The proposed system is built around an ESP32 microcontroller and integrates multiple sensors, including an MPU6050 motion sensor, a MAX30102 heart rate sensor, and an LM35 temperature sensor. These sensors continuously capture body movement, pulse rate variations, and temperature changes that may indicate abnormal physiological conditions. The collected data are analysed using predefined threshold logic to identify critical events. When abnormal patterns are detected, the system activates local alerts through a buzzer and vibration motor and automatically initiates an emergency call to caregivers using a GSM module (SIM800). A rechargeable battery enables portable and uninterrupted operation. The system is designed to be compact, cost-effective, and non-invasive, making it suitable for daily wearable use. Experimental evaluation confirms timely alert generation and reliable system response, demonstrating the potential of the proposed solution to enhance patient safety and caregiver awareness in real-world scenarios.

Keywords: Wearable Health Monitoring, Epilepsy Assist System, ESP32, Multi-Sensor Integration, Emergency Alert System.

ICFTSEMAP024

Mobile-Based Electronic Health Record Platform for ASHA Workers

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ABSTRACT

In rural and remote regions, ASHA workers play a crucial role in delivering primary healthcare services and maintaining health records of mothers, children, and households. However, most field-level data collection is still carried out using paper-based registers, which often leads to data loss, delayed reporting, duplication of records, and difficulty in accessing updated information. Limited or unreliable internet connectivity further restricts the effective use of existing digital health systems during field visits. To address these challenges, this project proposes a mobile-based Electronic Health Record (EHR) companion designed specifically for ASHA workers operating in low-internet and offline environments. The system allows health data to be entered directly through a mobile application during community visits without requiring continuous network access. All records are securely stored on the device and automatically synchronized with a central server once internet connectivity becomes available. A web-based interface is provided for supervisors and healthcare administrators to review collected data, monitor follow-up activities, and generate basic reports for planning and decision-making. The proposed system is developed using a mobile-first approach with standard web technologies for the user interface and a Python-based backend for data processing and synchronization. Structured databases are used to ensure data consistency and reliability. This project demonstrates a practical and scalable solution to improve digital health record management, reduce manual workload, and enhance the efficiency of community healthcare services in rural areas.

Keywords: ASHA Workers, Electronic Health Records, Mobile Health Application, Offline Data Entry, Community Healthcare, Data Synchronization.

ICFTSEMAP025

Adaptive Wearable Device for Real-Time Suppression of Hand Tremor in Parkinson's Disease

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ABSTRACT

Parkinson's disease is a progressive neurological disorder that significantly affects motor function, with hand tremor being one of its most disabling symptoms. Existing treatment approaches for tremor management include medication, invasive surgical procedures such as Deep Brain Stimulation, and a limited number of wearable assistive devices. Although these methods provide symptomatic relief, they are often associated with high cost, surgical risk, and limited adaptability to variations in individual tremor characteristics. Furthermore, many wearable tremor suppression systems operate using fixed control parameters, resulting in reduced effectiveness under changing tremor conditions. This project presents the design and development of an adaptive closed-loop wearable system for the suppression of Parkinsonian hand tremor. The proposed system continuously acquires hand motion data using an inertial measurement unit and performs real-time signal processing to extract tremor-specific features such as frequency and amplitude. Based on this analysis, an adaptive control strategy dynamically regulates a vibration actuator to generate counter-phase mechanical stimulation, thereby reducing tremor intensity. The system selectively targets pathological tremor while preserving voluntary hand movements. The proposed non-invasive, low-cost wearable solution aims to improve functional hand stability and support daily activities in individuals with Parkinson's disease.

Keywords- Parkinson's Disease, Hand Tremor, Tremor Suppression, Wearable Device, Closed-Loop System, Adaptive Control, Inertial Measurement Unit (IMU), Vibration Actuator, Non-Invasive Therapy, Real-Time Signal Processing

ICFTSEMAP026

Blockchain-Based Academic Certificate Verification

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ABSTRACT

Traditional certificate verification processes are often time-consuming, costly, and vulnerable to fraud. This project proposes a blockchain-based certificate verification system to address these challenges by storing academic and professional credentials on a secure, decentralized ledger. Certificates are converted into digital identifiers using cryptographic hash codes and recorded immutably on the blockchain. Institutes can upload and verify certificates through a secure portal, while students can log in with their IDs and passwords to view their certificates. Organizations and employers can verify certificate authenticity directly using a unique verification ID, eliminating intermediaries. This approach ensures instant, transparent, and automated verification, reduces administrative overhead, and enhances trust in credential authenticity. The system provides a scalable and reliable framework for secure credential management in both educational and professional domains.

Keywords: Decentralized Ledger, Cryptographic Hash Function, Digital Certificates, Smart Contracts

Modeling & Analysis of Cylinder Block for V8 Engine

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ABSTRACT

Loss of heat is a significant factor in the performance of internal combustion engines. In addition, a heat transfer phenomenon causes mechanical stresses that are thermally induced, compromising the efficiency of engine components. In engine design, the capability to determine heat transfer in engines plays a vital role. Today, the simulations are progressively being made at a much earlier stage of engine production with numerical simulations In the current research V type multi-cylinder assemblage is modeled. This design is introduced to ANSYS and completed the consistent state thermal and constructional investigation for anticipating heat stress, heat transference, heat flux in contrasting and two distinct materials (FU 4270, FU 2451) from presented material (Aluminium). Heat transfer is the significant part of power change in internal ignition engines. Finding problem areas in a strong wall is utilized as a driving force makes a plan a superior chilling system. Quick transitory heat fluxes with the ignition chamber and the strong divider have to be explored to comprehend the impacts of non-consistent temperatures.

Keywords: Cylinder Block, V8 Engine, Design, Analysis.

ICFTSEMAP028

Lost and Locate Centralized Smart Recovery System

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ABSTRACT

The loss and misplacement of personal and valuable items is a frequent problem in both urban and rural environments, often leading to stress, time loss, and financial impact. Traditional methods of locating lost items such as manual inquiries, notice boards, and social media posts are inefficient, unstructured, and lack a centralized tracking mechanism. These limitations reduce the probability of successful recovery and delay the identification of matching lost and found reports. The system enables users to report lost items, register found items, and track recovery status in real time. Advanced data processing and matching techniques are applied to analyse item attributes such as category, location, date, and description to improve identification accuracy. The application is developed and simulated at an early design stage to evaluate system performance, data flow, and reliability under different user scenarios. Multiple data models and storage strategies are analysed to optimize system efficiency, security, and scalability.

Keywords: Lost and Found System, Item Recovery, Real-Time Tracking, Data Matching, Centralized Tracking.

ICFTSEMAP029

Survey on Real Time Wireframe Pothole Detection System

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ABSTRACT

Road surface irregularities like potholes present major risks to vehicle safety, passenger comfort, and the efficiency of traffic flow. These issues have significant implications in the transportation domain, as they directly affect road safety, traffic management, and overall driving experience. With the growth of autonomous vehicles and smart city initiatives, the need for intelligent systems that can accurately detect and analyze road conditions in real time has become increasingly important. Traditional approaches for pothole detection have been enhanced through the integration of computer vision and artificial intelligence. These systems utilize cameras to capture road images, which are then processed using deep learning models to detect and classify potholes. By applying AI-based image analysis, the existing methods effectively automate the identification of road surface irregularities, offering a faster and more efficient alternative to manual inspections and sensor-based techniques. The limitation in existing systems is their reliance on two-dimensional bounding boxes or simple classification, which lack depth and shape information. To address this limitation, my project proposes a Real-Time Wireframe Pothole Detection System that generates a three-dimensional wireframe of road surfaces using camera input. The proposed model aims to integrate depth estimation with wireframe modeling for precise analysis of surface irregularities, enabling better decisions for autonomous vehicles and improving road safety.

Keywords: Real-Time Detection, Pothole Detection, Wireframe Modeling, Computer Vision, Deep Learning.

Sensor Driven Leak Detection and Automatic Valve Shutdown System for Oil and Gas Infrastucture

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ABSTRACT

Leakage in oil and gas pipelines is a major safety concern, as it can lead to fire hazards, environmental pollution and financial loss. This proposed system presents a sensor based leak detection system with automatic emergency valve closure to provide early warning and immediate control during leakage conditions. The aim is to improve safety by continuously monitoring pipeline conditions and responding without human delay. In the existing system, leak detection is mainly carried out through manual inspection and periodic monitoring which may not detect sudden or small leaks in time. These methods are often result in delayed response and increased risk. To overcome these issues, the proposed system introduces real time sensing and automated decision making for faster and more reliable leak management. It uses gas, pressure, flow and temperature sensors connected to a microcontroller that continuously evaluate operating conditions. When abnormal valves are detected alarms are activated and an emergency shut off valve is automatically closed to stop further leakage. This system enhances operational safety, reduces environmental impact and offers a practical solution for the oil and gas industry.

Keywords: Leak Detection, oil and Gas Industry, Sensor Based System, Emergency Valve Closure, Pipeline Safety, Industrial Automation.

ICFTSEMAP031

Crack Detection System Using Drone-Captured Images

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ABSTRACT

Structural cracks in concrete infrastructures such as buildings, bridges, and pavements pose significant safety and durability risks if not detected at an early stage. Conventional crack inspection methods rely heavily on manual visual assessment, which is time-consuming, subjective, and unsafe for large-scale or hard-to-reach structures. To address these limitations, this project presents a Drone-Based Concrete Crack Detection System using Machine Learning, integrated with an interactive Stream-lit web application. The system utilizes drone-captured images as input and employs a Convolutional Neural Network (CNN) to automatically detect the presence of cracks. In addition, image processing techniques such as grayscale conversion, thresholding, and pixel analysis are applied to estimate crack severity quantitatively. The system also generates downloadable CSV and PDF reports and provides audio-based result announcements to enhance usability. The proposed solution offers an efficient, scalable, and non-invasive approach for automated structural health monitoring and early crack detection.

Keywords: Concrete Crack Detection, Drone-Captured Images, Machine Learning, Convolutional Neural Network, Image Processing Techniques, Crack Severity Analysis, Stream-lit Web Application.

ICFTSEMAP032

Bianchi Type Perfect Fluid Cosmological Model in Modified Theory of Gravity

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ABSTRACT

A Bianchi type III Universe with a perfect fluid source is studied in f(R) gravity. We obtain exact solutions for the metric coefficients by assuming a power-law relationship between F(R) and the average scale factor and a proportionality between the shear and expansion scalars. We derive corresponding expressions for R, f(R), ρ, and p, and analyse key physical quantities. During the analysis, DEC was satisfied, but NEC and SEC were violated, indicating accelerated expansion. Despite the ω-ω ′ phase evolution, the equation of state remains in the phantom region. Stability is confirmed by perturbative analysis. A kinematic test such as the luminosity distance, the angular diameter distance, and the distance modulus is also consistent. It successfully describes late-time acceleration in an anisotropic framework.

Keywords: f(R) Gravity; Perfect Fluid; Bianchi Type III Space-Time; Modified Theory of Gravity;

Food Bridge AI-powered Food Redistribution Using Distance-Based ML Matching

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ABSTRACT

Food Bridge is a tech-driven platform that reduces food waste and combats hunger by connecting surplus food providers—such as restaurants, hotels, events, and households—with NGOs, shelters, and volunteers in need. Through mobile and web apps, providers post excess food, while nearby organizations receive real-time alerts for quick collection and distribution, ensuring efficiency, transparency, and food safety. The platform replaces unorganized manual methods with a structured digital workflow, offers smart matching to pair surplus food with the closest recipients, and provides analytics on food saved, emissions reduced, and community impact. Beyond addressing immediate hunger, Food Bridge promotes environmental sustainability, encourages social responsibility, and engages communities through awareness campaigns and volunteer participation. Scalable and adaptable, it can be implemented across institutions, cities, and regions to support long-term goals like zero hunger and responsible consumption.

Keywords: Community Participation; Environmental Sustainability; Food Redistribution; Hunger Reduction; Real-Time Notifications.

Real-Time Vision Explainer: Object Recognition and Multilingual Voice Narration

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ABSTRACT

Understanding visual information is a challenge for many users, especially when images contain multiple objects, people, or complex scenes. Existing solutions often provide limited explanations, depend heavily on proprietary systems, or lack multilingual and voice-based accessibility. This creates a gap for users who require detailed, understandable, and inclusive interpretation of visual content. To address this problem, this project proposes a real-time vision explanation system that analyzes images and provides meaningful explanations. The system allows users to authenticate securely and upload or capture images for processing. It detects objects and recognizes people using computer vision techniques, generates a descriptive summary of the image, translates the description into multiple languages, and produces voice narration for better accessibility. The system also stores user history in a structured text format for future reference. By combining vision analysis, language translation, and speech synthesis using free and open-source tools, the proposed solution enhances accessibility, learning, and interaction with visual data for a wide range of users.

Keywords: Real Time Vision, Object Detection, Multilingual Translation, Voice Narration, Artificial Intelligence, Computer Vision.

Modeling & Analysis of Cylinder Block for V8 Engine

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ABSTRACT

Loss of heat is a significant factor in the performance of internal combustion engines. In addition, a heat transfer phenomenon causes mechanical stresses that are thermally induced, compromising the efficiency of engine components. In engine design, the capability to determine heat transfer in engines plays a vital role. Today, the simulations are progressively being made at a much earlier stage of engine production with numerical simulations In the current research V type multi-cylinder assemblage is modeled. This design is introduced to ANSYS and completed the consistent state thermal and constructional investigation for anticipating heat stress, heat transference, heat flux in contrasting and two distinct materials (FU 4270, FU 2451) from presented material (Aluminium). Heat transfer is the significant part of power change in internal ignition engines. Finding problem areas in a strong wall is utilized as a driving force makes a plan a superior chilling system. Quick transitory heat fluxes with the ignition chamber and the strong divider have to be explored to comprehend the impacts of non-consistent temperatures.

Keywords: Cylinder Block, V8 Engine, Design, Analysis.

RiceShield: AI–IoT-Based Early Pest and Spoilage Prediction for Grain Storage

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ABSTRACT

Post-harvest grain storage often suffers significant losses from unnoticed pest infestations and spoilage. These issues usually start internally and go undetected until serious damage happens. To tackle this problem, we introduce RiceShield, a smart grain protection system using AI and IoT technology. This system predicts and identifies early signs of infestations in stored rice. RiceShield monitors important environmental factors like gas concentration, temperature, humidity, and monitoring by Camera controlled by an ESP32 platform. An increase in carbon dioxide levels is a warning sign of pest activity, while ammonia presence points to spoilage caused by microbial growth. By continuously tracking these conditions and sending timely alerts, the system allows for early action before visible damage occurs. This approach provides a non-invasive, cost- effective, and scalable solution that improves grain safety, reduces post-harvest losses, and supports sustainable agricultural storage methods.

Keywords: Smart Grain Storage, IoT, Pest Detection, CO_2 Monitoring, Spoilage Prediction, ESP32.

ICFTSEMAP037

A Student-Centric Mobile Platform for Extracurricular Talent Recognition (Skill streak)

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ABSTRACT

Extracurricular activities play a significant role in the holistic development of students by enhancing creativity, communication skills, leadership qualities, and practical knowledge. Despite their importance, most academic evaluation systems continue to emphasize examination scores and grades, offering limited scope for students to receive structured recognition for their extracurricular achievements. As a result, many student talents remain undocumented, underrepresented, and difficult to showcase in a professional manner. This paper proposes a Student-Centric Mobile Application for Extracurricular Talent Recognition, aimed at providing a dedicated digital platform where students can systematically record, organize, and present their extracurricular skills and accomplishments. The system allows students to create personalized talent profiles, categorize skills based on activity domains, upload achievement details, and showcase certifications and participations. The platform is designed to be intuitive and accessible, ensuring ease of use for students from diverse educational backgrounds. The application is developed using the Flutter framework for cross-platform user interface development, with Dart for application logic and an SQL-based database for structured and secure data management. The system architecture ensures efficient data flow and responsive interaction. Initial usage and analysis indicate that the proposed platform improves visibility of student talents, promotes confidence in skill presentation, and supports institutions and evaluators in identifying capable individuals beyond academic performance. The proposed system effectively bridges the gap between academic-focused evaluation and extracurricular talent recognition, contributing to a more balanced and inclusive student assessment approach.

Keywords: Extracurricular Talent Recognition, Student-Centric Mobile Application, Skill Profiling, Holistic Student Development, Flutter Framework, Talent Management System.

ICFTSEMAP038

Mental Health Dashboard Monitoring with Predictive Analytics

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ABSTRACT

Mental health problems such as stress, anxiety, and sleep disorders are increasing rapidly due to modern lifestyle, academic pressure, and work-related stress. Early identification a continuous monitoring of mental health conditions are essential to prevent severe psychological issues. However, traditional mental health assessment methods are mostly manual, time-consuming, and depend on physical consultations, which limits regular monitoring and early detection. This project proposes a Mental Health Monitoring Dashboard with Predictive Analytics, a web-based system designed to monitor, analyse, and predict mental health risks effectively. The system collects user-reported data such as mood, stress levels, and sleep patterns through a user-friendly interface. The collected data is processed and analysed using data analytics and predictive techniques to identify potential mental health risks at an early stage. The backend of the system is developed using Python and Flask, while the frontend is implemented using HTML, CSS, and JavaScript. An interactive dashboard visually presents insights, trends, and risk levels, making the results easy to understand for users. This system helps users to become aware of their mental health status and supports preventive care through early alerts and data-driven insights. Overall, the proposed system provides an efficient, accessible, and scalable solution for continuous mental health monitoring and early risk prediction.

Keywords: Mental Health Monitoring, Early Risk Prediction, Mood–Stress–Sleep Analysis, Predictive Analytics, Dashboard Visualization, Alerts and Insights.

ICFTSEMAP039

Weather Prediction from Satellite Images Using Machine Learning

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ABSTRACT

The current mini project aims at implementing the concept of Machine Learning to forecast weather conditions based on the satellite images, and its main objective is to assist agriculture. The system starts with a secure login module in which the users can enter the application with the help of a username and a password. Once a user has logged in successfully, he or she can use three major features. The initial one is Image-Based Weather Prediction, in which the user will post a picture pertaining to weather. The image is analyzed with the help of a pre-trained Convolutional Neural Network (CNN) model that classifies the weather as one of four, such as cloudy, sunny, foggy, or rainy. The system inquires the user on the crops that are being cultivated in their agricultural land based on the forecasted weather condition. This input may be done by typing or a speech-to-text option that uses a microphone which is provided to the user. The system proposes a proper level of irrigation or water consumption, according to the weather condition and the crops that have been chosen. The second characteristic is GPS-Based Weather Predication. When the user presses the button of My Location, the system retrieves the current location of the user and then forecasts the weather of the place. Just like the first one, the user will be required to indicate the crops under farming and according to the estimated weather, appropriate water usage or irrigation recommendations are given. This also has the speech to text feature to help the user. The third and last is an Agricultural Chatbot that responds to queries in agriculture. They can query on what crops to grow during a particular season, how to use pesticides and how to take care of their crops and any other related farming aspect and the chatbot gives the information accordingly. On the whole, the project will assist farmers in making sound decisions, using a lot of water optimally, and applying the latest machine learning technologies in agriculture.

Keywords: Weather prediction, Satellite Images, Machine Learning, CNN, Agriculture.

ICFTSEMAP040

Influence of Tunnel Geometry on Collapse Behaviour: A Numerical Study

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ABSTRACT

Tunnel collapse is a common concern in underground construction, especially when tunnels are driven through weak or jointed rock. Although ground conditions and support systems are often studied in detail, the effect of tunnel geometry on collapse behaviour is still not fully understood. This paper examines how different tunnel shapes influence collapse mechanisms using numerical modelling. Numerical analyses are carried out for circular, horseshoe, and D-shaped tunnels under varying overburden depths and rock mass quality conditions. Additional parametric studies are performed by changing joint orientation, groundwater pressure, and support conditions for each tunnel geometry. Collapse behaviour is evaluated by analysing stress distribution, development of plastic zones, deformation patterns, and failure modes obtained from the simulations. The results indicate that tunnel geometry has a noticeable influence on stress concentration and deformation, leading to different collapse patterns under similar ground conditions. A comparative assessment of tunnel shapes is presented to identify their relative stability. The study highlights the importance of considering tunnel geometry explicitly during design and provides useful insight for safer and more informed tunnel planning.

Keywords: Tunnel Geometry, Tunnel Collapse, Numerical Modelling, Tunnel Stability.

ICFTSEMAP041

Drug Free Steam and Heat Therapy System for Sinus Congestion Relief

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ABSTRACT

Sinus congestion is a widespread upper respiratory condition that causes nasal blockage, mucus accumulation, facial pressure, and discomfort. Conventional treatments often rely on decongestant medications and antibiotics, which may lead to side effects or dependency with prolonged use. This paper presents the design and development of a Drug-Free Steam and Heat Therapy System aimed at providing safe, non-invasive symptomatic relief for sinus congestion. The proposed system combines controlled steam inhalation with localized heat therapy to loosen mucus, improve sinus drainage, and reduce facial pain. An ESP32 microcontroller serves as the central control unit, regulating therapy duration and temperature through sensor-based monitoring and relay switching. User interface components such as an LCD display and buzzer provide real-time status updates and safety alerts. The prototype demonstrates consistent therapy delivery with automatic shutoff mechanisms to prevent overheating, making it suitable for repeated home-based use. The system offers a low-cost, user-friendly, and medication-free alternative for sinus congestion management. Future enhancements may include advanced feedback control and clinical validation for broader healthcare applications.

Keywords: **Sinus Congestion, Steam Inhalation Therapy, Localized Heat Therapy, Drug-Free Treatment, ESP32 Microcontroller, Temperature Monitoring, Relay Control, Assistive Healthcare Device, Non-Invasive Therapy, Home-Based Symptom Relief.**

An IoT-Integrated RFID System for Smart Canteen Payment Management

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ABSTRACT

People really need a way to pay for food in school and office cafeterias. The old ways of paying with cash or tokens are causing a lot of problems. They make people wait in lines and sometimes the person taking the money makes mistakes. This project is about making a system for paying in cafeterias. It uses technology to make paying faster and more secure. The system is called a integrated RFID system for smart canteen payment management. It automates the billing and monitoring processes. In this system each person gets a special card with an RFID tag. When they want to buy something they just tap the card at the counter. The payment is made. The payment is made without touching anything. It is very convenient and easy to use. The integrated RFID system for smart canteen payment management is a good solution, for this problem. The RFID reader figures out who the user is. The IoT module then connects the system to a computer server on the internet. This server stores information about the food the user picks and the money they spend. The system adds up the cost of the food away using prices that were set before. It then takes this amount out of the users stored money .People in charge can see what is happening with transactions, user money and sales, on the internet at the time. This helps them make choices and take care of things better. The system makes waiting times shorter. It gets rid of the need to add up numbers by hand. The RFID system reduces the need to handle cash. This makes the whole service work better and faster. The system improves the service efficiency of the RFID system. Additionally, stored data helps analyze consumption patterns and supports inventory planning. Therefore, the proposed solution provides a reliable, scalable, and user-friendly digital payment platform that enhances operational efficiency and promotes smart canteen management using IoT and RFID technologies.

Keywords: RFID, SmartCard, ESP32-Microcontroller, Fare Optimization, Cashless Payment, Canteen Management System, Real Time Monitoring.

ICFTSEMAP043

Drug Free Steam and Heat Therapy System for Sinus Congestion Relief

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ABSTRACT

Sinus congestion is a widespread upper respiratory condition that causes nasal blockage, mucus accumulation, facial pressure, and discomfort. Conventional treatments often rely on decongestant medications and antibiotics, which may lead to side effects or dependency with prolonged use. This paper presents the design and development of a Drug-Free Steam and Heat Therapy System aimed at providing safe, non-invasive symptomatic relief for sinus congestion. The proposed system combines controlled steam inhalation with localized heat therapy to loosen mucus, improve sinus drainage, and reduce facial pain. An ESP32 microcontroller serves as the central control unit, regulating therapy duration and temperature through sensor-based monitoring and relay switching. User interface components such as an LCD display and buzzer provide real-time status updates and safety alerts. The prototype demonstrates consistent therapy delivery with automatic shutoff mechanisms to prevent overheating, making it suitable for repeated home-based use. The system offers a low-cost, user-friendly, and medication-free alternative for sinus congestion management. Future enhancements may include advanced feedback control and clinical validation for broader healthcare applications.

Keywords: **Sinus Congestion, Steam Inhalation Therapy, Localized Heat Therapy, Drug-Free Treatment, ESP32 Microcontroller, Temperature Monitoring, Relay Control, Assistive Healthcare Device, Non-Invasive Therapy, Home-Based Symptom Relief.**

ICFTSEMAP044

Multi Modal Sleep Triggering System with Adaptive Audio and Visual Feedback for Insomnia Patient

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ABSTRACT

Sleep has an important part in maintaining the organ’s function of the body. Sleep dislocation would affect a human's health and life quality. Wakefulness is a kind of sleep complaint that frequently occurs in society due to stress. Wakefulness treatment is generally carried out by pharmacological remedy which has high side goods. This study aims to give an wakefulness treatment result by using an audio-visual stimulation (AVS) which is suitable to stimulate the brainwave to enter the relaxed state, as an early stage toward the sleep process. The project was designed as a prototype that provided both visual and auditory stimulation. Visual stimulation was delivered through a flashing blue light brace with a wavelength of 463.92 nm. Auditory stimulation consisted of binaural beats with a frequency that continuously decreased from 8 Hz to 1 Hz.The prototype performance was tested by collecting electroencephalogram(EEG) quantitatively and Brunel Mood Scale(BRUMS) qualitatively, which were attained from two adult subjects. EEG analysis results showed that there was an proliferation of theta surge and diminishment of beta surge in both subjects which indicated drowsy state. BRUMS results show that the subjects came relaxed when using the prototype. From the test conducted, it could be concluded that the prototype has the implicit to improve the sleep quality to overcome wakefulness.

Keywords: Sleep quality, Wakefulness, Insomnia, Audio-Visual Stimulation (AVS), Brainwave Entrainment, EEG, Theta Waves, Non-Pharmacological Therapy.

React.JS And Express.JS Based Automated Pharmacy Management System with Expiry Control and Drug Interaction Alert

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ABSTRACT

A Pharmacy Management System is designed to simplify and automate the daily operations of a pharmacy. The system manages medicine inventory, sales, billing, and supplier information in an organized manner. It provides real time updates on stock availability, helping pharmacists avoid shortages and overstocking. The application improves accuracy in billing and record maintenance by reducing manual work. Patient and prescription details can be stored securely for easy access and reference. A user- friendly dashboard enables pharmacists to manage stock levels, generate reports, and integrate supplier data seamlessly. The system also features mobile compatibility, enabling remote access and management of inventory from anywhere, ensuring convenience and flexibility. When a doctor enters a prescription for a patient, the system automatically checks if the medicine is available, updates stock quantities in real time, and generates alerts if stock levels fall below a set limit. It also helps in generating reports for sales and stock analysis. By digitizing pharmacy activities, the system enhances productivity and operational efficiency. Errors caused by manual handling are significantly minimized.. The system keeps a complete history of patient medications and basic insights into frequently prescribed medicines. It generates automated daily usage reports and audit logs to track system activities.

Keywords: Pharmacy Management System, React.js, Express.js, Expiry Control, Drug Interaction Alert.

ICFTSEMAP046

Wearable Gait Assessment System for Enhancing Post Stroke Mobility

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ABSTRACT

Stroke is a leading cause of long-term disability, often resulting in impaired gait patterns, poor balance control and a higher risk of falls. These gait impairments greatly affect an individual's independence and overall quality of life. For effective post-stroke rehabilitation, regular and objective evaluation of walking performance is essential. However, most existing gait assessment techniques depend on laboratory-based motion capture systems, force platforms or subjective clinical observation methods. Such approaches are costly, lack portability and are not practical for continuous monitoring in home environments. In addition, many systems focus only on either kinematics or kinetics parameters independently, which limits complete gait analysis. To overcome these challenges, this project proposes a wearable gait assessment system utilizing advanced sensor and IoT technology. An inertial measurement unit (MPU6050) is used to capture lower-limb motion and a force-sensitive resistor (FSR) captures plantar force variations during gait cycle. The sensor data are processed using an ESP32 microcontroller and transmitted wirelessly for real-time monitoring. The gait parameters are visualized via a user-friendly smartphone interface. The proposed system is low-cost, portable and suitable for both clinical and home-based rehabilitation, supporting safer and more effective post-stroke mobility recovery.

Keywords: Gait Assessment; Inertial Measurement Unit; IoT; Kinematics; Kinetics; Post Stroke Mobility; Rehabilitation; Wearable System.

ICFTSEMAP047

Video Call Intercom Based on IP System with Vibration Sensor

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ABSTRACT

Security at entry points is a critical requirement in residential and commercial environments. Traditional intercom systems provide only basic audio communication and depend on manual interaction, making them inefficient in detecting unauthorized access or intrusion attempts. Moreover, the absence of real-time video communication limits the ability of users to visually verify visitors. This paper presents a Video Call Intercom Based on IP System with Vibration Sensor, designed to enhance security through automatic intrusion detection and real-time video communication. The proposed system integrates a vibration sensor with an ESP32 microcontroller to continuously monitor abnormal vibrations such as door knocks, tampering, or forced entry. Upon detection, an alert is immediately transmitted to a backend server using IP-based communication. The system enables instant notification and supports live two-way video communication using WebRTC through a web browser. By eliminating the need for dedicated intercom hardware and providing browser-based access, the system offers a cost-effective, scalable, and efficient solution for modern security applications.

Keywords: IP Intercom System, WebRTC, ESP32, Vibration Sensor, IoT Security, Real-Time Video Communication.

ICFTSEMAP048

Multi-Modal Real Time Surveillance System Architecture for Intelligent Crowd Control Using YOLOv8

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ABSTRACT

High-density gatherings such as the Kumbh Mela pose severe crowd-safety challenges, motivating advanced real-time surveillance solutions. This paper presents a novel multi-modal crowd-control system that integrates video-based detection using YOLOv8, embedded pressure sensors, and thermal imaging to provide robust crowd metrics through sophisticated sensor fusion techniques. Our architecture employs spatial-temporal alignment of heterogeneous sensor streams, followed by confidence-weighted Kalman filtering (60% visual, 30% pressure, 10% thermal) to generate reliable crowd state estimates. The system incorporates a comprehensive analytics engine performing density estimation, flow analysis, bottleneck detection, route optimization, and predictive forecasting. Experimental validation demonstrates superior performance over single-modality approaches, achieving 94.7% detection accuracy and sub-100ms response times. The framework addresses critical limitations in conventional surveillance systems while providing actionable insights for proactive crowd management in ultra-large-scale events.

Keywords: Crowd Control, Multi-Sensor Fusion, YOLOv8, Kalman Filtering, Real-Time Surveillance, Thermal Imaging, Pressure Sensing.

ICFTSEMAP049

Diagnostic Device to Detect Infection in Urine

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ABSTRACT

Urinary Tract Infection (UTI) is a prevalent issues that requires attention as early and timely detection helps prevent serious complications, early infection detection plays a crucial role in effective diagnosis and treatment, conventional diagnostic methods are time consuming and often takes several days to provide result which leads to various problems that may lead to increased medical expenses, prolonged discomfort. The aim is to create a very basic, low cost design that allows for the initial, preliminary detection of a UTI before causing into a major issues and makes early screening more accessible and affordable. The design will contain a transparent test tube and a urine sample placed within the tube. A light source will be transmitted through the urine sample being tested for UTI, the presence of bacteria or other foreign materials in the urine will obstruct or scatter the light being transmitted through the tube. As a result, the intensity of light passing through the LDR will be very low or high according to the presence of infection. In the prototype, the Light Dependent Resistor (LDR) which is integrated with microcontroller test the amount of light intensity passing through the urine sample based on the preset threshold value. The results can be viewed through LCD screen and saved in cloud for further analysis, this is simple, affordable and non invasive, making it easy to prevent UTI before causing it into severe condition.

Keywords: UTI, Early Detection, Diagnosis, Low Cost Design, Preliminary Test, Test Tube, Light Dependent Resistor (LDR), Intensity, Threshold Value, LCD.

Changing Food Consumption Patterns: A Study of Quick-Service Restaurants (QSRs) in Patna, Bihar

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ABSTRACT

The rapid expansion of Quick-Service Restaurants (QSRs) in Tier-2 cities such as Patna signifies a substantial transformation in consumer eating habits, lifestyle preferences, and socio-economic dynamics. As urbanization accelerates and disposable incomes rise, consumers increasingly favor convenient, affordable, and standardized food options offered by QSRs. This study examines the evolving food consumption patterns in Patna, focusing on the factors that shape consumer preference for QSRs, including demographic variables, digital influence, brand perception, and service convenience. Using a mixed-method research design, primary data were collected from 100 respondents through a structured questionnaire, supplemented by an extensive review of contemporary literature. Analytical techniques such as frequency distribution, percentage analysis, mean scoring, and chi-square testing were applied to interpret consumer behavior trends. The findings reveal a pronounced shift toward QSR consumption, particularly among individuals aged 18–35, driven by convenience, taste consistency, hygiene expectations, and the growing reliance on digital platforms such as Zomato, Swiggy, and social media channels like Instagram. These platforms significantly impact decision-making by offering promotional incentives, peer reviews, and effortless accessibility. The study concludes that QSRs play an increasingly pivotal role in Patna’s food ecosystem and highlights the importance of digital engagement, value-based offerings, and localized menu strategies. Practical recommendations and future research directions are proposed to assist QSR operators, marketers, and policymakers in leveraging consumer trends effectively.

Keywords: Quick-Service Restaurants, Consumer Behavior, Digital Influence, Food Consumption Patterns, Patna, Urban Lifestyle, QSR Preferences, Marketing Strategies.

ICFTSEMAP051

Low Cost Optical Sensing Device for Non- Invasive Hemoglobin Measurement

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ABSTRACT

Focuses on the design and development of a low-cost optical sensing device for the noninvasive measurement of hemoglobin levels in the human body. Hemoglobin estimation is a crucial diagnostic parameter for identifying conditions such as anemia, blood loss, and other hematological disorders. Conventional hemoglobin testing methods require blood samples, which can cause discomfort, risk of infection, and are not suitable for frequent monitoring. The proposed system uses optical sensing principles based on light absorption characteristics of hemoglobin. By transmitting specific wavelengths of light through the fingertip and analyzing the reflected or transmitted signal using photo detectors, the device estimates hemoglobin concentration without the need for blood extraction. A microcontroller processes the sensor data and displays the results in a simple and user-friendly format. This low-cost, portable, and noninvasive solution is especially beneficial for remote healthcare settings, routine health monitoring, and point-of-care diagnostics. The device aims to improve accessibility, patient comfort, and early detection of hemoglobin-related disorders.

Keywords: Non Invasive Diagnosis, Anemia Detection, Hemoglobin Measurement, Optical Sensing, Embedded System.

ICFTSEMAP052

A Multimodal Deep Learning Approach for PCOS Detection

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ABSTRACT

Polycystic Ovary Syndrome (PCOS) is a common hormonal disorder affecting women of reproductive age and is associated with infertility, metabolic disorders, and long-term health risks. In clinical practice, PCOS is commonly diagnosed through clinical examination, hormonal analysis, and ovarian ultrasound imaging. In recent years, deep learning based diagnostic systems have been developed to automatically analyse ovarian ultrasound images using Convolutional Neural Networks (CNNs). Although these systems reduce manual effort and improve consistency, most existing deep learning approaches rely only on imaging data and do not consider important clinical parameters such as menstrual history and metabolic indicators, which may affect diagnostic reliability. To address this limitation, this work proposes a fully deep learning based multimodal framework for automated PCOS detection by integrating clinical data with ovarian ultrasound images. CNN models are used to process encoded clinical attributes and to extract relevant structural features from ultrasound images associated with PCOS. The learned feature representations from both modalities are combined using a feature-level fusion strategy and passed through fully connected layers to perform final classification. The proposed system is implemented using Python, OpenCV, and NumPy. The proposed solution demonstrates the potential of multimodal artificial intelligence in advancing smart and scalable healthcare diagnostics, particularly in environments with limited access to specialised medical expertise.

Keywords: Polycystic Ovary Syndrome, Deep Learning, Convolutional Neural Network, Feature Level Fusion, Multimodal AI.

ICFTSEMAP053

Machine Learning Based Early Detection of Autism and Severity Classification

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ABSTRACT

Autism Spectrum Disorder (ASD) is a neurodevelopmental condition that appears in early childhood and affects social interaction, communication, and behavior. Early identification of ASD in toddlers is crucial. Intervention during the early developmental window, which spans from 12 to 36 months, has been shown to significantly improve learning, communication, and adaptive skills. The current ASD screening system mainly relies on manual questionnaires and clinical observation; these methods are time-consuming. The machine learning based early autism detection model focuses on creating an automated early screening system for ASD. The proposed system aims to identify children at high risk of ASD by analyzing behavioral screening responses and related demographic information in a reliable and objective way. This system is developed using Python programming with relevant machine learning algorithms to support early and trustworthy risk identification. Multiple supervised machine learning models, including Multilayer Perceptron (MLP), Naive Bayes, Linear Discriminant Analysis (LDA), Quadratic Discriminant Analysis (QDA), and AdaBoost, are trained and evaluated. These algorithms work together to detect autism and indicate severity levels such as mild, moderate, and high. This approach reduces human error and delays in detection. It also supports early clinical referral and timely intervention, leading to better developmental outcomes and quality of life for children and their families. The proposed system removes the need for repeated clinical visits and costly diagnostic procedures, thereby lowering the overall costs of screening and assessment.

Keywords: Autism Spectrum Disorder, Early Autism Screening, Toddler Autism Detection, Machine Learning Models, Timely-Intervention.

ICFTSEMAP054

Smart Retail 360: Inventory and Profit Optimization

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ABSTRACT

The current project titled **Smart Retail 360: Intelligent Inventory and Profit Optimization System** focuses on developing an intelligent retail management platform that automates inventory monitoring and enhances profit optimization using machine learning and data analytics techniques. The main objective of the system is to improve efficiency in retail businesses by managing inventory data, monitoring stock levels, predicting restocking requirements, and analyzing sales performance within a centralized digital dashboard. The system begins with a secure login module to ensure authorized access and data protection. After successful authentication, users can enter product details through an Excel-like interface or upload inventory data directly using Excel files. Once the data is stored in the database, the system continuously monitors stock levels throughout the product lifecycle, identifies low-stock items based on predefined thresholds, and generates real-time alerts to prevent shortages. It also tracks product expiry dates and provides notifications to reduce wastage and financial loss. In addition, the system analyzes historical sales data using machine learning models to forecast future demand and recommend optimal restocking quantities. The platform automatically generates daily sales reports, calculates revenue and profit margins, and presents visual analytics through an interactive dashboard. Furthermore, an integrated business advisory chatbot allows store owners to receive intelligent suggestions regarding pricing strategies, promotional offers, and product performance improvement with proper justification based on sales insights. The system is designed to be scalable and adaptable for small, medium, and large retail businesses. Overall, Smart Retail 360 minimizes manual errors, enhances operational efficiency, supports data-driven decision-making, and increases profitability through intelligent automation and predictive analytics.

Keywords: Retail Management, Inventory Optimization, Sales Prediction, Machine Learning, Data Analytics, Profit Analysis, Business Intelligence.

ICFTSEMAP055

Click to Wear: Smart Fabric Drape & Style Studio

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ABSTRACT

The fashion and retail industry is rapidly shifting toward digital platforms, creating a growing need for smarter and more personalized shopping experiences. However, one major challenge in online shopping is that customers cannot clearly see how a garment will look before purchasing it, which often leads to uncertainty, incorrect buying decisions, and high return rates. To address this issue, this paper presents an AI-based Virtual Try-On system that allows users to digitally visualize clothing in realistic 2D and 3D formats. Using computer vision and deep learning techniques, the system detects garment regions, aligns them accurately, and preserves fabric texture, color, and structure to create natural-looking results with minimal distortion. Designed with a flexible and scalable architecture, it supports various clothing styles and can be integrated into web and mobile retail platforms. Experimental results show improved visual accuracy and better detail preservation compared to traditional methods, while user feedback highlights increased purchase confidence and engagement. Overall, the proposed system offers a practical and innovative solution that bridges the gap between physical and online shopping, reduces return rates, and promotes a more convenient and sustainable fashion experience.

Keywords: Artificial Intelligence (AI), Virtual Try-On, Computer Vision, Deep Learning.

Plug AI: A Unified Intelligent Platform for Conversational, Sentiment and Predictive AI Integration

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ABSTRACT

Artificial Intelligence (AI) technologies are increasingly used across domains such as education, healthcare, business analytics, and customer support. However, most AI systems are developed as standalone applications, resulting in fragmented architectures, high integration cost, and limited scalability. This paper proposes Plug AI, a unified AI platform that integrates Conversational AI, Voice Interaction, Sentiment Analysis, and Predictive Analytics into a single centralized system. The platform is built using modern web technologies and follows a microservices architecture to ensure scalability and modularity. The system enables real-time interaction, multi-dimensional sentiment evaluation, and predictive insights based on historical and live data. Experimental analysis shows that the proposed system reduces integration complexity, improves response efficiency, and enhances user experience compared to traditional isolated AI systems. The Plug AI platform provides a scalable and user-friendly solution for intelligent decision-making and multi-domain AI applications.

Keywords: Artificial Intelligence; Conversational AI; Predictive Analytics; Sentiment Analysis; Unified Platform.

ICFTSEMAP057

Multimodal Diagnosis Prediction Through Neuro-Symbolic Integration

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ABSTRACT

Alzheimer's disease (AD) is a progressive neurodegenerative disorder that gradually impairs memory and cognitive function. Early diagnosis, particularly at the Mild Cognitive Impairment (MCI) stage, is essential for enabling timely intervention and effective disease management. Although deep learning–based diagnostic systems achieve high classification accuracy using MRI data, their black-box nature limits interpretability and reduces clinical trust. Most existing frameworks focus primarily on single-modality imaging data, neglecting the integration of complementary clinical and genetic information. Recent neuro-symbolic approaches, such as NeuroSymAD and Logical Neural Networks (LNNs), have introduced explainable reasoning by combining neural perception with symbolic logic. However, these systems remain limited in multimodal integration and fine-grained staging of Alzheimer's disease. To address these limitations, this paper proposes a Multimodal Diagnosis Prediction framework through Neuro-Symbolic Integration. The proposed system integrates MRI images, clinical assessments, and genetic biomarkers within an interpretable architecture that combines deep neural networks with symbolic reasoning. The framework generates both diagnostic predictions and transparent rule-based explanations, enhancing accuracy, interpretability, and clinical reliability for Alzheimer's disease detection.

Keywords: Alzheimer's Disease, Neuro-Symbolic AI, Explainable AI, Logical Neural Networks, Multimodal Learning, Mild Cognitive Impairment.

ICFTSEMAP058

CLIPMIND: The Intelligent Video Shrinker

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ABSTRACT

The increasing volume of long-form educational and lecture-based videos has created a need for efficient summarization methods that preserve semantic meaning and narrative coherence. Many existing techniques rely mainly on visual features and overlook spoken content, which carries the core structure of knowledge-oriented videos. To address this gap, we propose ClipMind, a speech-driven video summarization framework that integrates transcription, semantic analysis, and video editing into a unified pipeline. The system uses automatic speech recognition (ASR) to generate time-aligned transcripts, applies topic-aware ranking and transformer-based abstractive summarization to identify key content, and maps selected segments back to video intervals to produce coherent highlight summaries. Experimental results show that ClipMind significantly reduces video length while maintaining high semantic relevance and improving user comprehension compared to baseline methods. The framework provides a scalable solution for intelligent summarization of long-form, knowledge-rich videos.

Keywords: Video Summarization; Speech-Driven Analysis; Automatic Speech Recognition; Abstractive Summarization; Transformer Models; Educational Videos.

Automated Healthcare & Emergency Response System for Lone Elderly and Pregnant Women (A-HERS A Smart, Continuous Health Monitoring & Auto-Alert System Using IoT

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ABSTRACT

This project introduces an IoT-based healthcare and emergency response solution for vulnerable individuals such as elderly people living alone and pregnant women. The system continuously monitors physiological vitals like heart rate, oxygen saturation, body temperature, and motion/fall patterns. Using Wi-Fi/GSM connectivity, the data is transmitted to a cloud platform where caregivers or doctors can track real-time health status. During abnormal conditions such as sudden fall, abnormal vitals, or prolonged inactivity, the system automatically: Activates a buzzer alarm locally. The collected data is sent to a cloud platform via Wi-Fi or GSM for real-time tracking by caregivers and doctors. During emergencies (fall, abnormal vitals, no movement for long duration), the system automatically: Activates a buzzer alarm, Sends SMS alerts to registered doctor/family, sends location data, Triggers automated ambulance notification.

Keywords: Artificial Intelligence, Smart Health Care, IoT, Lone Elders, Emergency Response System.

ICFTSEMAP060

A Review of Current Techniques for Debris Flow Susceptibility Assessment

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ABSTRACT

In the mountain region, debris flows are among the most recognized geological natural hazards, due to their unpredictability, rapid movement, and extensive runout distribution, which have been documented for several centuries. For reliable and efficient mitigation measures, debris flow susceptibility assessment (DFSA) is an essential approach. Over the past few decades, numerous methods have been developed to evaluate debris flow susceptibility. In this article, the authors study and discuss current techniques for DFSA in detail. These techniques are grouped into the following approaches: Qualitative, Semi-quantitative, and Quantitative. 30 research articles published between 2016 and 2025 were analyzed. Across the studies, 25 susceptibility assessment techniques were identified; among these, random forest (RF), logistic regression (LR), Support vector machine (SVM), and extreme gradient boosting (XGBoost) are most frequently employed. Overall, this article provides a comprehensive discussion of debris flow susceptibility assessment and serves as a valuable reference for scientists and researchers.

Keywords: Debris Flow, Susceptibility, Machine-Learning.

ICFTSEMAP061

Rail Madad with AI-Powered Complaint Redressal System

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ABSTRACT

Rail Madad is an essential platform for passenger grievance redressal in Indian Railways, but its current manual categorization and routing system often causes delays, inefficiencies, and inaccurate resolutions. These challenges are more critical when passengers submit complaints in the form of photos, videos, or audio, which require significant manual effort to interpret. To address this, the proposed project introduces an AI-powered Complaint Management System that integrates automation, intelligence, and predictive analytics into Rail Madad. Using computer vision, the system analyses images and videos to automatically categorize complaints such as cleanliness, staff behaviour, or infrastructure damage, while urgency detection models prioritize critical issues like safety hazards. Optical Character Recognition (OCR) extracts relevant text from visual data, and metadata analysis enhances context with time and location details. AI chatbots provide instant acknowledgments, gather additional inputs, and reduce response time. Smart routing algorithms ensure accurate forwarding of complaints to the right departments for quick action. Predictive maintenance models further analyse complaint trends to identify recurring issues and suggest proactive interventions. Sentiment analysis of passenger feedback and AI-driven performance monitoring ensure continuous improvement. By automating classification, prioritization, and routing, the system enhances resolution speed, accuracy, scalability, and passenger satisfaction, ultimately transforming Rail Madad into a faster, smarter, and more reliable grievance redressal system.

Keywords: Rail Madad, Artificial Intelligence, Complaint Management, YOLOv3, Chatbot, NLP, Indian Railways, Deep Learning, Grievance Redressal, Passenger Satisfaction.

Assistive Braille Communication System Using Wearable Glove

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ABSTRACT

Communication barriers faced by individuals with deaf-blindness severely limit their access to information, social interaction and independent living. To overcome these challenges, a wearable two-way communication glove is proposed to enable effective interaction between deaf-blind users and normal individuals. Existing communication methods for deaf-blind individuals mainly rely on human assistance, tactile sign language or static Braille tools, which lack portability, real-time interaction and user independence. This proposed system integrates a GSM module (SIM800L) to support wireless text communication over cellular networks. Incoming English text messages are received by an embedded microcontroller, which translates the characters into standard Braille patterns. These patterns are conveyed to the user through six miniature vibration motors mounted on the glove, with each motor corresponding to one Braille dot, thereby providing clear and accurate tactile feedback. For message transmission, the glove is equipped with four palm-mounted push buttons that allow the deaf-blind user to send predefined messages quickly and reliably. To enhance personal safety, a GPS module (NEO-6M) is incorporated so that emergency alerts automatically include the user's real-time location. The proposed prototype operates in real time, is portable, energy efficient, and requires minimal user training. Overall, the system provides an affordable, reliable and practical assistive solution that significantly improves communication, safety and independence for deaf-blind individuals in daily life.

Keywords: Deaf-Blind Communication, Assistive Technology, Braille Translation, GSM Module, GPS Module, Tactile Feedback.

ICFTSEMAP063

Sustainable Use of Treated Recycled Concrete Aggregates

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ABSTRACT

Structural Enhancement of Recycled Aggregates is a study focused on eliminating the performance deficits that prevent Recycled Concrete Aggregates (RCA) from being widely used in load-bearing structures. RCA traditionally suffers from high porosity and weak bonding due to old residual mortar, which creates a quality gap compared to natural stone. To overcome this limitation, this research critically evaluates varying beneficiation protocols, ranging from the physical stripping of cement layers to the chemical reinforcement of the material surface. The core of our analysis rests on the modification of the Interfacial Transition Zone (ITZ) to ensure structural integrity. We examine the efficacy of mechanical abrasion using the Los Angeles (LA) drum and advanced treatments like Graphene Oxide (GO) infusion to convert raw experimental data into actionable engineering solutions. The findings indicate that integrating optimized abrasion cycles with process changes, such as the Double Mixing Approach (DMA), can successfully reduce water absorption and elevate concrete strength to levels competitive with virgin aggregates.

Keywords: Recycled Concrete Aggregate, Surface Treatment, Pretreatment Methods, Mortar Removal, Mortar Strengthening.

ICFTSEMAP064

Digital Transformation of the India Tourism Industry Through Smart Service Integration

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ABSTRACT

Tourism plays a vital role in preserving cultural heritage and promoting cross-cultural understanding, particularly in a culturally diverse country such as India. With the rapid advancement of digital technologies, there is an increasing demand for intelligent and user-friendly platforms that provide comprehensive tourism information to international visitors. This paper presents the design and development of a web-based India Tourism website that delivers an integrated digital experience highlighting Indian culture, historical monuments, UNESCO World Heritage sites, and traditional cuisine. The proposed system emphasizes intuitive navigation and accessibility through a structured interface supported by an interactive India map that enables state-wise exploration. Augmented Reality (AR) models are incorporated to provide immersive visualization of selected monuments, enhancing user engagement and offering virtual previews of destinations to support informed travel planning. In addition, a visitor support module is integrated to assist users with essential travel-related guidance, thereby improving usability and overall satisfaction. The platform is developed using modern web technologies with a focus on responsiveness, scalability, and ease of use. The proposed system demonstrates how the integration of web-based interfaces and AR technologies can support smart tourism initiatives by delivering an informative, interactive, and culturally immersive experience for global tourists.

Keywords: India Tourism, Web Development, Augmented Reality, Cultural Heritage, UNESCO Sites, Interactive Map.

ICFTSEMAP065

Blockchain Integrated Secure Image Steganography using IPFS and Ethereum Sepolia Testnet

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ABSTRACT

The increasing use of digital communication has made it essential to maintain the confidentiality, integrity, and authenticity of sensitive information. Conventional image steganographic methods offer data hiding in digital images, but they fail to offer effective tamper proofing and secure ownership verification. To overcome these issues, this paper presents a Blockchain-Integrated Secure Image Steganography system using IPFS and Ethereum. In the proposed system, secret data is hidden within digital images using a Least Significant Bit (LSB) image steganographic method developed in Python. The stego images are then stored in the Inter Planetary File System (IPFS) for efficient and decentralized data storage. To ensure data integrity and secure access, the cryptographic hash values of the stego images and their corresponding IPFS Content Identifiers (CIDs) are securely stored on the Ethereum blockchain using smart contracts. The use of blockchain technology provides immutability, transparency, and tamper resistance, and IPFS provides decentralized storage without depending on centralized storage servers. The proposed system is validated to offer high image quality with negligible distortion and robust data security and traceability. This system is applicable for secure data sharing in confidential communication, digital forensics, and secure document transfer.

Keywords: Image Steganography, Blockchain, Ethereum, IPFS, Data Security, SHA-256.

ICFTSEMAP066

An Eco-Friendly Odor-Free Composting Technique for Households

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ABSTRACT

The proper disposal of home garbage has recently emerged as a *pressing* issue on a global scale. Also, the city government has been trying to get people to separate their dry and moist trash. But there is still a problem with trash management and a big domestic population that isn't helping out. This paper details the process of creating a waste management system that separates organic kitchen trash into dry and moist categories. Both models are part of the proposed system. The first step is to design a bin that can separate dry and moist trash. Additionally, a composting facility is showcased for the purpose of transforming organic waste into compost. Twenty homes' worth of garbage was fed into the suggested model, and the results show promise for trash segregation and composting. It is an eco-friendly and sustainable way to manage organic waste and it can have a number of benefits for the environment. Composting can help to reduce greenhouse gas emissions, improve soil quality, and conserve water. It can also help to reduce the amount of waste that goes to landfills, which can help to protect human health and the environment.

Keywords: Wet Waste, Dry Waste, Compost, Sorter Bin, Trash Management.

ICFTSEMAP067

An Efficient Tenant Led Virtual Machine Scheduling Using Machine Learning

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ABSTRACT

Cloud data centers often face the dual challenge of maintaining workload balance and maximizing resource utilization due to the highly dynamic and heterogeneous nature of hosted applications. To address these limitations, this study presents a hybrid VM scheduling framework that combines Bayes-based clustering with Particle Swarm Optimization (PSO). The model first applies clustering to group tasks based on workload similarities and resource demand characteristics, while Bayesian probability reasoning is employed to refine host selection and minimize overload risk. Subsequently, PSO is utilized to search iteratively for an optimal scheduling solution by employing a fitness function that considers response time, energy efficiency, and resource utilization. A matrix-based allocation model is introduced to represent scheduling states and guide final deployment decisions in each iteration. Experimental outcomes show that the proposed approach enhances load balancing, reduces execution delays, and improves system scalability, thereby ensuring more efficient and adaptive performance in multi-tenant cloud environments.

Keywords: Virtual Machine Scheduling, Load Balancing, Resource Allocation.

ICFTSEMAP068

Survey on AI-Powered Hybrid Model for Heart and Lungs Disease Detection Using Acoustic Signals

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ABSTRACT

Recent advancements in Artificial Intelligence have significantly improved automated analysis of heart sound signals for cardiac screening. Phonocardiograms (PCG) provide a non-invasive and cost-effective means of identifying abnormalities such as coronary artery disease (CAD), but publicly available datasets remain limited and severely imbalanced. These constraints hinder the performance of deep learning models that rely on large, diverse samples. To overcome this challenge, this study employs a Progressive Wasserstein GAN (PWGAN) to generate high-fidelity synthetic CAD-like PCG segments that capture clinically relevant spectral and temporal characteristics. Evaluation using the Fréchet Audio Distance and post-processing techniques, including bandpass filtering, confirms the realism and diagnostic value of the generated samples. Augmenting training datasets with these synthetic signals leads to substantial improvements in the sensitivity, specificity, and overall performance of multiple heart sound classification models. The results demonstrate that GAN-based augmentation is an effective and scalable solution for addressing data scarcity in biomedical audio analysis, supporting the development of more accurate and accessible PCG-based diagnostic systems. Future work highlights the need for enhanced evaluation metrics, reduced computational complexity, and integration with multimodal large language models to enable clinically deployable AI-driven cardiac screening tools

Keywords Coronary Artery Disease**,** Data Augmentation**,** Generative Adversarial Networks Heart Sound Classification, Phonocardiogram.

AuraVote: A Secure, Decentralized Voting System with AI-Powered Liveness and Identity Verification Using ArcFace and Blockchain

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ABSTRACT

Secure and transparent electronic voting remains a major challenge due to vulnerabilities in centralized infrastructures and insufficient voter authentication methods. Existing systems frequently depend on weak biometric checks or static credential verification, making them susceptible to spoofing, identity duplication, and manipulation of centralized databases. AuraVote introduces a next-generation decentralized voting framework that integrates real-time artificial intelligence with blockchain-based immutability to guarantee voter authenticity and system integrity. The proposed model departs from traditional pipelines based on MediaPipe, MTCNN, and FaceNet by adopting a more robust architecture that includes RetinaFace for face localization, ArcFace for generating highly discriminative 512-dimensional embeddings, and cosine-similarity-based identity verification. Alongside facial verification, the system executes multi-modal liveness detection incorporating blink dynamics, natural micromovement estimation, and texture-frequency analysis to counter photo, video, and digital screen presentation attacks. A FastAPI-based verification backend processes live video streams from the browser and communicates verification results to a Next.js decentralized application. Verified users are then permitted to cast votes on an Ethereum Proof-of-Authority (PoA) blockchain using MetaMask, ensuring tamper-proof vote recording and cryptographically signed voter participation. This paper details the architecture, AI algorithms, blockchain workflow, system security properties, and implementation strategy, demonstrating how combining modern deep-learning–based authentication with decentralized ledgers establishes a secure, scalable, and trustworthy e-voting ecosystem.

Keywords: ArcFace, FastAPI, Blockchain, PoA Network, Liveness Detection, Face Recognition, Solidity, Next.js, Decentralized Voting, RetinaFace.

BMI – Personalized Smart Wearable Knee Brace with Sensor-Driven Adaptive Exoskeleton Assistance

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ABSTRACT

The Knee Brace employs sensor-driven adaptive exoskeleton technology to assist individuals who experience knee pain and reduced mobility due to medical conditions like Osteoarthritis. Unlike traditional knee braces, exoskeleton-based systems provide active assistance that improves movement and relieves pain. The project combines a load sensor, flex sensor and pressure sensor with an additional lightweight frame made out of soft material, NodeMCU, a rechargeable battery, a Servo motor and a vibration motor. By monitoring the user's knee angle, load and stress in real-time and allowing for on-demand adjustment by the user of the DC motor, it reduces the amount of stress on the knee. The user’s Body Mass Index is calculated using a mobile app and used to set safe load limits. Users with higher Body Mass Index receive increased knee support to reduce joint stress, while vibration feedback alerts them when unsafe thresholds are reached. The creation of a low-cost solution that is both user-friendly and mobile by using data logging as a tool for rehabilitation, enhanced by the use of the IoT in combination with the study of the role that Soft Robotic Systems have on reducing Injuries and shifting the Gait of Users to minimize the risk of injury.

Keywords: Sensor-Based Assistance, Adaptive Motor Control, Wearable Technology, Osteoarthritis Support, Vibration Haptic Feedback and Real-Time Monitoring.

ICFTSEMAP071

IoT-Based Biometric Medicine Distribution System for Smart Healthcare

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ABSTRACT

The rapid growth of population and increasing demand for healthcare services have created significant challenges in the safe and efficient distribution of medicines, especially in home-care environments. Manual Medicine dispensing often leads to medication errors, unauthorized access, and drug misuse, which can seriously affect patient safety. To address these issues, this project proposes a Biometric Medicine Distribution using Arduino Uno and NodeMCU (ESP8266). The system employs fingerprint-based biometric authentication to ensure secure, controlled access to medicines for authorized users, including patients and caregivers. The Arduino Uno acts as the primary controller, handling user authentication, motor-driven dispensing, and interfacing with an LCD and buzzer for real-time feedback and alerts. Upon successful verification, the prescribed medicine is automatically dispensed, reducing human intervention and minimizing dosage errors. The NodeMCU enables IoT-based connectivity, allowing real-time data on medicine usage. This feature is particularly beneficial for home-based patient care. The proposed system is cost-effective, compact, and energy-efficient, offering a reliable solution for secure medicine management in smart healthcare and home-use applications.

Keywords: Biometric Authentication, Automated Medicine Dispensing, Arduino Uno, Internet of Things, Fingerprint Sensor, Home-Based Patient Care, Medication Safety.

ICFTSEMAP072

Automatic Waste Segregation System and Environment Awareness Application

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ABSTRACT

Waste management is one aspect of Environmental Sustainability which is an increasing global challenge and maintaining waste on a global scale, Waste management is a key part in achieving Sustainable Development Goals. This paper proposes an Artificial Intelligence based Waste Classification System using Deep Learning Technologies in the derivation of an Interactive Waste Classification System with a purpose, to enhance users' awareness of their responsibilities regarding their own waste disposal practices through the use of gamification principles incorporated into this system. The Waste Classification System uses a Convolutional Neural Network in conjunction with MobileNetV2 architecture to classify and distinguish between 12 different categories of waste: plastic, paper, metal, glass, cardboard, biological waste, batteries, textiles, footwear, and general rubbish. Our proposed Intelligent Waste Classification System achieved 95.4% classification accuracy when assessed using the sample validation data set. All of these gamification components combined to create a platform capable of building long-term user engagement. In conjunction to the vast number of Eco-Points users can accumulate and accumulate progressively achieved Badge levels users can receive, Users gain virtual status when they compete on leaderboards, and track their progress visually using a user-friendly interface. The Intelligent Waste Classification System is accessible to all users through the use of the web browser, while the Intelligent Waste Classification System also provides Personalized and tailored recycling guidance via an Intelligent Conversational Chatbot. The Intelligent Waste Classification System contains an Administration Dashboard feature to monitor user behavior and to gather system usage statistics, and an Automated Email Notification System for community engagement. Our findings and extensive evaluation indicate that the application of artificial intelligence combined with principles of behavioral psychology will lead to a much more effective way of educating users about Waste Management Practices, and developing a practical method for implementing these practices.

Keywords: Waste Classification, Deep Learning, MobileNetV2, Gamification, Environmental Sustainability, Transfer Learning, Convolutional Neural Networks, Web Application.

ICFTSEMAP073

IoT-Based Real-Time Wild Animal Detection and Alert System Using Deep Learning and Multi-Sensor Integration

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ABSTRACT

Throughout the world - especially those areas close to Forest Ecosystems - humans often face difficulties or challenges created as a result of conflict with various types of animals around them. Such conflicts arise when livestock have been attacked by wild animals, or when there is a direct threat to the safety of humans from an attacking wild animal(s) that may cause harm or even death. This article introduces a new IoT based intelligent wildlife monitor/detection system. It will use the YOLOv8 deep learning algorithm/architecture combined with Environmental Sensor Networks (ESNs), as well as ESP32 microcontrollers for Real-time Data Acquisition (RDA) from 3 types of sensors: DHT11 Temperature & Humidity Sensor, MQ135 Air Quality Sensor and HC-SR04 Ultrasonic Distance Sensor; and provide the user with a way to monitor wildlife movement through actual real-time webcam images using access control (RBAC) based website (Flask Framework), receive alerts about detected wildlife on multiple channels (Telegram API, Google Text-to-Speech audio notifications and LCD displays) and maintain historical records of previous detection events and all user accounts through an SQLite Database. By testing the prototype and analyzing its findings against the traditional methods used; it offers a much better response time to classify detected animals (under 2 seconds) and provides a much longer period of time that the user has to respond than traditional methods of monitoring/watching for wildlife attacks.

Keywords: Wildlife Detection, YOLOv8, IoT, ESP32, Deep Learning, Human-Wildlife Conflict, Computer Vision, Multi-Sensor Integration.

Bilateral Trade in a Changing Global Economy: Economic Impacts of India–US Trade Agreements

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ABSTRACT

The global economy is undergoing structural changes driven by geopolitical tensions, supply-chain realignments, and evolving trade policies. In this context, India–United States bilateral trade relations have acquired growing economic and strategic significance. This paper analyses the economic impacts of India–US trade agreements and structured negotiation frameworks, focusing on tariff adjustments, sectoral competitiveness, non-tariff barriers, and trade creation effects. The study is based exclusively on official and authentic government sources, including releases from the Press Information Bureau (Government of India), the Office of the United States Trade Representative (USTR), and White House Joint Statements. The findings indicate that tariff rationalisation and regulatory cooperation can enhance trade efficiency and export competitiveness, particularly in pharmaceuticals, textiles, gems and jewellery, and selected manufacturing sectors. However, sensitive sectors such as agriculture may experience adjustment pressures, necessitating phased liberalisation and safeguards. The paper concludes that modern bilateral trade agreements increasingly extend beyond tariffs to encompass standards cooperation and supply-chain alignment.

Keywords: Bilateral Trade, India–US Trade Agreements, Tariffs, Competitiveness.

ICFTSEMAP075

AR-Based Study Helper: An Interactive 3D Learning System for Education

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ABSTRACT

In the modern era of digital education, students often encounter challenges in comprehending complex theoretical concepts through conventional text-based learning approaches. To overcome these limitations, this paper proposes an Augmented Reality (AR) - Based Study Helper that enhances the learning experience by integrating interactive 3D visualization, voice- based narration, and automated quiz assessment. The proposed system is designed as a web-based platform consisting of three main modules: Student, Expert, and Admin. The student module allows learners to log in, select topics using search or dropdown options, view interactive 3D models with synchronized voice and text explanations, and assess their understanding through quizzes. The expert module enables subject experts to add topics, upload 3D models with explanations, and create quiz questions. The admin module is responsible for approving experts, monitoring student performance, tracking expert contributions, and managing system statistics. All modules are supported by an integrated database for efficient data management. This interactive approach greatly improves student interest, understanding of concepts, and long-term learning compared to traditional methods.

Keywords: Interactive 3D Visualization, AR Learning, Web-Based Education System, Voice-Assisted Learning, Online Quiz Assessment.

ICFTSEMAP076

Survey On Tomato Maturity & Disease Prediction and Fertilizer Recommendations Using Deep Learning

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ABSTRACT

Artificial Intelligence (AI) is the domain that empowers machines to perform tasks such as learning, reasoning, and decision-making. Within AI, Deep Learning (DL) is a sub-domain that uses multi-layered neural networks to automatically extract features and patterns from complex data such as images. This has made DL a powerful tool in fields like agriculture, healthcare and automation. In existing system, they have been carried out on the precise and rapid prediction of tomato maturity using deep learning-based image recognition. The existing study successfully classified tomatoes into different maturity stages, enabling faster and more objective decision making compared to manual inspection. The work demonstrated the potential of computer vision for supporting the agricultural industry by improving harvesting efficiency and product quality. The main disadvantages of the existing system are that it only focused on tomato maturity detection and did not address plant health issues or provide guidance for disease management. To overcome these limitations, our proposed work integrates YOLO-based real-time image detection to not only identify tomato maturity but also detect common tomato plant diseases. Furthermore, the system recommends suitable fertilizers, pesticides, and organic manure, making it a comprehensive decision-support tool for farmers.

Keywords: **Artificial Intelligence, Deep Learning, YOLO, Tomato Maturity Prediction, Plant Disease Detection, Smart Agriculture.**

ICFTSEMAP077

Development of an Analog Pressure Monitoring and Haptic Feedback System for Transtibial Prostheses

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ABSTRACT

Below-knee amputees often suffer from discomfort, pain, and skin irritation because of improper pressure distribution between the residual limb and prosthetic socket during extended use. Walking activity, limb swelling, and residual limb volume changes further exacerbate discomfort and may diminish consistent prosthesis use. Traditional transtibial prosthetic sockets are passive and generally static, relying on manual adjustments, offering no real-time indication of excessive limb–socket interface pressure. The project proposes the design and development of a low-cost, Analog-based pressure monitoring and haptic feedback approach to improving prosthetic socket comfort and user safety. FSRs are placed at pre-identified high-pressure areas inside the socket to constantly monitor changes in pressure at the limb–socket interface. Sensor outputs are post-processed using an adjustable threshold level Analog comparator circuit to realize unsafe pressure conditions. There is no recourse to microcontrollers or complex programming in this system to retain cost-effectiveness for low resource-level prosthetic clinics.

Keywords: Below-Knee Amputees, Transtibial Prosthesis, Limb–Socket Interface Pressure, Force Sensitive Resistor, Analog Pressure Monitoring, Low-Cost Prosthetic.

ICFTSEMAP078

Machine Learning in Biomedical Implants: MATLAB Applications and Future Directions

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ABSTRACT

Using machine learning (ML) with biomedical implants has greatly improved patient care, diagnosis, and treatment. MATLAB is a popular tool for creating ML algorithms in this field because it has strong computing power and many features. This review looks at how ML is used with MATLAB in different biomedical implants like neural prosthetics, heart implants, orthopedic devices, and systems that monitor patients continuously. We talk about important ML methods like support vector machines, neural networks, random forests, and deep learning. These methods are used in MATLAB for tasks like signal processing, recognizing patterns, and making predictions in implant devices. The review also discusses challenges like data quality, computing limits, following regulations, and the need for real-time processing. Future plans focus on using edge computing, federated learning, and making ML models easier to understand in clinical settings. This analysis helps researchers and practitioners use MATLAB to improve smart biomedical implant technologies.

Keywords: Machine Learning, MATLAB, Biomedical Implants, Neural Networks, Signal Processing, Medical Devices, Artificial Intelligence.

ICFTSEMAP079

Wireless Power Transfer (WPT) Enabled Iot Sensors Using Ultrasonic, Soil Moisture, And Temperature Sensors with 16×2 I²C LCD Display

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ABSTRACT

This project focuses on the design and development of a Wireless Power Transfer (WPT)-enabled Internet of Things (IoT) sensor node that operates without batteries. The system harnesses energy from electromagnetic fields using ultra-thin electrically small antennas and converts it into DC power through a rectifier circuit. The Arduino UNO microcontroller uses this energy to power sensors such as temperature (LM35), soil moisture, and ultrasonic sensors. Real-time sensor data is displayed on a 16×2 I²C LCD display. The proposed system demonstrates a sustainable, battery-free, and maintenance-free approach for smart agriculture and environmental monitoring applications.

Keywords: Wireless Power Transfer (WPT), Internet of Things (IoT), Ultrasonic Sensor, Soil Moisture Sensor, Temperature Sensor, Wireless Sensor Networks, Battery-Free Sensors, Smart Agriculture, Embedded Systems, 16×2 I²C LCD Display.

ICFTSEMAP080

Survey On Deep Learning-Based EEG Communication Aid for Paralyzed Patients

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ABSTRACT

Recent advances in Artificial Intelligence have enabled more effective analysis of brain signals for healthcare applications. Electroencephalography (EEG) provides a non-invasive means of capturing brain activity, but its complex and noisy nature makes interpretation difficult. Deep learning approaches have shown strong potential in extracting meaningful patterns from EEG data, supporting the development of intelligent assistive systems. This survey reviews existing research on EEG-based deep learning methods for brain–computer interface (BCI) communication, emotion recognition, and pain or discomfort detection. BCI systems offer an alternative communication pathway for individuals with severe motor impairments by translating brain activity into commands without requiring physical movement. Both auditory and visual BCI paradigms are discussed, highlighting their role in enabling basic interaction and decision-making. In addition to communication, recent studies demonstrate the use of EEG signals to assess emotional states and pain levels, providing important insights into user well-being. The findings highlight the need for unified EEG-based assistive frameworks that combine communication and well-being monitoring to improve patient care and quality of life.

Keywords: Brain–Computer Interface, EEG, Deep Learning, Assistive Healthcare, Emotion and Pain Monitoring.

Survey On Deep Fake Detection in Multi-Model Using Deep Learning

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ABSTRACT

Artificial Intelligence (AI) is the domain that enables machines to perform tasks like learning, reasoning and decision-making. Deep Learning (DL), as a sub-domain of AI, uses multi-layered neural networks for automatic feature extraction, high accuracy and strong capability to handle complex data like images and videos, making it highly effective in deepfake detection. In existing system, they used deepfake detection which primarily rely on Convolutional Neural Networks (CNNs) to analyze spatial artifacts in video frames and frequency-domain analysis to capture subtle inconsistencies. Optimization-based techniques such as Ant Colony Optimization (ACO)–Particle Swarm Optimization (PSO) have also been applied for spatial– temporal feature extraction, improving classification performance. These methods achieve strong results on benchmark datasets and form the technical foundation for current detection research. Even though existing systems demonstrate strong performance on controlled datasets, they often struggle with low-quality, compressed or rapidly evolving fake videos. To overcome the problem, we proposed CNN by using video pre-processing and continual retraining with updated datasets. CNNs are effective for analyzing video frames and extracting spatial artifacts, while RNNs can model temporal dependencies in audio. By integrating these components into a multi- modal framework, the robustness, adaptability and accuracy of deepfake detection can be significantly enhanced for real-world applications.

Keywords: Deep Fake Detection, Generative Adversarial Networks (GANs), Convolutional Neural Networks (CNNs), Multi-Modal Detection, Frequency-Domain Features, Digital Trust.

ICFTSEMAP082

Survey On Ai-Driven Stock Price Prediction Using Historical Data and Sentiment from News Sources with Deep Learning

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ABSTRACT

Data Science integrates statistics, programming, and domain knowledge to derive meaningful insights from complex and diverse datasets. In the financial sector, it plays a critical role in analysing market movements, forecasting stock prices, and supporting investment decisions. By uncovering hidden patterns within both historical and real-time data, Data Science equips investors with tools to minimize risks, optimize portfolios, and make informed financial choices. Its ability to handle large datasets, extract key features, and apply advanced analytics has transformed traditional methods of market analysis into data-driven, intelligent systems. In existing system they used PLSTM-AL is a deep learning-based stock market forecasting model that combines Pareto-like sequential sampling optimization, LSTM, and an Attention Layer to improve prediction accuracy. It has been tested on daily data from multiple global indices (2003-2024) and its shown to outperform other advanced models like Convolutional Neural Network (CNN) and Gated Recurrent Unit (GRU). These models are proving its robustness and effectiveness in forecasting stock market. Even though the existing system introduces a PLSTM-AL is a deep learning uses GRU and CNN models for stock price prediction, they often lack of financial news integration and stock alerts results in incomplete analysis, they depend only on limiting real-time adaptability. To overcome the problem we proposed hybrid deep learning model that captures both short-term and long-term patterns in stock market data. Additionally, it integrates real-time sentiment analysis using Newspaper3K, which extracts and analyses financial news to evaluate its effect on market fluctuations. By combining historical data with real-time sentiment insights, the system improves prediction accuracy, reduces errors, and provides more reliable decision support for investors in dynamic financial environments.

Keywords: Data Science, Stock Market Prediction, Deep Learning, Hybrid Model, LSTM, CNN, GRU, Attention Mechanism, Sentiment Analysis, Financial News Analysis, Newspaper3K, Time Series Forecasting, Real-Time Data, Investment Decision Support.

ICFTSEMAP083

Survey On Predicting Traffic Accidents and Generating Intelligent Reports Using DeepLearning Techniques

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ABSTRACT

Recent advancements in Artificial Intelligence have enabled more effective analysis of traffic environments for intelligent transportation applications. Deep learning techniques have shown strong potential in extracting meaningful patterns from video data, supporting the development of intelligent systems for accident monitoring and safety enhancement. This survey reviews existing research on deep learning–based methods for accident prediction, vehicle behavior analysis, and traffic rule violation detection. Accident detection systems commonly employ multimodal data sources, combining structured traffic records with unstructured incident descriptions to improve forecasting accuracy through advanced deep learning architectures such as ensemble networks and hybrid BiGRU-CNN models. These works collectively highlight the growing capability of AI systems to support automated traffic monitoring and road-safety assessment. The survey emphasizes current methodologies, presented in the literature, providing insights into emerging trends in intelligent transportation research.

Keywords: Artificial Intelligence, Deep Learning, Traffic Rule Violation Monitoring, Road Safety, Traffic Management, Road Safety Assessment, Smart City Applications.

ICFTSEMAP084

Style AI: An Integrated AI-Driven Framework for Virtual Wardrobe Management and Fashion Try-On

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ABSTRACT

The rapid expansion of e-commerce has funda- mentally altered the fashion retail landscape, yet a significant" visualization gap" remains for consumers unable to assess the fit and appearance of garments digitally. This gap frequently leads to high return rates and customer dissatisfaction. This paper introduces StyleAI, an innovative platform that utilizes computer vision and machine learning to offer a comprehensive virtual try-on and wardrobe management solution. By leveraging pose detection and image processing, StyleAI allows users to visualize clothing on their own photos without specialized hardware. Built on a React-TypeScript and FastAPI architecture, the system achieves a 90

Keywords: Virtual Try-On, Computer Vision, E-commerce, Machine Learning, Digital Wardrobe.

ICFTSEMAP085

Smart Campus Transit System Using Real-Time Face Recognition and Flask-Based Automated Access Monitoring

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ABSTRACT

The Integrated Smart Campus Bus Management System (ISC-BMS) has been developed to provide a comprehensive and effective approach to managing transportation operations through the use of intelligent automation and real time monitoring. The various functional modules are combined into one system including driver log-in based attendance; Student Bus Pass Application Workflows, and Centralized Oversight Administrator Dashboard. It is proposed that using a face recognition subsystem will automate the process of taking attendance as students board the buses thereby improving accuracy in seating arrangements, reducing the opportunity for proxies, and increasing overall safety and security on campus. Real-time GPS location tracking will provide constant visibility of bus routes, as well as notification of the movement of the bus, delays, and whether the student is on board or not to parents. Several key limitations are addressed by the proposed framework including the limitations of current systems that only track or schedule services separately without incorporating any behavioral verification or automating identity authentication process(es). By integrating the above with biometric recognition, digital pass processing, and multi-role communication capabilities all into a single platform; school districts will be able to improve the safety, operational transparency, and user experience of their transportation operations thereby creating a more reliable and sustainable campus transportation management system.

Keywords: Amart Campus, Face Recognition, OpenCV, Flask, Histogram Features, CLAHE, Haar Cascade, Real-Time Authentication, Biometric System, Transit Monitoring.

ICFTSEMAP086

UniConnect: A Framework for Smart University Automation Using MCP and LLM

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ABSTRACT

Managing a university involves handling various interconnected tasks like student admissions, academic scheduling, resource allocation, and communication across departments. Traditional systems often struggle because they operate in isolated silos, making it difficult to share information seamlessly. This fragmentation leads to inefficiencies and frustration for students, faculty, and administrators alike. Our paper explores a new approach to university management through UniConnect, a system that brings together Large Language Models and the Model Context Protocol to create a more unified experience. By using natural language processing, UniConnect allows users to interact with university systems conversationally, making complex tasks simpler and more intuitive. The Model Context Protocol acts as a bridge, connecting different university databases and services without requiring them to be rebuilt from scratch. This means that existing systems can work together more effectively. We examine how this integration can improve daily operations, from course registration to resource management, while maintaining data security and user privacy. Our review discusses both the potential benefits and practical challenges of implementing such a system in real university environments. We look at how artificial intelligence can make administrative processes smoother while still keeping the human element central to education. The goal is to show how technology can support, rather than replace, the essential relationships that make universities thrive.

Keywords: AI in Education; Large Language Models; Model Context Protocol; Smart Campus; University Automation

AI Based Transaction Anomaly and Fraud Detection System

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ABSTRACT

The fast growth of digital payment systems has led to a substantial rise in both the volume and complexity of fraudulent financial transactions. Traditional rule-based fraud detection methods lack flexibility and often fail to recognize new or previously unknown fraud patterns. This paper presents a hybrid explainable artificial intelligence (XAI) framework for real-time financial fraud detection. The proposed approach combines supervised learning using the XGBoost algorithm with unsupervised anomaly detection through Isolation Forest, enabling the system to identify both known fraud types and emerging suspicious activities. To enhance detection performance, the framework incorporates behavioral profiling, temporal transaction analysis, merchant risk evaluation, and device-level consistency features. Additionally, explainable AI techniques based on SHAP are applied to generate clear and interpretable explanations for each fraud prediction. Experimental results show that the hybrid model achieves higher recall and a lower false-positive rate compared to individual models, demonstrating its effectiveness and suitability for deployment in modern banking systems and digital payment platforms.

Keywords: Financial Fraud Detection, Anomaly Detection, XGBoost, Isolation Fores, Explainable, AI, Digital Payments.

ICFTSEMAP088

Survey On Enhancing Early Parkinson's Disease Detection Using Deep Learning for Image-Based Diagnosis

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ABSTRACT

Parkinson's disease (PD) is a progressive neurodegenerative disorder in which early diagnosis is essential for effective treatment and disease management. Recent advances in deep learning have demonstrated significant potential in medical image analysis through automated feature learning. Existing PD detection approaches primarily employ Convolutional Neural Networks (CNNs) to classify handwritten spiral images of patients and healthy individuals. Architectures such as VGG16, LeNet, DenseNet121, and InceptionV3 have been widely used, with InceptionV3 achieving comparatively higher detection accuracy. However, reliance on a single imaging modality limits the ability to capture comprehensive motor and clinical characteristics of Parkinson's disease. To overcome this limitation, this paper proposes a **multi-modal deep learning framework** that integrates MRI brain images, clinical assessment data, and handwritten spiral drawings. By fusing complementary information from multiple sources, the proposed approach improves diagnostic accuracy and reliability. Experimental results indicate that the multi-modal model outperforms single-modality methods, demonstrating its effectiveness for early Parkinson's disease detection and its potential as a clinical decision-support tool.

Keywords: Parkinson's Disease, Deep Learning, Convolutional Neural Networks, MRI, Clinical Datasets, Handwritten Spirals, Early Diagnosis, Prediction Accuracy.

Smart Irrigation System Using IoT

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ABSTRACT

Efficient water utilization is a critical challenge in modern agriculture due to increasing water scarcity and unpredictable climatic conditions. Traditional irrigation practices often result in excessive water consumption and low crop productivity due to the absence of real-time monitoring and intelligent decision-making. This paper proposes a Smart Irrigation System that integrates Internet of Things (IoT) technology with Machine Learning techniques to optimize irrigation scheduling and water usage. The system employs soil moisture, temperature, and humidity sensors to continuously monitor field conditions. Collected data is transmitted through a Wi-Fi-enabled microcontroller to a cloud platform, where a Random Forest–based machine learning model predicts irrigation requirements. Based on these predictions, irrigation is automatically controlled to supply water only when required. Experimental evaluation demonstrates improved irrigation efficiency, reduced water wastage, and minimized human intervention. The proposed system provides a scalable and cost-effective solution for sustainable agriculture and smart farming applications.

Keywords: Smart Irrigation, Internet of Things, Machine Learning, Random Forest, Precision Agriculture.

ICFTSEMAP090

QUANTUM KEY - Based Secure Image Steganography with Aes Encryption

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ABSTRACT

We are building a highly secure way of sending messages by combining the use of Quantum Key Distribution (QKD) to produce encryption keys and encrypting messages with a variant of AES-256 (a substitution cipher) and Least Significant Bit (LSB) steganography or hiding them in images. We are employing the BB84 protocol to produce these keys based on the quantum properties of the photonic (light) particles; we can detect any third parties attempting to intercept the key and measure it by the Quantum Bit Error Rate (QBER). Once we have produced a secure key, we then use it to encrypt the actual message with AES-256, and we hide that in a digital image (the cover image) with LSB steganography. Thus, if a third party were to capture the Stego image (the digital image containing the hidden encrypted message), they would not be able to decrypt the hidden message without the key that was produced using quantum cryptography. An analytics dashboard monitors the success of the key generation, the QBER value, and other performance metrics, giving a high degree of transparency and trustworthiness of the whole process.

Keywords: Quantum Key Distribution, BB84 Protocol, LSB Steganography, Secure Communication.

ICFTSEMAP091

The Evolving Landscape of IT Recruitment, Challenges and Innovations

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ABSTRACT

The rapid evolution of the information technology (IT) has significantly transformed recruitment practices, introducing complex challenges in identifying, evaluating, and hiring skilled professionals. It emphasizes the role of Artificial intelligence in optimizing acquisition processes, improving hiring efficiency, and enhancing candidate-job fit. In addition to skill assessment and developing recommendations, the system offers real-time job listings that matches the user's skill profile. It provides detailed information about available positions, including job descriptions, company details and application recruitment. The proposed study attempts to address these issues, with an intelligent recruitment system that integrates automation, data analytics, and generative artificial intelligence. The scheme streamlines resume screening, skill matching, and candidate shortlisting by leveraging AI-driven analysis and structured data management. The aggregation highlight that AI-driven recruitment scheme importantly reduce time to hire, better screening consistency, and support fairer hiring practices. This research emphasizes the role of intelligent technologies in reshaping IT recruitment and provides a scalable framework for organizations seeking efficient, unbiased, and a future ready talent acquisition spaces. To handle the unstructured resume and to generate structured candidate profiles, the Generative AI models are employed, which are further ranked using similarity scoring and specify recruitment criteria.

Keywords - Recruitment, Resume, AI, Skills.

ICFTSEMAP092

Evaluating the Sustainability and Scalability of Java Virtual Threads and R2DBC in High Concurrency Web Portal Architecture

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ABSTRACT

Java's Virtual Threads, introduced through Project Loom, represent a significant shift in how Java handles concurrency. This study evaluates the performance of Virtual Threads in a Spring Boot web portal integrated with a reactive PostgreSQL database. A data-intensive student management dashboard was developed to benchmark the transition from the traditional thread-per-request model to a lightweight task-based concurrency model. Experimental results show a 75% reduction in memory overhead and noticeably improved system responsiveness under peak loads of 2,000 concurrent users. Compared to platform threads, Virtual Threads provided better throughput, stable response times, and improved memory efficiency without increasing development complexity. These findings demonstrate that Virtual Threads offer a practical and scalable solution for building modern high-performance web applications.

Keywords: Virtual Threads, Project Loom, R2DBC, Spring Boot, Sustainable Computing, High-Performance Computing, Web Portal Scalability.

ICFTSEMAP093

AI-Driven Predictive Maintenance Framework for Industrial IoT Using Hybrid Deep Learning Models

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ABSTRACT

Predictive maintenance (PdM) is a cornerstone of Industry 4.0 strategies to reduce unplanned downtime and maintenance costs. The rapid proliferation of Industrial Internet of Things (IIoT) has transformed traditional manufacturing and industrial operations by enabling real-time monitoring, data-driven decision-making, and automation. This paper proposes an end-to-end AI-driven predictive maintenance framework for industrial IoT (IIoT) environments that combines a hybrid 1D-Convolutional Neural Network (CNN) for spatial/feature extraction with Long Short-Term Memory (LSTM) layers for temporal modelling (CNN–LSTM). The proposed framework integrates advanced sensing, edge computing, cloud-based analytics, and hybrid deep learning techniques to enable real-time condition monitoring and early fault detection in industrial machinery. The system leverages data collected from heterogeneous IIoT devices, including vibration sensors, temperature sensors, pressure gauges, and acoustic monitors, which are transmitted through a secure communication network for processing and analysis. The hybrid model is optimized using adaptive training strategies and validated through extensive experiments using publicly available industrial datasets. The paper highlights the practical feasibility of deploying the model at the edge, enabling real-time decision-making with reduced bandwidth consumption and lower dependence on cloud infrastructure.

Keywords: Predictive Maintenance, Industrial IoT (IIoT), Hybrid CNN–LSTM, Industry 4.0, Edge Computing.

Survey on Interpretable Breast Cancer Prediction Using Deep Learning and XAI

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ABSTRACT

Artificial Intelligence (AI), especially Deep Learning, is transforming healthcare by enabling automated analysis of medical images. These techniques handle complex medical data efficiently, supporting early detection and diagnosis of diseases. Breast cancer, one of the most deadly diseases worldwide, requires timely detection to improve survival rates. Existing systems primarily utilize convolutional neural network models such as VGG16 and VGG19 for the classification and prediction of breast cancer from medical images. While these models have demonstrated significant success in image classification tasks. They have also contributed to advancing research in medical diagnostics by offering automated feature extraction and reducing manual intervention in analyzing medical images. They often lack precise segmentation, which can reduce diagnostic accuracy and lead to incomplete interpretations. To overcome this, the proposed system uses YOLOv12, which combines detection and segmentation in one framework. It localizes and segments tumors simultaneously in ultrasonic breast images, ensuring accurate identification of cancerous regions. This approach improves diagnostic precision, enables real-time analysis, and enhances overall reliability. Ultrasonic imaging adds a non-invasive, cost-effective, and widely accessible solution, supporting early detection and better treatment planning.

Keywords: YOLOv12 Architecture, Ultrasonic Breast Cancer Imaging, Integrated Detection–Segmentation, Real-Time Medical Diagnosis, Precision Tumor Segmentation.

ICFTSEMAP095

AI-Powered Deaf Companion System for Inclusive Communication Between Deafand Hearing Individuals

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ABSTRACT

The AI-Powered Deaf Companion System is designed to bridge the communication gap between deaf and hearing individuals using cutting-edge deep learning technologies. Leveraging Temporal Convolutional Networks (TCNs), the Sign Recognition Module (SRM) accurately interprets sign language gestures in real time. Meanwhile, the Speech Recognition and Synthesis Module (SRSM), powered by Hidden Markov Models (HMMs), converts spoken language into text, ensuring seamless bidirectional communication. To enhance accessibility, the Avatar Module (AM) visually translates spoken text into sign language gestures, making interactions more intuitive. By integrating computer vision, natural language processing (NLP), and speech synthesis, this system fosters inclusive communication, empowering the deaf community with greater independence and interaction opportunities.

Keywords: **Artificial Intelligence (AI), Deep Learning, Temporal Convolutional Networks (TCNs), Hidden Markov Models (HMMs), Neural Networks, Computer Vision, Natural Language Processing (NLP), Sign Language Recognition, Speech-to-Text (STT), Gesture Recognition, Speech Recognition.**

ICFTSEMAP096

Design of AN IoT-Enabled Intelligent Breath Analysis System for Early Prediction of COPD

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ABSTRACT

Smart Breath Analyzers have been designed to support doctors in remotely monitoring patients suffering from Chronic Obstructive Pulmonary Disease (COPD) using telemedicine technology. These analyzers serve as intelligent sensing devices capable of evaluating the quality of a patient's exhaled breath. They incorporate multiple sensors, including CO_2, smoke, volatile organic compounds (VOCs), relative humidity, temperature, heart rate and respiration sensors, which collect vital parameters through a bi-tube breathing circuit. The acquired sensor data are transmitted in real time to healthcare providers or National Health Service units via an Internet of Things -based cloud communication system using TCP/IP protocols. This system allows for continuous patient monitoring, enabling early detection of breathing difficulties or disease exacerbation that may require timely medical attention. The integration of telemedicine and IoT provides a cost-effective, efficient approach to managing chronic respiratory conditions from a distance. To evaluate prototype performance, a gas-mixing setup was used to test responses to various gases, smoke concentrations and acetone levels in exhaled breath collected in sampling bags. The prototype utilizes an IoT for data collection, processing, and control. Additionally, the system integrates Health Tags for continuous indoor and outdoor monitoring of high-risk COPD patients. Overall, this work serves as a proof-of-concept for an IoT-enabled respiration monitoring system that enhances chronic disease management, continuous assessment and remote healthcare delivery.

Keywords: Smart Breath Analyzer, Chronic Obstructive Pulmonary Disease (COPD), Internet of Things (IoT), Remote Health Monitoring, Telemedicine.

ICFTSEMAP097

Deep Learning–Based Real-Time Speech Emotion Detection and Suicide/Self-Harm Prevention System

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ABSTRACT

Speech, as a medium, is powerful, carrying not just linguistic content but also a multifarious emotional and psychological content of strong value in understanding a person's emotional status. Early identification of emotional issues, depression, suicidal thoughts, and intent, among others, using speech-based communication is crucial for providing timely support services. In this regard, this paper seeks to propose a deep learning-based Real-Time Speech Emotion Detection and Suicide/Self-Harm Prevention System. It utilizes a hybrid CNN-LSTM model, which comprises Convolutional Neural Networks (CNNs) in detecting vocally as well as linguistically related cues of a person's emotional and psychological status. Specific features, which are imperative in emotion detection and identification, are Mel Frequency Cepstral Coefficients (MFCCs), Pitch variation, Speech Energy, Zero-crossing rates, and Spectral Features, among others.In terms of enhancing reliability, a speech-to-text module is used to convert speech into text, which is then further analyzed using Natural Language Processing (NLP) techniques to detect suicidal tendencies in the text. By using both audio-based emotion recognition and text-based semantics, a more complete analysis of the mental health risk is possible. This framework is designed to work with real-world data sources such as voicemail, phone calls, and recorded speech messages. This allows for continuous, real-time monitoring. Ultimately, this proposed system aims to provide a useful role in terms of aiding in suicide prevention by providing a source of intelligent, automated solutions in this area.

Keywords: LSTM, CNN, MFCC

ICFTSEMAP098

Goal-Oriented Social Feed Optimizer

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ABSTRACT

In recent years, social media platforms such as YouTube, Instagram and Facebook have become indispensable sources of information, learning, and entertainment. However, the recommendation algorithms used by these platforms are optimized for engagement rather than productivity, often exposing users to irrelevant and distracting content. This results in loss of focus, reduced learning efficiency, and increased screen time without meaningful outcomes. This paper proposes **Goal-Oriented Social Feed Optimizer (GOSFO)**, an AI-based system that aligns digital content consumption with user-defined learning or career objectives. The system integrates Natural Language Processing (NLP), semantic similarity models, and browser automation to identify, filter, and promote goal-relevant content while suppressing irrelevant distractions. Experimental usage demonstrates improved relevance in content recommendations and increased productive screen time.

Keyword: Social Media, NLP, Recommender Systems, Browser Automation, Digital Well-being, Machine Learning.

ICFTSEMAP099

Real-Time Smoke Removal and Rescue System Using HSRDN

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ABSTRACT

Fire accidents have become increasingly severe in recent years due to heavy smoke, which significantly reduces visibility and delays rescue operations. Poor visual conditions limit the effectiveness of vision-based rescue systems, making real-time smoke removal and accurate victim detection essential. Existing smoke removal and image enhancement methods, including traditional dehazing and deep learning models such as FFA-Net, mainly focus on visual enhancement and suffer from high computational complexity, reduced performance in dense smoke, and lack of intelligent rescue decision-making, while object detection models like YOLO experience accuracy degradation under low-visibility conditions. To overcome these limitations, this paper proposes an AI-driven Real-Time Smoke Removal and Rescue System using HSRDN, which integrates real-time smoke removal, enhanced object detection, and fuzzy logic–based rescue prioritization. The performance of the proposed system is evaluated using PSNR and SSIM to measure image quality and structural restoration, mAP to assess detection accuracy, and FPS to validate real-time processing efficiency. Experimental results demonstrate that the proposed approach provides improved visibility, higher detection accuracy, and reliable real-time performance, making it suitable for practical fire rescue applications.

Keywords: Smoke Removal, RT-HSRDN, Fuzzy Logic, YOLOv8, Deep Learning, Computer Vision, Fire Rescue System.

ICFTSEMAP100

Determinants of Female Consumers' Online Shopping Behaviour: An Integrated TPB–TAM Approach in a Bottom-of-the-Pyramid Context

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ABSTRACT

The study examines the factors influencing female consumers' online shopping behaviour, combining the theory of planned behaviour (TPB) with the technology acceptance model (TAM) within the Bottom of the Pyramid context. This study investigates the connections among attitude, subjective norm, perceived behavior control, perceived usefulness, and perceived ease of use with online shopping intention, with online trust examined as a mediating variable. The integration of TPB and TAM is applied to explain the female consumers' online shopping behaviour in a resource-constrained regional setting. The research was conducted on a sample of 226 female consumers residing in various semi-urban and rural regions of Bihar, India, representing the BOP geographical unit, who possessed some experience using social media platforms and engaging in online buying. The collected data were analyzed using regression analysis and the Hayes PROCESS model to determine the direct and indirect relationships among the studied variables. The study observed a significant shift in buying behavior to online platforms from offline among consumers. The finding reflected that attitude, subjective norms, perceived control behavior, ease of use, and perceived usefulness have a positive influence on online shopping intention and behaviour. Online trust plays a crucial role in shaping the final behaviour. The study contributes to a deeper understanding of female customers' active adoption of online platforms across different product categories, which can assist e-commerce firms and policymakers in designing user-friendly platforms and digital initiatives for not-so-privileged regions. The researchers could not find relevant literature covering female consumer buying behavior for low income group and digitally emerging regions.

Keywords: TPB, TAM, Bottom of the Pyramid, Online shopping behavior, Online Trust, Hayes Process, Moderation.

ICFTSEMAP101

Autonomous Climate Mitigation System

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ABSTRACT

The menace of flood disasters has continued to threaten lives, infrastructure, and economies of the people around the world. The problem of traditional disaster management systems is usually slow prediction and poor coordination in responding to the disaster. In this paper, I have proposed a multi-agent artificial intelligence platform based on an autonomous climate disaster management system (ACDMS), which will predict, monitor, and control the flood disaster events in real time. The system encompasses deep learning architecture to predict floods and monitor disasters, reinforcement learning to allocate resources dynamically and graph-based algorithms to optimize evacuation paths. It further integrates real time weather information, satellite images, geospatial data, and social media contributions to increase situational awareness and decision making. Experimental assessments reveal that ACDMS is more efficient in response, resource optimization and makes coordination better than traditional methods of disaster management. The framework proposed offers a comprehensive and robust solution that is capable of sustaining proactive disaster preparedness, emergency real-time operations and post-disaster recovery planning. The paper also adds to the body of knowledge on intelligent and adaptive flood disaster management with a single architecture that is powered by artificial intelligence.

Keywords: Artificial Intelligence; Climate Disaster Management; Flood Prediction; Multi Agent Systems; Reinforcement Learning.

ICFTSEMAP102

Development of Essential Oil Nano Emulsion Based Coating Solution to Enhance the Shelf-Life of Custard Apple

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ABSTRACT

The present study investigated the postharvest preservation efficacy of edible coatings derived from fenugreek seed mucilage (FSM) enriched with essential oil-based nanoemulsions (EO-NEs) on custard apples (Annona squamosa L.) during 7 days of ambient storage. Custard apples are highly perishable climacteric fruits prone to rapid senescence and decay, limiting their marketability and shelf life. To address this, FSM was employed as a natural polysaccharide-based matrix, combined with EO-NEs to enhance barrier and antimicrobial properties. Physicochemical (TSS, TA, AA), biochemical (TAA, TFC, TPC), physical (weight loss, firmness), microbial (decay index), and sensory attributes were evaluated on Day 1, Day 4, and Day 7. Results indicated that FSM + EO-NE coatings effectively delayed weight loss, retained firmness, and preserved antioxidant parameters compared to uncoated control fruits. Notably, treated fruits showed significantly lower decay incidence and superior sensory scores even on Day 7. The synergistic action of FSM and EO-NEs was attributed to the semi-permeable film, antioxidant activity, and antifungal properties. These findings suggest that FSM-based edible coatings enriched with essential oils can serve as natural, sustainable alternatives to synthetic postharvest treatments, potentially extending the shelf life of custard apples and enhancing fruit quality during storage.

Keywords: Custard Apple, Fenugreek Seed Mucilage, Essential Oil Nanoemulsion, Edible Coating.

ICFTSEMAP103

Survey on A Multivariate Kalman-Filtered AI Framework for Emotion-Aware Multimodal Biosensor Fusion in Intelligent Sports Performance Monitoring

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ABSTRACT

We have all heard a lot about the potential of data enabled by artificial intelligence (AI) to improve performance. This progress has seen the steady advancements of wearable biosensors capable of generating live data and feedback on an athlete's physiological state, allowing for smarter training and enabling consistent performance improvement. However, wearable biosensors often struggle with noise and signal interference caused by various factors, including muscle movements, sweat, and environmental conditions. This study proposes the Smart Performance Analysis and Real-time Tracking Algorithm (SPARTA) for enhancing athletic performance using AI techniques. SPARTA leverages AI algorithms to analyse real-time physiological data- including heart rate, oxygen saturation, skin conductance, and cortisol levels- enabling dynamic adjustments to training loads and recovery protocols. Experimental evaluations using the Biosensor-Student Health Fitness Dataset (n = 500 input samples) demonstrated SPARTA' s capability to achieve 91.34 % accuracy in SpO 2 monitoring, 88.72 % precision in skin conductance detection, 82.64 % correlation with laboratory assays for sweat electrolyte analysis, and 78.65 % accuracy in non-invasive cortisol level tracking. With more advances in artificial intelligence, wearable biosensors will greatly help boost athletic performance, further dominating the sports & fitness globe.

Keywords: AI Techniques, SPARTA, Signal Interference, Data Fusion, Heart Rate, SpO2, Cortisol, Skin Conductance, Sports Science, Human-Computer Interaction (HCI), Biomechanics.

ICFTSEMAP104

Survey On Hybrid Deep Learning Approach for Brain Tumor Detection

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ABSTRACT

Recent advancements in deep learning have significantly improved automated medical image analysis, particularly in the diagnosis of brain tumors using Magnetic Resonance Imaging (MRI). Accurate brain tumor diagnosis requires the detection of tumor presence, identification of tumor type and grade, and precise localization of the affected region. However, many existing approaches rely on multiple independent models for these tasks, leading to increased computational complexity and reduced efficiency. To address this challenge, this study presents a CNN-based framework for brain tumor detection, classification, and location identification using MRI images. A multi-task learning approach based on a Convolutional Neural Network with a shared architecture is employed to simultaneously classify tumor presence, tumor grade, and tumor type. In addition, a CNN-based U-Net model is utilized for brain tumor segmentation to accurately identify the tumor location. The system is trained and evaluated using publicly available MRI datasets containing multiple tumor categories and grades. Experimental results demonstrate that the proposed multi-task classification model achieves an overall accuracy of 92%, while the tumor segmentation module attains an average Dice coefficient of 0.89. The findings indicate that the proposed approach effectively reduces computational complexity while providing comprehensive diagnostic information, making it suitable for assisting clinicians in accurate and efficient brain tumor diagnosis.

Keywords: Brain Tumor Detection, MRI, Convolutional Neural Network, Multi-task Learning, Tumor Segmentation.

ICFTSEMAP105

Performance Evaluation of Hybrid Power Filters under Unbalanced and Distorted Voltage Conditions

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ABSTRACT

The proliferation of non-linear loads in modern power systems has escalated power quality concerns, primarily due to harmonic currents and reactive power demands. While **Hybrid Power Filters (HPF)**—combining the cost-effectiveness of passive filters with the precision of active filters—offer a promising solution, their performance often degrades under non-ideal grid conditions. This paper presents a comprehensive **performance evaluation** of a shunt hybrid power filter specifically designed to operate under **unbalanced and distorted supply voltage conditions**. A robust control strategy, based on Clark Transformation **[p-q Theory]**, is employed to ensure accurate harmonic extraction despite the presence of negative-sequence components and background harmonics in the grid. The study utilizes **MATLAB/Simulink** simulations and experimental validation to analyze Total Harmonic Distortion (THD) and power factor correction across varying load levels.

Keywords—Harmonic Compensation, Hybrid Power Filter (HPF), Power Quality, Unbalanced Voltage, Total Harmonic Distortion (THD), Robust Control.

ICFTSEMAP106

Production, Thermodynamic Evaluation and Emission Analysis of Composite Biodiesel Derived from Michelia champaca and Amoora rohituka in a CRDI Diesel Engine

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ABSTRACT

This research undertook a comprehensive analysis of the thermodynamics, combustion, and emissions associated with biodiesel, specifically a 50:50 blend derived from Michelia champaca and Amoora rohituka seed oils. The biodiesel was produced through a two-step process: esterification followed by transesterification, achieving an 85% conversion rate. Experiments were conducted using a CRDI diesel engine, maintaining a constant speed of 1500 rpm under four different load conditions. The results indicate that the blends with a lower concentration performed rather similarly to traditional diesel fuel, all the while reducing CO and HC emissions. To ensure the reliability of these findings, statistical methods were employed for result validation, coupled with uncertainty quantification.

Keywords: Composite Biodiesel; CRDI Engine; Combustion Analysis; Emission Reduction; Sustainability.

ICFTSEMAP107

Machine Learning Enabled Personalized Career Guidance for School Students

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ABSTRACT

After the tenth grade, students must make a critical decision about their careers, but many are confused by the lack of appropriate and individualized guidance. It can be challenging for students to select the best course of action because traditional career counseling techniques are frequently generic and offer the same recommendations to every student without taking into account their unique interests, abilities, and strengths. As a result, a student-centered career counseling system that offers tailored suggestions is required. With the help of machine learning (ML) techniques, this proposed work seeks to create a customized career guidance platform for schoolchildren, particularly those finishing the tenth grade. Through assessments, the system examines students' interests, skills, subject knowledge, and personalities. It suggests suitable career paths and streams such as Science, Commerce and Arts.

Keywords: Machine Learning, Personalized Career Recommendation, 10th Grade Students, Stream Selection.

ICFTSEMAP108

Survey On Deep Enhancing of Classification and Predictions of Gastric Cancer by Using Lime AI

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ABSTRACT

Artificial Intelligence is improving healthcare by allowing more accurate prediction and identification of diseases. Deep Learning, which is a part of AI, is good at examining complicated medical information. Stomach cancer is a serious health issue that often leads to high death rates. Detecting this disease early and correctly categorizing it are key to effective treatment. The use of AI and Deep Learning increases the accuracy of diagnosis and helps doctors take promptly. In existing systems, stomach cancer prediction relies on deep learning models like EfficientNet-B5 for medical image classification. These models often suffer from overfitting, reducing accuracy and lack explainable AI, making results hard to trust. Techniques such as data augmentation, regularization, and explainable AI are applied to improve accuracy, reliability and interpretability for clinical use. Even though existing models like EfficientNet-B5 provide cancer prediction, they face issues such as overfitting and lack of explainable AI, reducing accuracy and trust. These limitations make it difficult for doctors to rely on automated predictions. The proposed system uses a hybrid model combining machine learning and deep learning techniques to improve accuracy. It integrates LIME (Local Interpretable Model-agnostic Explanations) for interpretable and transparent predictions. This approach enhances diagnostic reliability and supports effective clinical decision-making.

Keywords: Artificial Intelligence, Deep Learning, Stomach Cancer Detection, Medical Image Classification, EfficientNet-B5, Explainable AI, LIME, Hybrid Model, Diagnostic Accuracy, Clinical Decision Support.

ICFTSEMAP109

Veritas Net: Hybrid Fake News Detection and Fact Verification System

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ABSTRACT

The rapid growth of digital media and social networking platforms has significantly accelerated the spread of fake news, leading to misinformation that affects public opinion, social harmony, and decision-making processes. Traditional fake news detection approaches primarily rely on binary classification and handcrafted features, which are often insufficient to handle semantic complexity, partial truths, and evolving misinformation patterns. To address these challenges, this project proposes an intelligent fake news detection platform based on VeritasNet, integrated with Automated Machine Learning (AutoML) and fact verification mechanisms. The proposed system employs advanced Natural Language Processing (NLP) techniques for semantic understanding of news content. Transformer-based deep learning models such as BERT are utilized to extract rich contextual representations from text, which are further processed by VeritasNet to perform deep pattern learning and uncertainty-aware classification. AutoML is incorporated to automatically select optimal model architectures and tune hyperparameters, ensuring robust performance while minimizing manual intervention. In addition to content-based analysis, the system includes a fact verification module that extracts key claims from news articles and validates them against trusted fact-checking sources and knowledge bases. To enhance transparency and user trust, Explainable Artificial Intelligence (XAI) techniques such as SHAP, LIME, and attention visualization are employed to interpret model predictions. Experimental results demonstrate that the proposed approach achieves improved accuracy, balanced precision–recall performance, and reliable confidence estimation compared to conventional models. The integration of VeritasNet, AutoML, and fact verification makes the system suitable for real-world deployment in combating digital misinformation.

Keywords: Fake News Detection, VeritasNet, AutoML, Natural Language Processing, BERT, Fact Verification, Explainable AI.

ICFTSEMAP110

Survey On Weapon Based Crime Monitoring System

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ABSTRACT

The rapid growth of urban environments has increased the demand for intelligent and reliable crime prevention systems. Traditional surveillance methods rely heavily on continuous human monitoring, which is often inefficient due to fatigue, delayed response, and limited attention. To address these limitations, this work presents a real-time crime monitoring system that integrates deep learning and computer vision techniques for automated surveillance analysis. The proposed system processes live video streams captured through CCTV cameras to identify suspicious activities and potential criminal incidents. The system employs a multi-stage framework consisting of weapon detection, abnormal behavior (violence) detection, and face recognition. Advanced deep learning models such as YOLO-based object detection and lightweight convolutional neural networks are utilized to ensure fast and accurate analysis under real-time constraints. Facial recognition techniques are incorporated to distinguish authorized personnel from suspects, reducing false alarms and improving decision reliability.Experimental results demonstrate that the proposed system achieves high accuracy in detecting weapons, violent actions, and identifying individuals across diverse real-world scenarios. The integrated alert mechanism enables timely notification to law enforcement authorities, enhancing public safety and rapid response capabilities. Overall, the system provides an efficient, scalable, and intelligent solution for modern crime monitoring applications. Future work aims to improve robustness under low-light conditions, optimize computational performance, and integrate multimodal data sources for enhanced situational awareness.

Keywords: Crime Monitoring System, Deep Learning, Video Surveillance, Weapon Detection, Violence Detection, Face Recognition.

CHATSEC: Conversational Security Intelligence

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ABSTRACT

Security Operations Centers face growing operational strain due to increasing alert volumes, complex threat patterns, and heavy reliance on manual investigation workflows. This paper presents CHATSEC, an AI and ML driven cybersecurity framework designed to enhance efficiency, resilience, and analytical depth within enterprise SOC environments. The proposed system integrates a Conversational SIEM Assistant powered by large language models, a secure Model Context Protocol middleware, automated incident response and watchdog agents, and behavioral analytics for insider and pattern-based threat detection. The framework is implemented over existing SIEM platforms including Elastic SIEM and Wazuh, ensuring compatibility with established security infrastructures. Evaluation within a VMware-based virtual SOC demonstrates significant reductions in detection and investigation time exceeding fifty percent, alongside improved threat visibility and automation of repetitive SOC tasks. The results indicate that conversational intelligence combined with autonomous response mechanisms can substantially improve the speed, accuracy, and adaptability of modern SOC operations.

Keywords: Security Operations Center, Conversational SIEM, Artificial Intelligence, Machine Learning, Incident Response Automation, Behavioral Analytics, Cybersecurity.

Survey on GraphRXInsight

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ABSTRACT

Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL) are widely used in healthcare for analyzing molecular and clinical data to support decision-making. AI builds intelligent systems, ML enables data-driven learning, and DL, especially Graph Attention Networks (GAT) and Deep Neural Networks (DNN), extract complex features for accurate drug–drug interaction (DDI) prediction. In existing systems to predict DDI, the models using Deep Neural Networks (DNN), Random Forest, and XGBoost with DrugBank's structural similarity profiles are applied. They use the Synthetic Minority Over-sampling Technique to handle class imbalance in clinical datasets. These approaches aim to improve the prediction of rare but critical adverse drug interactions and have demonstrated accuracy around 93.80%. Even though current models provide useful insights, they have limitations in fully capturing complex drug interactions and real-world usage. They struggle with handling imbalanced data and do not fully use Graph Neural Networks (GNN) to model multi-drug networks. We propose a hybrid model, CLINENSEMBLE, combining Graph Attention Networks (GAT), Deep Neural Networks (DNN), and CatBoost. Using chemical structure data and graph features along with clinical information, it improves rare event detection by over 5% and overall accuracy. This helps doctors and researchers better assess drug interaction risks and improve patient safety.

Keywords: Drug–Drug Interaction (DDI) Prediction, Graph Neural Networks (GNN), Graph Attention Networks (GAT), Deep Neural Networks (DNN), Healthcare Artificial Intelligence.

Human Resource Management in the Gig Economy: A Study On Sustainable HRM Practices in Gig Workers in The Era of Algorithmic Control

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ABSTRACT

The rapid expansion of gig economy has indeed been able to transform the traditional market. The businesses today are bound to function in an extremely dynamic environment, which we can call the VUCA world. Volatility, Uncertainty, Complexity and Ambiguity is no more a buzz world but many adaptive responses to it are growing. One such kind of economy is the gig economy where the workers are paid “per task” or “per gig”. This comes as a contrast to the traditional economy where the Human Resources’ jobs were relatively stable, compensation regulated and they were protected by some social security that saved them in various contingencies of life. Not just this, the shift from traditional approach to algorithm-based control from human control has also caused structural changes in this sector in the Human Resource Management practices. As Gig economy, both platform and non- platform continues to grow, the Human resource management practices have to be introspected minutely for its inclusivity and sustainable viability in both economic and social terms.

Keywords: Gig Economy, Sustainability, Human Resources, Human Resource Practices.

ICFTSEMAP114

Digital Transformation, Cyber Risk, Governance in Global Digital Trade: Challenges and Resilience Strategies

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ABSTRACT

The swift in Digital Trade has revolutionised global trade and commerce in the post-pandemic era, restructuring the way businesses manage their transactions, supply chains and engage with cross-border consumers. The rise in cyber fraud and increasingly sophisticated cybersecurity threats has exposed as digitalisation has enhanced efficiency, connectivity and market access. The study investigates the interconnection of digital transformation, cybersecurity and governance within digital trade ecosystems, highlighting the growing complexity of managing digital risk in a highly interconnected global economy. The study also examines how the increased digitalisation of transactions, cloud-based platforms, cross-border data flows and artificial intelligence-driven systems have created new vulnerabilities, challenging the existing framework for fraud prevention, data protection, and digital compliance. The interdependence of online transactions and digital payment systems increased the scope for cybercriminal activities, which intensified the need for stronger regulatory supervision and control. The study evaluates the effectiveness of regulatory responses, including cybersecurity laws, digital trade agreements, and institutional policy mechanisms in mitigating these risks by using secondary data from government reports, journals, articles, etc. The study also explores the importance of integrated resilience strategies that combine regulatory coordination, ethical data, governance, organisational preparedness, and technological safeguards. The research underscores the necessity of building trust, transparency, and long-term sustainability within the evolving global digital economy by aligning technological innovation with secure and responsible trade practices.

Keywords: Digital Transformation, Cyber Fraud, Digital Trade, Cybersecurity, Regulatory Framework, Data Protection.

Blockchain Powered E-Voting with AI-Based Facial Recognition and Multi-Factor Authentication

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ABSTRACT

The integrity of democratic voting systems is increasingly threatened by security vulnerabilities, lack of transparency, and trust deficits, making electoral processes susceptible to manipulation. To address these concerns, Binance Smart Chain (BSC) introduces a blockchain-powered voting framework that leverages the Proof of Staked Authority (PoSA) consensus protocol to enhance security and decentralization. To further fortify the system, ResNet-101, a deep learning-based convolutional neural network (CNN), is integrated for facial recognition authentication, ensuring voter legitimacy and eliminating identity fraud. Additionally, one-time password (OTP) authentication and live location tracking strengthen the system against unauthorized access and proxy voting. By combining blockchain technology, biometric verification, and AI-driven facial authentication, BSC establishes a highly secure, transparent, and tamper-proof voting system. This approach aims to restore public trust in electoral processes, setting a new benchmark for secure and verifiable digital voting systems in democratic governance.

Keywords: Blockchain, PoSA, ResNet-101, Facial Recognition, OTP, Voting Security, Transparency, Authentication.

ICFTSEMAP116

Automated Legal Entity Extraction with Legal-BERT and NER

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ABSTRACT

The rapid growth of digital legal documents creates opportunities for automated information extraction while introducing challenges such as domain-specific terminology, complex sentence structures, and class imbalance. Named Entity Recognition (NER) plays a crucial role in legal text analytics; however, conventional rule-based and generic machine learning approaches often struggle with contextual ambiguity and nested entities. This study proposes a hybrid Legal-BERT-based framework integrated with a semantic similarity filtering mechanism to improve entity boundary precision and extraction reliability. The model leverages domain-adapted contextual embeddings and semantic validation to reduce false positives and enhance prediction consistency. Experimental results demonstrate strong performance, achieving a precision, recall, and F1-score of 0.92 with an overall accuracy of 0.99. A Streamlit-based web application is developed for document upload, entity visualization, and statistical analysis, enabling practical deployment in legal workflows. The proposed framework provides a scalable solution for automated legal entity recognition and intelligent legal text analytics.

Keywords: Legal Named Entity Recognition, Legal-BERT, Transformer Models, Semantic Similarity Filtering, Natural Language Processing, Legal Text Mining, Contextual Embeddings.

ICFTSEMAP117

Freelancer Skill Gap Analyzer and Upskilling Pathfinder

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ABSTRACT

The rapid growing of the digital tools and the growth of freelancer job have effectively reshaped the competencies require in the current workforce. Freelancer now a days struggling to find which skills are currently valued or which capabilities they should develop to stay competitive. Without access to data-driven guidance, many face ineffective learning investments and inconsistent income progression. Addressing this issue calls for a systematic method to assess skill mismatches and propose targeted development pathways. This study introduces a Freelancer Skill Gap Analyzer and Upskilling Pathfinder designed to measure the divergence between individual skill profiles and real-time market expectations. The system leverages structured data encompassing freelancer earnings, job categories, and skill associations to generate gap scores through a comparative analysis of labor supply and demand. Earnings indicators are used as economic signals to approximate market valuation, while estimated return on investment informs the practicality of recommended learning interventions. The proposed solution says SQL-based data manipulation alongside visual analytics to deliver clear, actionable insights such as ranked skill demand, comparative income potential, and customized learning trajectories. Designed to accommodate periodic data refreshes, the system remains responsive to shifting labour market conditions. By translating raw labor data into concrete guidance, the framework supports strategic career decisions within the freelance economy and provides an interactive model for future skill analytics research.

Keywords: Freelancer Analytics, Skill Gap Assessment, Career Path Recommendation, Labor Demand Analysis, Earnings-Based Guidance.

ICFTSEMAP118

Structural Transformation in India's Wealth Management Industry: Trends, Growth Drivers, and Emerging Challenges

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ABSTRACT

Rapid economic expansion, increasing financialization, and the growing base of affluent households have significantly reshaped India's wealth management landscape. India has emerged as one of the leading nations globally in terms of private wealth accumulation, accompanied by a steady rise in High-Net-Worth Individuals (HNIs), ultra-HNIs, and billionaire entrepreneurs. This structural shift has intensified competition and compelled wealth management firms to move beyond traditional product-centric advisory services toward integrated, client-focused financial solutions. The industry is progressively adopting technology-enabled platforms that enhance transparency, personalization, and efficiency in portfolio management. Contemporary wealth advisory services now incorporate holistic financial planning, tax efficiency strategies, estate structuring, risk management, portfolio diversification, and alternative investment avenues. Increased participation of retail and high-value investors in capital markets, supported by digital access and improved financial literacy, has further accelerated this transformation.. The study is based on secondary data sources, industry reports, and trend analysis to examine growth patterns, structural drivers, and emerging challenges within the sector. Findings indicate strong projected growth in individual wealth and expanding demand for sophisticated advisory models in the coming years. Although limited by reliance on secondary data, the study offers analytical insights into the evolving architecture of India's wealth management ecosystem and its future trajectory.

Keywords: Wealth Management, High-Net-Worth Individuals, Structural Transformation, Financial Advisory, Digital Finance.

Characterisation of Soil Quality in Highway-Influenced Areas of Uttarakhand

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ABSTRACT

Soil is a composite of biological and physical structures that give vital support to plants. The rocks' weathering process forms it. Soil plays a major role in providing facilities to our society, most importantly, food production. Soil comprises minerals, organic matter, water, and air that provide micronutrients for the finest growth and reproduction of animals and plants. Soil quality and health are defined as the volume of soil that functions as a vigorous living system within land-use borders. Highways navigate many agricultural areas, and they often have destructive effects on agricultural soils, including the modified stability of SOM. Here, we analyse the physicochemical properties of the soil of Udham Singh Nagar district of Uttarakhand, India. Few samples were collected along the road in the Udham Singh Nagar district, based on distance from the highway, and analysed for their physicochemical properties. The current study will assist the farmers in understanding soil quality and what fertilisers or manures need to be added to their soil for worthwhile productivity.

Keywords: Agricultural Land; Highway Construction; Organic Carbon; Soil Chemistry, Soil Pollution.

ICFTSEMAP120

Dual-Prompt Text–Image Matching Framework Using CLIP for Real-Time Authenticity Detection

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ABSTRACT

The rapid spread of digital information through social media and online platforms has increased the risk of misinformation, including false incident reports and manipulated visual content. Ensuring the authenticity of such multimodal information is a significant challenge, especially in real-time scenarios where timely verification is essential. Conventional verification methods rely on static datasets and manual fact-checking, which limits scalability and response speed. In this research, a Dual-Prompt Text–Image Incident Verification System (DP-TIVS) is proposed to automatically authenticate incident reports using multimodal analysis. The system integrates Natural Language Processing using spaCy for Named Entity Recognition, real-time image retrieval using Bing Image Search API, and vision–language models such as CLIP and BLIP-2 for semantic similarity analysis and image captioning. The framework operates in two modes: Text-to-Image verification, where incident descriptions are matched with retrieved images, and Image-to-Text verification, where uploaded images are validated through generated captions and similarity comparison. The system performs cross-modal similarity computation, authenticity classification, and confidence scoring to generate structured verification reports. Experimental evaluation demonstrates that the proposed system achieves high accuracy and efficiency in detecting authentic and fabricated incident reports, providing a scalable and reliable solution for real-time incident verification and misinformation detection.

Keywords: Multimodal Verification, CLIP, BLIP-2, Natural Language Processing, Incident Authentication, Image-Text Matching, Real-Time Verification, Vision–Language Models.

ICFTSEMAP121

Profiling Consumer to Consumer e-Commerce using Facebook

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ABSTRACT

Modern retailing is being shaped by consumer-to-consumer (C2C) electronic commerce, or e-commerce. Although social media as an economic vehicle can result in varying customer value and, therefore, has the potential to create customer interactions that lie outside the reach for conventional C2C e-commerce actors, it has remained a relatively uncharted field of research despite the fact that a significant portion of C2C e-commerce takes place in social media-driven platforms (e.g. Facebook). Therefore, this paper's goal is to discover and investigate various C2C e-commerce consumer profiles on social media. Confirmatory factor analysis and cluster analysis are used to analyze the data acquired from a quantitative investigation which concentrates on C2C e-commerce. We offer and analyze four distinct customer profiles: apathetic, salvagers, bargain seekers, and enthusiasts. These profiles show how valuable it is for consumers to trade used things with other Facebook members. The data was collected from one country only. Cultural differences in perceptions of C2C online shopping and Facebook's role may affect the generalizability of the findings. The cross-sectional information set was based on self-reported data. The study found that social media can offer a unique platform for consumer-to-consumer (C2C) e-commerce, potentially resulting in unique and unique consuming experiences. Second, companies should carefully consider how their present segments fit with the customer characteristics outlined in the study (enthusiasts, value seekers, salvagers, and apathetic) in order to assess future value creation potential and challenges. Third, traditional stores should think about the possibility of acting as a venue for cross-selling or other types of C2C contact in order to offer their customers the benefits that are distinctive of C2C e-commerce. This study is among the first attempts to locate C2C consumers of online stores in the context of social media. It's fascinating to see that rather than typical customer demographics, the profiles differ based on the kind of value that people perceive. Collectively, they offer fascinating empirical access to explore the potential and implications of C2C e-commerce powered by social media.

ICFTSEMAP122

Smart Safe Dryer for Handloom Sarees with Controlled Humidity, Temperature and Airflow

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ABSTRACT

Handloom weaving involves repeated wetting of yarn during saree production to maintain stiffness and alignment. However, the subsequent drying process is typically performed using natural air or handheld devices, leading to uneven moisture removal, increased drying time, and potential fabric damage. This work proposes a compact and safe drying assistance system designed specifically for handloom environments. The system utilizes controlled airflow generated by a cooling fan combined with a low-temperature Positive Temperature Coefficient (PTC) heating element to enhance evaporation without exceeding safe thermal limits for silk and cotton fabrics. Temperature and humidity sensors continuously monitor the fabric microenvironment, enabling automatic cut-off once the desired dryness threshold is achieved. The prototype demonstrates uniform drying, reduced manual effort, and improved operational efficiency compared to traditional methods. The proposed solution provides a low-cost, loom-integrated approach that enhances productivity while preserving fabric integrity and traditional weaving practices.

Keywords: Handloom Weaving, Saree Drying, Textile Processing, Temperature Control, Humidity Monitoring, Embedded Systems.

ICFTSEMAP123

Adaptive Dust Suppression System for Open-Cast Mining Using Wind-Based Trajectory Prediction

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ABSTRACT

Open-cast mining poses significant environmental and occupational health risks due to the dispersion of dust by wind, which degrades air quality and increases pollution in mining areas and surrounding communities. Conventional dust suppression systems, which are either manually operated or run continuously, often suffer from inefficient water usage and limited control over dust dispersion. To overcome these limitations, this paper proposes an adaptive dust suppression system based on an ESP32 embedded controller integrated with wind direction and particulate matter (PM) sensors. The system continuously monitors real-time dust concentration and wind conditions to estimate the most likely dust dispersion path and automatically activates directional water spray mechanisms for targeted dust control. The system architecture includes an ESP32 microcontroller, wind direction sensor, dust sensor, relay-controlled water pump, spray nozzles, and an LCD module for local display of environmental parameters. The ESP32's built-in Wifi enables real-time data transmission to a cloud-based IoT platform for remote monitoring and data logging. Experimental results demonstrate improved dust suppression efficiency and optimized water consumption compared to conventional fixed spraying methods. The proposed solution is scalable, cost-effective, and suitable for smart mining environments.

Keywords: Open-Cast Mining, Adaptive Dust Suppression, ESP32, Particulate Matter (PM), IoT Monitoring.

Hybrid Deep Learning – Based Email Spam Detection and De-Duplication

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ABSTRACT

Rapid growth of digital data has created serious storage inefficiency due to duplicate files and increasing email overflow, particularly under Gmail storage limits, which disrupt communication. Deep learning enables intelligent automation for large-scale data management through accurate pattern recognition in emails and files, improving spam detection and duplicate identification. However, existing solutions depend on separate tools for local duplicate file removal and manual email cleanup, with no unified platform connecting local file management and Gmail monitoring, increasing the risk of accidental data loss. To address these challenges, this paper proposes D-SPARC — Spam Detection and Redundancy Control Framework, a deep learning–based integrated system that performs efficient spam filtering and duplicate data detection within a single platform. The system continuously monitors Gmail storage, automatically removes redundant data, filters spam emails, and optimizes storage utilization to ensure uninterrupted and efficient communication.

Keywords: Spam Detection, Duplicate Detection, Gmail Monitoring, Storage Optimization, D-SPARC.

ICFTSEMAP125

Artificial Intelligence Based Mathematical Education in Sustainable Development for Academics

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ABSTRACT

Artificial intelligence (AI) used for education at different courses, the study of comparison and view for implementing sustainable awareness using AI is targeted in this article. The concepts of NEP2020 intertwined with sustainability motive for mathematics education is the key area of investigation. It includes adaptive and personalized learning system, intelligent tutoring and conceptual understanding, AI supported assessment and learning analytics. We focus on the contribution of AI in achieving sustainable goals such as advancing inclusive and quality education, skill development and scalability in education system, challenges and ethical dimensions. A review of NEP2020 for sustainability is included for the completeness of the concepts. Observations are tabulated and analysed using the present research carried in this direction. A detailed summary is presented for addressing the specific research questions such as what are the sustainability goals in education in general, what are the mathematical educational inclusions for the sustainability goals using machine learning, what are the researches done during the period of 2020 to 2025 to achieve SDG4. The paper reviews the recent literature and indicates the research gaps to be addressed. It explores the tools which can be used for handling certain topics in classrooms. It was observed that a data driven academic planning is to be focused for the valid implementation of sustainability goals in education.

Keywords: Mathematical Education, Artificial Intelligence, AI Supported Sustainability Applications, NEP2020, SDG4.

ICFTSEMAP126

Survey on Data Fingerprinting and Visualization for AI-Enhanced Cyber-Defence Systems

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ABSTRACT

Artificial intelligence (AI)-assisted cyber-attacks have evolved to become increasingly successful across all phases of the cyber-defence life cycle. For example, during the reconnaissance phase, AI-enhanced tools such as MalGAN can be deployed to automatically exploit vulnerabilities in cyber-defence systems. However, existing countermeasures are often unable to detect attacks launched by most AI-enhanced tools. The solution presented in this paper represents an initial step toward using data fingerprinting and data visualization to defend against AI-enhanced cyber-attacks. The AIECDS methodology for the development of AI-Enhanced Cyber-Defence Systems is introduced and discussed. This methodology incorporates dedicated tasks for data fingerprinting and visualization to address limitations associated with traditional data-driven defense approaches. The use of fingerprinted data and visualized representations in cyber-defence systems has the potential to significantly reduce the complexity of decision boundaries and simplify the machine learning models required to improve detection efficiency, even when dealing with malicious threats represented by very small sample datasets. The effectiveness of the proposed approach is validated by demonstrating how the resulting fingerprints enable clear visual discrimination between benign and malicious events. This validation is performed through a use case focused on the discovery of cyber threats using fingerprinted network sessions.

Keywords: Artificial Intelligence–Enhanced Cyber-Attacks, Cyber-Defence Lifecycle, Data Fingerprinting, Data Visualization, AI-Enhanced Cyber-Defence Systems (AIECDS), Machine Learning, Anomaly Detection, Adversarial Attacks, Network Security, Threat Detection.

ICFTSEMAP127

A Simulation Study for Power Quality Enhancement in EV Chargers Using SCP-NLMF Method

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ABSTRACT

Power quality enhancement in EV chargers is essential to reduce harmonics, improve power factor, and maintain voltage stability due to the nonlinear nature of power electronic converters. It ensures grid reliability, regulatory compliance and efficient operation under high-power and fast-charging conditions. This research work focus on the design of multifunctional charging system of electric vehicles (EVs) comprising grid connection, EV batteries and residential loads. The system is designed with two way converter to control the charging and discharging of batteries and to keep the DC bus voltage constant in Electric Vehicles. A voltage source converter (VSC) is connected to the grid. It operates smoothly as it is able to control nonlinear local loads that are non-uniform. The discharging and charging process regulates the voltage and the current profiles depending on the battery state of charge (SOC). A sparse constrained proportionate normalized least mean fourth (SCP-NLMF) algorithm is used in this research work to gain better control over the system. The strategy minimizes the mean square error (MSE) by minimizing the convergence slowdown at the later stages and enhancing the estimation of the active load current component. The accuracy of SCP-NLMF approach and the effectiveness of the suggested charging system is demonstrated using MATLAB/SIMULINK

Keywords: Bidirectional Converter, Voltage Source Converter, Electric Vehicles, Multipurpose EV Charger, Nonlinear Loads.

ICFTSEMAP128

Driving Electric Mobility: Classification and Application Perspectives of Battery Technologies

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ABSTRACT

The rapid pace of the electric vehicle (EV) adoption has increased the importance of high-performance, safety and cost-effective battery technologies. Even with significant advancements, no single battery chemistry can satisfy the ever-increasing demands for low cost, environmental sustainability, rapid charging, thermal stability, high energy density and extended cycle life. This poses a critical problem to researchers and manufacturers in choosing and optimizing battery systems in various use in EVs. In this paper, a systematic and comparative review of prominent battery technologies utilized in electric vehicles, such as lead-acid, nickel-metal hydride (NiMH), lithium-ion (Li-ion) variants of nickel-manganese-cobalt (NMC) and lithium iron phosphate (LFP) as well as new solid-state and sodium-ion batteries are discussed. The comparative metrics are key evaluation parameters, which include the energy density, power density, cycle life, charging characteristics, safety performance, thermal behaviour, cost factors and material sustainability. A performance-based evaluation system is created to evaluate the trade-offs between the various chemistries and their applicability to various EV types such as hybrid, passenger and commercial vehicles. However, there are still problems with end-of-life management, thermal runaway and reliance on raw materials. Emerging technologies like solid-state and sodium-ion batteries have potential. In order to support the next generation of electric mobility, this work provides a thorough technological foundation for battery selection with an emphasis on material innovation, battery management systems and sustainable lifespan approaches.

Keywords: Electric Vehicles, Lithium-Ion Batteries, Lithium Iron Phosphate (LFP), Solid-state Batteries, Energy Density, Battery Management System.

ICFTSEMAP129

A Structured Adaptive Framework for Technology-Enhanced Cognitive Rehabilitation for Preadolescent Children

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ABSTRACT

Cognitive rehabilitation is widely used to improve memory, attention, and executive functioning in children with developmental and neurological conditions. The preadolescent i.e. age group of 8 to 15 years is particularly important, as this period involves significant cognitive growth and academic development. Although digital tools and serious games are increasingly applied in rehabilitation for this population, many existing systems lack structured adaptability and systematic evaluation methods. Most current solutions use fixed difficulty levels and limited personalization, which may reduce long-term effectiveness and engagement in children. This study proposes a structured adaptive framework for technology-enhanced cognitive rehabilitation systems specifically designed for children between the age group from 8 to 15 years. The framework integrates baseline cognitive assessment, adaptive progression mechanisms, performance monitoring, and structured feedback layers that are suitable for this developmental stage. The proposed model aims to provide a systematic approach for designing rehabilitation technologies that respond to individual performance levels and cognitive needs of school-aged children. Through comparative analysis of existing digital cognitive tools and framework validation using theoretical mapping and expert evaluation, this research aims to establish a structured model that enhances personalization, accessibility, and measurable cognitive outcomes for preadolescent children.

Keywords: Framework, Cognitive, Rehabilitation.

ICFTSEMAP130

Survey On Intelligent Latency Oriented Encryption for Balancing Security and Quality of Service in Cloud Systems

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ABSTRACT

Cloud computing demands robust data security while maintaining acceptable Quality of Service (QoS). Several encryption mechanisms, including profile-based, rule-based, and attribute-based approaches, have been proposed to secure cloud data. However, many existing techniques face challenges in ensuring strong tamper resistance and maintaining low latency, which directly impacts service performance. Achieving an effective balance between security strength and QoS remains a significant research concern. This survey reviews existing cloud encryption strategies with particular attention to latency-aware and adaptive encryption mechanisms. It analyzes how different schemes address security requirements and evaluates their impact on QoS parameters such as latency and efficiency. Special focus is given to time-oriented and latency approximation-based approaches that dynamically select encryption techniques based on performance metrics. The study highlights current limitations, compares existing models, and identifies research gaps in adaptive encryption selection for cloud environments. The survey aims to provide insights for developing efficient, QoS-aware encryption frameworks in future cloud systems.

Keywords: Cloud Computing, Adaptive Encryption, Latency Optimization, Quality of Service (QoS), Data Security.

The Role of Artificial Intelligence in Enhancing Learning Among College Students

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ABSTRACT

Artificial Intelligence (AI) is no longer just a concept; it is quickly becoming a standard tool in college classrooms. This paper, titled *"The Role of Artificial Intelligence in enhancing learning among college students,"* examines how education is shifting due to tools ranging from chatbots to grading software. The primary objective of this study is to determine if these tools genuinely improve the learning process or if they simply serve as shortcuts for completing assignments. There is no gainsaying the fact that Technology is everywhere in the landscape and the future belongs to it. Despite the popularity of Artificial Intelligence, there is a significant gap in current research. While experts often discuss the advanced capabilities of the technology, there is limited evidence regarding how the average student interacts with it daily. To address this issue, this research conducted a survey of undergraduate students to track their study habits and academic grades. The results indicate that AI creates a highly personalized learning environment. Students reported that receiving instant feedback allows them to correct mistakes immediately, which keeps them engaged significantly longer than waiting for traditional grades from a professor. However, the findings also present a warning. While AI improves efficiency, there is a risk that students may rely on it too much, potentially weakening their critical thinking skills. They may rely too much on technology underutilizing their human skills in conceptual and analytical things.

Keywords: **Artificial Intelligence, Student Learning, Technology, College Students.**

ICFTSEMAP132

From Sacred Space to Tourist Place? The Impact of Commercialization on Cultural Authenticity in Varanasi

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ABSTRACT

This study investigates how increasing commercialization shapes the perceived cultural authenticity of spiritual tourism in Varanasi, one of India's most significant sacred cities. Although prior research has highlighted the economic benefits of religious and spiritual tourism, relatively few empirical studies have examined how commercialization, overtourism and mediated representations jointly influence tourists' authenticity perceptions in the Indian context. Addressing this gap, the study develops and tests a multidimensional framework linking commercialization, over-tourism and crowding, media, and digital influence, and economic and policy interventions to perceived cultural authenticity and sustainability outcomes. Grounded in authenticity theory and sustainable tourism discourse, the research conceptualizes six latent constructs: Commercialization, over-Tourism, Media & Digital Influence, Economic & Policy Interventions, Perceived Cultural Authenticity and Sustainability Outcomes. A structured questionnaire has been designed, incorporating Varanasi-specific indicators such as VIP Darshan, paid Aarti viewings, crowding at ghats and temples, online representations of spiritual sites, and perceptions of regulatory and community involvement in tourism management. Data has been collected from domestic, international, and local visitors to capture diverse perspectives on spirituality, commercialization, and authenticity, and analysed using PLS-SEM. The proposed model anticipates that commercialization, over-tourism, and media influence negatively affect perceived cultural authenticity, whereas supportive economic and policy interventions enhance it. Perceived cultural authenticity, in turn, is expected to positively influence perceived sustainability of Varanasi as a spiritual destination. The paper presents preliminary empirical insights and discusses implications for destination managers and policymakers seeking to balance economic gains with the preservation of spiritual and cultural integrity. By focusing on Varanasi as a critical case, the study contributes to debates on responsible spiritual tourism in India and offers actionable recommendations for commercialization and regulatory strategies that safeguard authenticity while supporting local livelihoods and long-term destination sustainability.

Keywords: Spiritual Tourism, Commercialization, Perceived Cultural Authenticity, Over-Tourism.

Integrated Water Quality and Heavy Metal Assessment of Mezierü and Dzü Tributaries of the Doyang River Northeast India.

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ABSTRACT

Small and medium tributaries in Northeast India are increasingly exposed to anthropogenic pressures, yet systematic assessments integrating physicochemical and heavy metal contamination remain limited. This study presents an integrated evaluation of water quality in the Mezierü and Dzü tributaries of the Doyang River, Nagaland, using Water Quality Index (WQI) and Metal Pollution Index (MPI) approaches. Water samples were collected from multiple sites representing varying degrees of anthropogenic influence to examine spatial and seasonal variations. Physicochemical parameters and trace metal concentrations were analyzed following standard methods. WQI values show a progressive increase downstream, indicating deterioration of water quality, with slightly elevated values during the summer season. MPI results reveal a similar spatial pattern, suggesting increasing heavy metal accumulation in downstream sections. Correlation analysis demonstrates a very strong positive relationship between WQI and MPI ($r \approx 0.99$–1.00, $p < 0.01$), confirming that metal contamination is a dominant factor influencing overall water quality in the study area. The findings highlight that spatial variability, particularly along the forest–urban gradient, exerts a stronger control on river health than seasonal fluctuations. This integrated index-based assessment provides critical scientific evidence to support targeted water management strategies and sustainable protection of ecologically and socially important tributaries in Northeast India.

Keywords: WQI, MPI, Trace Metals, Spatial-Seasonal Variation; Tributaries. Doyang River; Northeast India.

Assessing the Effects of Sustainable Banking Practices and Digital Innovation on Customer Satisfaction and Long-Term Customer Loyalty

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ABSTRACT

The study examines how sustainable banking practices and digital innovation influence customer satisfaction and long-term customer loyalty within the Indian banking sector. In this changing environment, the industry has emphasised environmental, social, and governance (ESG) responsibilities alongside rapid technological advancements. Sustainable practices—such as green loans, paperless transactions, and energy-efficient operations—shape public perceptions of corporate accountability. Concurrently, digital innovations such as mobile banking, AI-driven support, and e-statements enhance customer experiences through increased convenience and efficiency. A quantitative research design is employed in this study, collecting responses from 100 customers of major commercial banks through a structured questionnaire. Mean scores were calculated to assess customer satisfaction and loyalty levels. To study the relationship among variables, correlational analysis was performed. Results disclose strong positive correlations between digital innovation and customer satisfaction, and between sustainable practices and customer loyalty. Customers' opinion of sustainability initiatives is not merely as eco-friendly efforts but as indicators of ethical, transparent operations that build continuing trust. The findings emphasise the need for banks to integrate sustainability goals with digital transformation strategies to strengthen customer relationships and gain a competitive edge. This study contributes to the growing literature on sustainable finance and technology adoption by offering insights from an emerging economy context, emphasising ongoing innovation to sustain customer satisfaction and loyalty.

Keywords: Sustainable Banking, Digital Innovation, Customer Satisfaction, Customer Loyalty, ESG, FinTech.

ICFTSEMAP135

Micro-Replacement in Healthcare: How AI Decision Support Systems Invisibly Displace Human Clinical Judgment

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ABSTRACT

The emerging phenomenon that has been termed micro-replacement—the subtle and incremental manner in which AI technology is surreptitiously stealing small pieces of physician decision-making in the medical field. Unlike job replacement or full automation, micro-replacement occurs through small, almost undetectable increments, with physicians increasingly incorporating AI recommendations regarding diagnosis, patient classification, and treatment decision-making. The study describes how variables such as reliance on AI, becoming less vigilant, and the design of AI software, such as providing default options, notifications, or pre-formatted recommendations, can cumulatively influence physicians' decisions, diminish autonomous thinking, and impact their accountability. It tackles this shift in operations occurring through examples of recent hospital scenarios and staff utilizing AI to support diagnosis and prescription writing and how this shift occurs, why it occurs, and what the implications of this shift are regarding the loss of skills, lack of awareness, and unclear control over decisions. To offset the advantages of AI with a strong human judgment, the study proposes a framework for responsible use, which includes keeping humans in the loop, making AI decision-making transparent, and ongoing training. The study places great emphasis on having proper rules and guidelines to ensure the preservation of human judgment as part of human judgment

Keywords: Micro-Replacement, Artificial Intelligence, Clinical Decision-Making, Automation Bias, Human Judgment, Responsible AI, Healthcare Technology.

ICFTSEMAP136

Multi Model Remote Parameter Monitoring and Detection Using ML Algorithms for Pollution Prediction System

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ABSTRACT

Experimental work of Environmental pollution monitoring has become crucial for sustainable urban development and public health management. This paper presents a comprehensive real-time pollution monitoring system capable of simultaneously tracking multiple environmental parameters including air quality, carbon monoxide concentration, temperature, humidity, and dust. The system integrates an ESP32 microcontroller with multiple specialized sensors, GPS based geo-tagging capabilities, and cloud-based data visualization through Blynk IoT platform. A novel aspect of this work includes the implementation of machine learning algorithms for continuous predictive analytics, enabling forecasting of pollution trends based on historical data patterns. The system features local display capabilities via LCD screen and remote monitoring through IoT connectivity, making it suitable for both mobile and stationary deployment scenarios. Field deployment results demonstrate the system's effectiveness in providing accurate, real-time environmental data with predictive capabilities that can assist in proactive pollution management strategies. The compact design, powered by a regulated DC supply with buck boost conversion, ensures portability and reliability for diverse environmental monitoring applications. The proposed work employs Random Forest algorithm with neural network LSTM and provides a accuracy of R^2 =0.91% from different geotagged location along with validation parameters and the obtained is superior among conventional systems.

Keywords: Environmental Monitoring, IoT, Pollution Detection, Machine Learning Prediction, Multi-Sensor Integration, ESP32, Air Quality Monitoring, Geo-Tagged Sensing.

ICFTSEMAP137

AR-Driven Intelligent Food Ordering System

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ABSTRACT

Conventional restaurant menus primarily rely on static images and textual descriptions, offering limited insight into the actual appearance, portion size, and ingredient composition of dishes. This limitation often results in incorrect orders, communication gaps, and customer dissatisfaction, particularly for individuals with dietary restrictions or food allergies. To address these challenges, this study proposes a comprehensive three-dimensional (3D) menu system based on Augmented Reality (AR) integrated with modern web technologies to enhance the dining experience. The system utilizes technologies such as Three.js, React.js, AR.js/8thWall, Node.js, and MongoDB to provide real-time 3D dish visualization, allergen-aware filtering, multilingual support, QR-based table mapping, and live order tracking. Experimental evaluation demonstrates significant improvements in customer engagement, decision-making accuracy, and ordering satisfaction. Furthermore, the proposed solution serves as a scalable smart restaurant platform, contributing to the advancement of digital hospitality systems.

Keywords: Augmented Reality, 3D Menu, Allergen Filter, Web AR, Restaurant Automation, Smart Dining System.

ICFTSEMAP138

AI-Based Traffic Signal Controller Using Real-Time Traffic Density

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ABSTRACT

Traffic congestion has become a major challenge in urban areas due to the rapid increase in the number of vehicles and the limitations of traditional fixed-time traffic signal systems. This project presents the design and implementation of an AI-based traffic signal controller that dynamically adjusts signal timing based on real-time traffic density. Using computer vision techniques and artificial intelligence algorithms, the system analyzes live video feeds from cameras installed at intersections to detect and count vehicles in each lane. The collected data is processed to estimate traffic density, and the signal duration is automatically optimized to minimize waiting time and improve traffic flow efficiency. The proposed system is implemented using a Arduino controller integrated with cameras, sensors, and traffic lights, enabling adaptive control without manual intervention. Experimental results demonstrate that the AI-based controller effectively reduces congestion and enhances traffic management compared to conventional fixed cycle systems.

Keywords: Artificial Intelligence (AI), Traffic Signal Control System, Real-Time Traffic Density, Smart Traffic Management, Intelligent Transportation System (ITS), Machine Learning Algorithms, Computer Vision, Image Processing, Adaptive Signal Timing, Traffic Flow Optimization, Congestion Control, Vehicle Detection, IoT-Based Monitoring, Embedded Systems, Urban Traffic Automation.

ICFTSEMAP139

Enhanced YOLOv8 Architecture for Thermal Multi-Sensor Military Target Detection

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ABSTRACT

This study presents an optimized YOLOv8 detector engineered specifically for nocturnal and low-illumination military surveillance, utilizing integrated thermal, near-infrared (NIR), and visible-spectrum imaging. The proposed system incorporates thermal-aware preprocessing modules for adaptive noise mitigation, an attention-enhanced backbone architecture to amplify thermally significant regions, and hierarchical multimodal fusion mechanisms—operating at pixel, feature, and decision levels—to detect personnel, vehicles, and weapons in challenging operational environments. Experimental validation on more than 50,000 thermal images demonstrated superior performance across diverse weather conditions and temperature extremes ranging from -20°C to +50°C. The system achieved an 88.1% mean Average Precision (mAP) in overall target detection with less than 2.1% false alarm rates, representing a 33.7% performance improvement over single-modality baselines. Real-time processing capabilities of 23.8 fps at 1080p resolution with minimal GPU memory utilization (2.3 GB) validate the feasibility of field deployment on resource-constrained platforms.

Keywords: Thermal Imaging, Object Detection, YOLOv8 Optimization, Military Surveillance, Multi-Spectral Fusion, Real-Time Detection.

ICFTSEMAP140

Automated Coronary Artery Disease Severity Grading Using Deep Learning

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ABSTRACT

Coronary Artery Disease (CAD) is a common cardiovascular disease worldwide and requires accurate and early detection to avoid serious complications such as heart attacks and cardiac arrests. Invasive Coronary Angiography (ICA) remains the most reliable method for assessing the degree of arterial stenosis; however, its interpretation is often subjective and prone to human error. This project aims to overcome these limitations by utilizing a deep learning-based framework for automated CAD classification using ICA images. The classification of image patches into lesion and non-lesion categories across different lesion severity ranges is performed using five Convolutional Neural Network (CNN) architectures: DenseNet, ResNet, LeNet, AlexNet, and VGGNet, after implementation and evaluation. High-quality model training is achieved by preprocessing the dataset through vessel segmentation, patch extraction, class balancing, and data augmentation. Measurement of performance is done by using accuracy, precision, recall, F-measure, and area under the ROC curve (AUC). The experimental results demonstrate that DenseNet and ResNet perform better, with an F-measure that exceeds 90% and an AUC that exceeds 98%, effectively identifying severe lesion regions. By using the proposed framework, CAD diagnosis can be improved through the use of a robust, automated, and clinically applicable solution that improves diagnostic precision and reduces manual workload for cardiologists.

Keywords: Coronary Artery Disease (CAD), Deep Learning, Convolutional Neural Networks, Invasive Coronary Angiography, DenseNet, ResNet, LeNet, AlexNet, VGGNet.

ICFTSEMAP141

Smart E-Commerce App Using AR to Visualise Products in Real Time On Android

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ABSTRACT

More interactive and immersive product visualization techniques are needed by modern e-commerce platforms to improve user confidence and decision-making when purchasing online. However, static photos and videos are the mainstay of traditional e-commerce applications, which frequently fall short of giving a genuine knowledge of products, resulting in higher return rates and customer discontent. This study suggests a clever e-commerce program that uses Augmented Reality (AR) to show products on Android smartphones in real time. Before making a purchase, customers may accurately visualize a product's size, appearance, and location by utilizing smartphones to view and interact with three-dimensional (3D) replicas of the object in their natural setting. By leveraging AR technology, cloud-based product data, and real-time rendering, the application enhances user engagement, personalization, and shopping experience. While keeping the solution affordable and accessible for customers, it seeks to boost purchase satisfaction, decrease product returns, and promote customer trust. Comparing AR-based e-commerce to traditional e-commerce applications, user feedback and experimental results show that the former greatly enhances product knowledge, user engagement, and overall purchasing efficiency.

Keywords: Augmented Reality, Smart E-Commerce, Product Visualization, AR Core, Android Application.

ICFTSEMAP142

Analyzing the use of Blockchain in E-voting Systems

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ABSTRACT

Modernization of electoral systems is essential as digital platforms increasingly replace traditional processes. This paper presents a secure blockchain-based e-voting system implemented using Ethereum smart contracts to address transparency, trust, and tamper-resistance concerns. A commit–reveal voting protocol preserves ballot secrecy by concealing voter choices during the voting phase, while a Merkle tree–based whitelist authenticates eligible voters efficiently without storing identities on-chain. Each voter casts exactly one vote using a unique blockchain address, enforced through smart-contract logic. The system ensures immutability, integrity, and verifiability of election results and was implemented and tested using the Ganache Ethereum local blockchain and MetaMask, demonstrating a scalable and privacy-preserving e-voting solution suitable for real-world deployment.

Keywords: Blockchain, E-Voting, Commit–Reveal Protocol, Merkle Tree, Ethereum, Ganache.

Self-Supervised Semantic Prior GAN with Context Aware Feature Guidance and Topology-Aware Refinement for Heritage Image Restoration: Results and Performance Analysis

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ABSTRACT

Heritage images often contain large, irregular losses where exemplar filling leaves seams and diffusion/transformer restorers are computationally heavy. This work proposes an efficient GAN-based restoration framework that preserves brushstrokes and structural lines under irregular masks while keeping runtime practical. The model integrates three components: Adaptive Semantic Prior Injection (ASPI) for context, Topology-Aware Structure Refinement (TASR) for line/edge continuity, and Entropy-Guided Self-Supervision (EGSS) to emphasize uncertain regions. Training uses Damaged Paintings (Kaggle) at 256×256 for 50 epochs, WikiArt (masked) for self-supervised prior learning, and Indian Heritage Monuments for external validation, with comparisons against exemplar, cGAN (partial convolutions), diffusion, and transformer baselines. The framework achieves 32.6 dB PSNR, 0.92 SSIM, and 18.7 FID, surpassing diffusion and transformer models while improving texture continuity over exemplar/cGAN methods. Ablation indicates TASR drives structural fidelity, with ASPI and EGSS providing complementary gains. Efficiency is maintained with 34.8M parameters, 108.5 GFLOPs, and 0.118 s/img inference. Remaining gaps include faint seams on very fine cracks and only plausible (not exact) fills for very large missing regions; future work targets higher-resolution training and selective diffusion priors.

Keywords: Heritage Image Restoration; Image Inpainting; Generative Adversarial Network (GAN); Semantic Prior; Structure Refinement; Computational Efficiency.

ICFTSEMAP144

FootEase Pro: Inbuilt Insole Massager

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ABSTRACT

Foot fatigue and muscular discomfort are common problems caused by prolonged standing and walking. To overcome this issue, FootEase Pro: Inbuilt Insole Massager is designed as a lightweight, affordable, and compact massaging system integrated directly inside a shoe insole. The proposed system uses miniature vibration motors placed at key pressure points of the foot to improve blood circulation and reduce stress without affecting normal walking movement. The insole is designed using flexible and durable materials to ensure comfort, load-bearing capability, and long-term usage. The product focuses on cost-effectiveness, portability, and ergonomic design, making it suitable for daily use by working professionals, elderly individuals, and students. FootEase Pro enhances regular footwear by incorporating therapeutic functionality in a simple and economical way.

Keywords: Insole Massager, Foot Therapy, Ergonomic Design, Vibration Mechanism, Affordable Device.

ICFTSEMAP145

Currency Security Validation System Using Machine Learning Deep Learning Algorithms

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ABSTRACT

Counterfeit currency has become a global financial threat, undermining economic stability and public trust. Traditional detection systems, which rely on physical verification or simple pattern matching, are increasingly ineffective against sophisticated forgeries. This project proposes a **Multi-Currency Security Validation System** that integrates **Deep Learning and Machine Learning** techniques for the authentication of both **banknotes and coins**. The system uses **ResNet50-based transfer learning** to extract rich visual features such as holographic threads, colour shifts, watermarks, and embossing patterns, while **XGBoost** classifies these features to determine authenticity. Grad-CAM–based visual explainability highlights the regions used for decision-making, enhancing transparency. The hybrid CNN-XGBoost architecture supports multiple currencies (INR, USD, EUR, etc.), ensuring high scalability and accuracy exceeding 98%. The proposed solution offers a secure, intelligent, and real-time framework suitable for deployment in banks, ATMs, airports, and retail outlets worldwide.

Keywords: Currency Authentication, Counterfeit Detection, Machine Learning, Random Forest, Deep Learning, Neural Networks, Feature Engineering, Financial Security Systems.

ICFTSEMAP146

IoT Integrated Infant Cradle for Jaundice Care

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ABSTRACT

The Smart Cradle is an IoT-based infant care system designed to ensure the safety and well-being of newborns through automated monitoring and treatment. It features a wet sensor to detect urination, a cry sensor to identify prolonged crying, and a motion sensor that activates gentle cradle swinging. A built-in blue LED phototherapy unit provides effective treatment for neonatal jaundice. Sensor data is transmitted in real-time to caregivers via a mobile app or cloud-based system, enabling timely alerts and responses. Powered by microcontrollers and IoT modules, the system ensures seamless automation and communication. This cradle is especially beneficial for working parents, hospitals, and rural areas, offering a reliable, compact solution that reduces caregiver burden and improves infant health.

Keywords: Infant safety, Smart baby Cradle, Blue LED Therapy, Neonatal Monitoring, Mobile App.

ICFTSEMAP147

AI-Integrated Approaches in Microbial Degradation of Plastics

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ABSTRACT

Plastic pollution has emerged as a critical global environmental challenge due to the persistent nature of synthetic polymers such as polyethylene (PE), polypropylene (PP), polystyrene (PS), and polyethylene terephthalate (PET). Although certain microorganisms including *Ideonella sakaiensis*, *Pseudomonas putida*, *Bacillus subtilis*, and *Rhodococcus ruber* exhibit plastic-degrading capabilities, natural biodegradation processes remain slow and inefficient under environmental conditions. Integrating artificial intelligence (AI) with microbial biotechnology offers a promising strategy to accelerate and optimize plastic degradation. This study investigates AI-driven methodologies for the discovery, engineering, and optimization of plastic-degrading microbes and enzymes. Machine learning models, including Random Forest, Support Vector Machines, and deep neural networks, were employed to analyze genomic and metagenomic datasets for the identification of novel plastic-degrading enzymes such as PETase, MHETase, cutinases, and laccases. AI-based protein structure prediction tools, including AlphaFold, enabled detailed modeling of enzyme–substrate interactions, supporting rational mutagenesis to improve catalytic efficiency and thermostability. Furthermore, genetic algorithms and artificial neural networks were applied to optimize bioreactor conditions—such as pH, temperature, and nutrient concentrations—leading to enhanced PET degradation rates under controlled laboratory conditions. AI-guided enzyme variants demonstrated superior activity and stability compared to wild-type strains, resulting in significantly improved polymer hydrolysis efficiency. These findings underscore the value of integrating computational prediction with experimental validation to accelerate sustainable plastic waste management. Overall, AI-integrated microbial engineering represents a transformative and scalable approach for advancing eco-friendly solutions to mitigate global plastic pollution.

Keywords: Artificial Intelligence, Microbial Degradation, Plastic Pollution

ICFTSEMAP148

Decentralized Autonomous Organizations (DAO)- The Future of Corporate Governance

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ABSTRACT

Decentralized Autonomous Organizations (DAOs) represent a fundamental shift in collective action and business management, transitioning from traditional "top-down" hierarchies to blockchain-based distribution of power. This research explores how DAOs utilize smart contracts to establish autonomous, decentralized organizations governed by code rather than central leadership. By leveraging token-based voting and automated execution, DAOs address critical "pain points" in corporate governance, specifically transparency and the Principal-Agent Problem.Through a comparative analysis of traditional corporations and decentralized models like Maker DAO and Uni swap, the study highlights the benefits of public auditability and aligned financial incentives. However, the transition to this "future of management" faces significant hurdles, including regulatory uncertainty, security vulnerabilities in code, and voter apathy. This paper concludes that while DAOs offer a democratic, flat alternative to the modern firm, their ultimate success depends on evolving legal frameworks and robust technical security.

ICFTSEMAP149

Agile Work Environment

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ABSTRACT

This study explores how the adoption of a hybrid nature of work, i.e., an amalgamation of offline and online work, can enhance the effectiveness of the work manifold. The research paper is an exploratory study based on secondary data from authentic published sources. The study mainly concentrates on the Zoom online meeting platform, which emerged as one of the leading panaceas for resolving meeting issues and facilitating hassle-free in-person and group meetings. Various data fetching platforms like the Organization for Economic & Cooperation Development (OECD) and emailtooltester.com were referred to as elicit number-based evidence. This paper epitome in furnishing the conceptual outline of video conferencing application, its advantages, applications and acceptance with special reference to Zoom and a comparative analysis on a global insight.

Keywords: Hybrid Work Culture, Hybrid Work Model, Competitive Advantage, Sustainable Practices and Virtual Team Management.

ICFTSEMAP150

Survey On Real Time Helmet Violation and Detection Using Deep Learning

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ABSTRACT

Artificial Intelligence (AI) has become a transformative technology for addressing real-world societal challenges by enabling machines to perform tasks that traditionally require human intelligence. In particular, deep learning has gained significant importance in the field of computer vision, where it supports automatic recognition, classification, and analysis of visual data. These capabilities have proven highly effective in traffic surveillance systems, enabling improved monitoring, decision-making, and enforcement to enhance road safety. In existing research, an artificial intelligence–based multitier framework with a lightweight classifier was proposed for helmetless biker detection. The system follows a two-stage approach in which motorbike riders are first detected from traffic surveillance footage and then classified based on helmet usage. The use of a lightweight model helps reduce computational complexity and supports faster processing while maintaining acceptable accuracy. Custom datasets were also developed to evaluate the system under controlled conditions. However, the existing approach focuses primarily on helmet detection and does not address other critical traffic violations or challenging environmental scenarios. To overcome these limitations, the proposed system employs DetectNet and the YOLOv12 algorithm to enable comprehensive and real-time traffic violation detection. In addition to helmet detection, the system is designed to identify multiple violations such as triple riding and mobile phone usage, while also maintaining robustness under diverse conditions including night-time and rainy environments. Furthermore, the system integrates an automated alert mechanism that notifies authorized institutions and sends warning messages to riders. This holistic approach improves detection accuracy, supports real-time enforcement, and enhances practical deployment, ultimately contributing to safer and more intelligent traffic management systems.

Keywords: Artificial Intelligence, Deep Learning, Helmet Detection, Traffic Safety, Computer Vision.

Federated Learning Based Framework for Privacy-Preserving Employee Attrition Prediction in Human Resource Analytics

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ABSTRACT

Employee attrition represents a critical challenge for contemporary organizations, impacting productivity, elevating recruitment costs, and eroding institutional knowledge. While machine learning offers robust tools for predicting turnover, its application is frequently hindered by "small data" constraints within individual firms and the sensitive nature of human resource (HR) records, which complicates collaborative data-sharing. This paper introduces FedHR, a federated learning framework designed to enable multi-institutional collaboration without the exchange of raw data. In FedHR, predictive models are trained locally within participating organizations, with only model parameters aggregated centrally. This architecture ensures compliance with rigorous privacy standards like GDPR and CCPA, further reinforced by differential privacy mechanisms. Using a simulated consortium of four diverse enterprises, we demonstrate that FedHR achieves an F1-score of 0.88, significantly outperforming isolated local models (0.74) and approaching the efficiency of centralized baselines (0.90). These results establish federated learning as an ethically responsible and technically viable infrastructure for strategic talent analytics.

Keywords: Employee Attrition, Federated Learning, Privacy-Preserving Machine Learning, Human Resource Analytics, Differential Privacy, Distributed Model Training, Talent Analytics.

ICFTSEMAP152

An Analysis of Plastic Reuse Strategies for Road Construction in Bihar for a Sustainable Society

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ABSTRACT

Plastic waste poses a significant threat to the environment. On average, a person uses about 45 kg of plastic annually. Since plastics are non-biodegradable, they take a very long time to decompose naturally. The frequent use of plastics in our society harms soil fertility, clogs drainage systems, and causes many other problems. Managing plastic waste disposal is a major challenge today. Reusing plastic waste can address this critical issue. The Bihar Government has launched a project to build durable, water-resistant roads using single-use plastic waste under the Lohia Swachh Bihar Abhiyan, which emphasizes systematic waste collection and management. Through this program, single-use plastic waste is collected door-to-door from rural households and sent to a plastic waste management unit, where it is used in road construction. This research paper aims to analyze the government-led initiative to incorporate waste plastic into road construction in Bihar.

Keywords: Single-Use Plastic, Sustainable Society, Road Construction, Lohia Swachh Bihar Abhiyan.

ICFTSEMAP153

Artificial Intelligence as a Service (AIaaS) for Secure Cloud Computing: Threat Detection, Risk Mitigation, and Autonomous Defence

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ABSTRACT

The rapid expansion of cloud computing has revolutionized digital service delivery by enabling scalable, on-demand, and distributed infrastructures. However, the dynamic and multi-tenant nature of cloud environments has intensified security vulnerabilities, including advanced persistent threats, data breaches, insider attacks, and misconfiguration risks. Conventional rule-based security mechanisms often lack adaptability and real-time responsiveness required in modern cloud ecosystems. This paper investigates the integration of Artificial Intelligence as a Service (AIaaS) into cloud computing architectures to establish intelligent, autonomous, and scalable security frameworks. A security-driven AIaaS model is proposed that incorporates machine learning-based threat detection, behavioral anomaly analysis, predictive risk mitigation, and automated defense orchestration. By leveraging distributed intelligence principles and edge-cloud collaboration, the framework enhances detection accuracy, reduces response latency, and improves system resilience. The study further examines architectural design considerations, deployment challenges, and performance trade-offs associated with AI-enabled security services in hybrid and multi-cloud environments. The results demonstrate that AIaaS-based security mechanisms significantly strengthen cyber resilience by enabling adaptive monitoring and autonomous response strategies. The proposed framework contributes to the advancement of secure, intelligent, and self-optimizing cloud infrastructures.

Keywords: Artificial Intelligence as a Service (AIaaS), Cloud Security, Threat Detection, Risk Mitigation, Autonomous Defense, Edge-Cloud Computing, Cyber Resilience.

ICFTSEMAP154

AI-Based Retinal Disease Detection Using Deep Learning

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ABSTRACT

Eye diseases are a major cause of vision impairment worldwide, and early detection is essential to prevent permanent blindness and improve treatment outcomes. This paper proposes an Artificial Intelligence-based Eye Disease Detection System using Deep Learning techniques to automatically identify common eye disorders from retinal images. The system employs a Convolutional Neural Network (CNN) with the MobileNetV2 architecture through transfer learning to classify images into categories such as Cataract, Glaucoma, Diabetic Retinopathy, and Normal conditions. The proposed model achieved high classification accuracy when evaluated on a standard medical image dataset. The system is implemented as a user-friendly web application that allows image upload for instant prediction and provides basic medical guidance through an intelligent chatbot. This approach demonstrates that integrating artificial intelligence with medical image analysis can support ophthalmologists in early diagnosis and improve accessible eye healthcare services

Keywords: Waste Classification, Deep Learning, MobileNetV2, Gamification, Environmental Sustainability, Transfer Learning, Convolutional Neural Networks, Web Application.

ICFTSEMAP155

Automated Car Ventilation System Using Peltier Module

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ABSTRACT

The current paper discusses a new thermoelectric ventilating system that can operate on a 12V 7Ah Lead Acid-Battery using solid-state thermoelectric modules and which is capable of providing reliable and low noise cooling services without necessarily involving the use of traditional types of refrigerants in order to meet the international demand for environmentally friendly and energy-saving cooling systems. Integrating advanced control algorithms and a microcontroller platform, the system is dynamically adjusted to the real-time data related to the characteristics of temperatures and humidity and, accordingly, the cooling supply and airflow are optimally controlled to reduce the amount of consumed power and, consequently, the battery life. The experimental evaluations demonstrate the capability of the system to reduce the interior temperatures by less than 25 degrees Celsius under unfavorable ambient conditions, and maintain operational independent features with capable control of the battery. Although the natural cooling power and material stability of Peltier devices cause limitations to its usage, this off-grid, battery-powered cooling system is highly promising as a sustainable, off-grid cooling system. This paper suggests further developments in the thermoelectric material science and system design, to enhance its functionality and increase its utility in many, energy-prohibited settings.

Keywords: Thermoelectric Cooling, Peltier Modules, Eco-Friendly Cooling System, Off-Grid Cooling, Battery-Powered Ventilation, Lead-Acid Battery, Microcontroller-Based Control, Energy-Efficient Cooling, Sustainable Cooling Technology, Temperature And Humidity Control.

ICFTSEMAP156

Harnessing Artificial Intelligence for Climate Change Resilience: An Integrated Framework for Prediction, Adaptation, and Policy Support

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ABSTRACT

Climate change is progressing at an alarming rate, threatening ecosystems, agriculture, and socio-economic stability worldwide. Addressing these multifaceted risks requires advanced tools capable of analyzing vast and complex datasets. Artificial Intelligence (AI) has emerged as a powerful enabler in climate science, offering innovative approaches to modeling, impact assessment, and adaptive planning. This paper introduces an integrated AI-climate framework that leverages deep learning models, geospatial analysis, and socio-environmental indicators to enhance early warning systems, climate vulnerability assessments, and regional adaptation strategies. The framework is evaluated through a detailed case study in semi-arid India, where AI models are used to predict drought patterns, assess land degradation, and identify high-risk zones. The results demonstrate AI's potential in generating actionable insights for local governance and sustainable development. The study concludes with practical recommendations for deploying scalable, ethical, and transparent AI systems in climate risk management and policy planning, emphasizing the need for interdisciplinary collaboration and data-driven decision-making.

Keywords: Artificial Intelligence. Machine Learning, Climate Change Resilience, Climate Vulnerabilityassessment, Drought Prediction, Sustainable Development.

ICFTSEMAP157

Automated Machine Learning Deployment System Using MLOps

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ABSTRACT

The increasing adoption of Machine Learning (ML) in real-world applications has exposed significant challenges in deploying models reliably and reproducibly at scale. While most ML workflows primarily focus on model development and evaluation, operational aspects such as experiment tracking, version control, automated retraining, and continuous deployment are often inadequately addressed. This paper presents an Automated Machine Learning Deployment framework grounded in MLOps principles to bridge the gap between model development and production deployment. The proposed system automates the end-to-end ML lifecycle, including data ingestion, multi-model training, performance evaluation, and best-model selection. Data Version Control (DVC) is employed to manage dataset versions, while batch-based data drift monitoring is implemented using Evidently AI to analyze distributional changes across training iterations. MLflow is utilized for systematic experiment tracking and model versioning, ensuring reproducibility and traceability of results. Continuous Integration and Continuous Deployment (CI/CD) are implemented using GitHub Actions to enable automated retraining and seamless deployment upon code or data updates. The selected model is containerized using Docker and deployed as a FastAPI-based REST service on an AWS EC2 instance. The proposed framework demonstrates improved deployment reliability, scalability, and reproducibility through automated MLOps practices, making it suitable for real-world machine learning applications.

Keywords: Continuous Integration and Continuous Deployment (CI/CD); Data Drift Detection; Experiment Tracking; Machine Learning Deployment; MLOps.

ICFTSEMAP158

Survey on Early Heart Disease Prediction using Machine Learning Algorithms with Power-Bi Visualization

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ABSTRACT

Artificial Intelligence enhances predictive healthcare by enabling faster and more accurate cardiovascular disease risk estimation. By integrating AI with Machine learning and Data science, cardiology has been transformed through reliable, accessible, and timely diagnostics. Artificial Intelligence optimizes cardiovascular disease detection by supporting informed clinical decision-making, enabling real-time patient monitoring, promoting early detection and personalized treatment planning. Moreover, AI-driven predictive analytics and visualization techniques improve cardiovascular risk prediction and management. The existing studies uses classifiers such as Logistic Regression, Decision Trees, Random Forests, KNN, and SVM, including Adaptive Boosting, Bagging, Voting, and Extreme Gradient Boosting to improve accuracy. Data imbalance is addressed using SMOTE, while feature selection techniques like chi-square, ANOVA, and Mutual Information refine predictors. Explainable AI, through SHAP, increases model transparency and is incorporated in mobile applications. Eventhough advancements exist, AI-driven cardiovascular prediction faces significant challenges, including limited and imbalanced datasets that increase over-fitting, high implementation costs, and inadequate evaluation of mobile usability. Labor-intensive diagnostic procedures like CT scans, ECGs, and MRIs impede timely cardiovascular disease detection. To overcome these problems, I propose collecting a balanced dataset, applying multiple machine learning algorithms, Box-Cox for normalization and Kaplan-Meier analysis for survival analysis, enhancing feature selection, and deploying interactive real-time visualizations through Power BI and Streamlit to build a scalable, personalized cardiovascular decision support system.

Keywords: Artificial Intelligence, Machine Learning, Data Science, Real-Time Patient Monitoring, Predictive Analytics, SMOTE, Feature selection, Box-Cox, Kaplan Meier, Power BI, Streamlit.

ICFTSEMAP159

Smart Building Sites: Leveraging RSSI for Automated PPE and Safety Compliance

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ABSTRACT

In construction and building sites, workers are often at risk of injury or entrapment due to accidents such as building collapses, where individuals may become buried beneath debris and rendered unreachable. To enhance search and rescue efforts, we integrated Received Signal Strength Indicator (RSSI) technology, which uses signal strength from devices like smartphones or personal locator beacons to accurately triangulate the location of trapped individuals beneath rubble. By incorporating RSSI into our rescue protocol, we can rapidly assess the environment and streamline rescue operations, significantly reducing the time required to locate injured workers. Additionally, our system continuously monitors the RSSI of workers on-site, triggering an alert via a buzzer if a worker moves away from the designated construction area, ensuring real-time safety and preventing unauthorized departures. This innovative approach improves the overall effectiveness of rescue missions and workplace safety by enabling quick and accurate worker location both during emergencies and routine operations. The use of RSSI in high-risk environments highlights the potential of advanced technology to safeguard workers, ensuring timely rescues and proactive safety measures.

Keywords: Construction Safety, RSSI Technology, Search and Rescue, Worker Monitoring, Emergency Response, Signal Strength, Location Tracking, Building Collapse, Debris Recovery, Real- Time Alerts, Workplace Safety.

ICFTSEMAP160

AI Prediction and Optimization of Environment Monitoring System with IOT for Agriculture Farmers

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ABSTRACT

Farmers face challenges such as unpredictable climate conditions, water scarcity, soil degradation, and increasing production costs. Existing monitoring systems provide real-time data but lack predictive intelligence and automated optimization. There is a need for an intelligent system that not only monitors environmental parameters but also predicts future conditions and optimizes resource utilization to maximize crop yield and minimize operational costs. This paper presents an AI-based prediction and optimization framework for an IoT-enabled environmental monitoring system designed to support smart agriculture. The system integrates multiple field sensors to continuously monitor key environmental parameters such as soil moisture, temperature, humidity, pH, light intensity, rainfall, and CO_2 levels. Data collected through the Internet of Things (IoT) architecture is transmitted to a cloud or edge computing platform, where Artificial Intelligence (AI) and machine learning models analyze historical and real-time data to predict climatic variations, soil conditions, crop health, and yield outcomes. Advanced optimization algorithms are employed to automate irrigation scheduling, fertilizer application, and energy management, thereby reducing water consumption, operational costs, and human intervention. The proposed system enhances decision-making accuracy, improves crop productivity, and promotes sustainable and resource-efficient farming practices, making it highly suitable for precision agriculture and climate-resilient farming environments.

Keywords: AIoT, Agritech, Environmental Monitoring.

Achieving Zero Defects in The Production Line Through the Use of POKA-YOKE Techniques

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ABSTRACT

In an increasingly competitive industrial environment, organizations must consistently deliver defect-free products while maintaining cost efficiency. The evolution of Total Quality Management (TQM) and Total Productive Maintenance (TPM) across manufacturing and service sectors has introduced structured methodologies aimed at continuous quality enhancement. Techniques such as Kaizen, Six Sigma, Just-in-Time (JIT), Poka-Yoke, and Flexible Manufacturing Systems (FMS) contribute significantly to establishing a sustainable quality-oriented culture. Manufacturing processes frequently involve repetitive and routine tasks that may lead to operator fatigue and reduced concentration, thereby increasing the probability of human error. Given the inherent limitations of human performance, preventive quality control mechanisms become essential. The Poka-Yoke approach, based on mistake-proofing principles, offers practical and cost-effective solutions to eliminate inadvertent errors at their source. Its systematic implementation not only reduces defects and rework but also alleviates cognitive burden on workers. Consequently, employees are able to redirect their skills and creativity toward more value-adding and innovative activities, enhancing overall organizational performance and long-term operational efficiency.

Keywords: Operational Efficiency, Human Error Reduction, Six Sigma, Lean Manufacturing and Quality Culture.

ICFTSEMAP162

No-Code ML Dataset Evaluator: Empowering Experts with Intelligent Model Selection Using Web-Based Statistical Analysis

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ABSTRACT

This paper presents a comprehensive web-based system that automates machine learning dataset quality assessment and algorithm recommendation for non-expert practitioners. The platform implements a novel six-component trainability scoring framework (0-100 scale) that evaluates data completeness, feature quality, sample size adequacy, duplicate detection, outlier prevalence, and class balance to quantify dataset readiness. The system performs dual-layer statistical analysis: descriptive metrics (mean, median, standard deviation, skewness, kurtosis) for distribution characterization, and inferential tests (Pearson correlation, normality assessment) for relationship identification and transformation guidance. A multi-task evaluation engine independently scores datasets for classification, regression, and clustering suitability, while a context-aware recommendation algorithm matches datasets to specific models based on detected quality signatures—suggesting XGBoost with class weighting for imbalanced data or Huber loss regressors for outlier-heavy datasets. Users receive three to five prioritized improvement recommendations with quantified impact predictions, comparing baseline versus projected performance metrics. Interactive visualizations render statistical insights through histograms, correlation heatmaps, and quality dashboards. Implemented entirely client-side using JavaScript, Chart.js, and PapaParse, the system ensures data privacy while providing instant feedback, successfully.

Keywords: Machine Learning, Dataset Quality Assessment, Algorithm Recommendation, Trainability Scoring, Statistical Analysis, Data Profiling, Automated Model Selection, Web-Based Tool, Data Quality Metrics, Interactive Visualization.

Artificial Intelligence in the Indian Banking Sector: A Systematic Review and Future Directions

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ABSTRACT

Artificial Intelligence (AI) is transforming global banking, and India's sector is undergoing rapid digitalization fueled by regulatory reforms, financial inclusion goals, and innovation policies. This PRISMA-based systematic review synthesizes **50 peer-reviewed studies (2016–2025)** on AI applications in Indian banking. Findings are categorized into five thematic clusters: **customer service automation, fraud detection and prevention, credit risk assessment, regulatory compliance, and adoption barriers**. Results show strong adoption in retail-facing services (chatbots, biometrics, RPA) and fraud detection, while progress in compliance, explainable AI, and rural banking remains limited. Unique drivers—such as multilingual diversity, financial inclusion schemes, and RBI-backed digital initiatives—differentiate India's adoption trajectory from global patterns. Persistent challenges include **algorithmic bias, transparency gaps, digital literacy divides, and workforce readiness**. This study contributes a **thematic taxonomy of AI adoption** in Indian banking and a **forward-looking research agenda** focused on explainable and ethical AI, rural applications, cross-technology integration, and governance frameworks. The insights provide actionable guidance for policymakers, practitioners, and technology providers to enable **responsible, transparent, and inclusive AI adoption in emerging economies**.

Keywords: Artificial Intelligence, Machine Learning, Indian Banking Sector, FinTech and Digital Banking, Financial Inclusion.

ICFTSEMAP164
Decentralized identity Verification System for Forensic Management

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ABSTRACT

In modern decentralized environments, ensuring secure, efficient, and flexible data sharing without relying on centralized authorities remains a critical challenge. Traditional access control and authorization mechanisms often suffer from single points of failure, limited scalability, and vulnerability to unauthorized access. This paper presents a Blockchain-Based Intelligent Authorization and Secure Data Sharing Framework designed to address these limitations through two key innovations. First, the proposed system leverages blockchain's immutable and decentralized ledger to implement secure authorization, permission delegation, and revocation using smart contracts, ensuring transparency, trust, and non-repudiation. Second, the framework incorporates decentralized identifiers (DIDs) and verifiable credentials (VCs) to enable flexible, role-based access control while preventing misuse or unauthorized transfer of permissions. By distributing authorization enforcement across multiple nodes, the system enhances resilience against attacks and eliminates dependency on centralized servers. The proposed model is expected to improve security, scalability, and efficiency in sensitive applications such as healthcare data management and cloud-blockchain systems. Overall, the framework provides a robust and decentralized solution for secure data sharing in real-world blockchain environments.

Keywords: Blockchain, Secure Data Sharing, Smart Contracts, Decentralized Authorization, Permission Delegation, Verifiable Credentials (VCs), Decentralized Identifiers (DIDs).

ICFTSEMAP165

VitaML: Vitamin Deficiency Detection Using Machine Learning

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ABSTRACT

Due to a lack of easily accessible and non-invasive diagnostic techniques, vitamin deficiencies are becoming a serious public health concern that frequently go undiagnosed. Conventional testing can be costly, time-consuming, and equipment-intensive. VitaML presents a machine learning-based framework that uses visible biomarkers, skin characteristics, and discoloration to detect vitamin deficiencies. VitaML seeks to develop a rapid, inexpensive, and user-friendly diagnostic tool by training classification models on a dataset of visual cues associated with particular deficiencies. This system can encourage preventive care, increase awareness of general nutritional health, and assist medical professionals in identifying problems early.

Keywords: Deficiencies, Equipment Discoloration, Biomarkers.

AI-Based Landslide and Debris Flow Detection Using YOLO

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ABSTRACT

Landslides and debris flow are among the most destructive natural hazards, causing severe loss of life, infrastructure damage, and environmental degradation, especially in mountainous and hilly regions. Timely and accurate detection of these hazards is essential for effective disaster mitigation and emergency response. This project presents an AI-based landslide and debris flow detection system using YOLO, a deep learning object detection model, combined with remote sensing imagery. The proposed approach automates the detection process by analyzing satellite and aerial images in real time, overcoming the limitations of traditional manual surveys and rule-based methods. Image preprocessing, model training, optimization, and evaluation are performed to improve detection accuracy and reduce false positives. The system enables rapid identification of hazardous zones, supporting early warning systems and disaster management authorities. The proposed model demonstrates scalability, reliability, and efficiency, making it suitable for real-time geohazard monitoring and decision support applications.

Keywords: Landslide Detection, Debris Flow, YOLO, Deep Learning, Remote Sensing, Disaster Management.

ICFTSEMAP167

Survey on Lung Disease Prediction Using a Hybrid Capsule Network-VGG19 Model with Deep Learning

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ABSTRACT

Lung cancer screening using low-dose computed tomography (CT) plays a crucial role in reducing mortality by enabling early detection of malignant pulmonary nodules. However, accurate malignancy prediction remains challenging, particularly for incident nodules that develop over successive screening rounds. Many existing deep learning approaches rely on single time-point CT scans, limiting their ability to capture temporal changes in nodule characteristics. This study proposes a fully automated deep learning-based Computer-Aided Diagnosis system, DeepCAD-NLM-L, designed to improve lung cancer risk prediction through longitudinal and multimodal analysis. The model integrates nodule-level features, lung-level imaging information, and patient clinical metadata across multiple CT time-points. By leveraging time-series data, the framework captures subtle structural and morphological changes that may not be easily recognized through visual assessment alone. The proposed model was trained using the National Lung Screening Trial (NLST) dataset and evaluated on both internal and external datasets, including the SUMMIT lung cancer screening study. Experimental results demonstrate that incorporating longitudinal imaging and multimodal features significantly enhances predictive performance compared to single time-point models, achieving strong discriminative capability. Furthermore, DeepCAD-NLM-L shows complementary performance to experienced radiologists, particularly in challenging screening cases. Overall, the findings highlight the value of combining temporal imaging data and clinical information in automated lung cancer prediction systems, supporting their potential role as assistive tools in large-scale lung cancer screening programs.

Keywords: Lung Cancer Screening, Computed Tomography, Deep Learning, Longitudinal Analysis, Computer-Aided Diagnosis, Malignancy Prediction, Deep Convolutional Autoencoder Denoising – Non-Local Means – Large.

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ABSTRACT

The growing reliance on digital learning environments has created a strong demand for platforms that not only manage educational content efficiently but also allow instructors to generate revenue independently. However, many of the existing online learning solutions are designed as separate modules and do not have enough flexibility in terms of content management, pricing strategies, and effective learner engagement. This paper presents the design of Learnova, a comprehensive web-based platform developed to simplify the process of sharing, creating, and monetizing content in one place. Learnova enables the following features such as publishing courses, subscription-based models, package deals, live teaching, and digital product distribution. Developed on a full-stack framework, the system provides support for secure authentication systems, secure payment systems, restricted access to content, and analytics for teachers. To improve learner engagement, the system has been developed with assessment tools, tracking systems, and discussion forums. The above-mentioned features improve learning outcomes. System performance analysis confirms that Learnova is scalable, reliable, and appropriate for teachers and learners who require a flexible and robust e-learning system.

Keywords: E-Learning Platform, Learning Management System, Content Monetization, Online Education, Web Technologies, Full-Stack-Development.

AdeptiCode: An AI-Driven Adaptive Learning Platform Using Bayesian Knowledge Tracing

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ABSTRACT

The rapid growth of digital education has revealed the limitations of traditional e-learning platforms, which fail to address individual learner differences. This paper presents **AdeptiCode**, an AI-driven adaptive learning platform that provides personalized learning experiences using **Bayesian Knowledge Tracing (BKT)** to estimate learner mastery based on performance and Interactions. The platform dynamically adjusts learning content and difficulty according to learner progress. It includes a web interface, backend system, and AI module for adaptive decision-making, along with continuous assessment and progress tracking. By delivering personalized learning paths, AdeptiCode improves knowledge acquisition and engagement. The system demonstrates the effective application of artificial intelligence in education and provides a scalable solution for adaptive digital learning.

Keywords: Adaptive Learning, Artificial Intelligence, Bayesian Knowledge Tracing, E-Learning, Personalized Education.

ICFTSEMAP170

IoT-Enabled Multimodal Deep Learning Framework for Early-Stage Rice Leaf Disease Detection and Severity Assessment with Explainable AI

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ABSTRACT

Rice cultivation faces significant challenges from foliar diseases, which can reduce yields by up to 40% if not detected early. Traditional disease monitoring relies heavily on manual field scouting, which is time-consuming, subjective, and often results in delayed intervention. This research presents an innovative IoT-enabled multimodal deep learning framework that integrates visual imagery with real-time environmental data for early-stage rice leaf disease detection and severity assessment. The proposed system employs a three-tier architecture comprising sensing, edge processing, and cloud analytics layers. Our multimodal fusion approach combines image features extracted through EfficientNet-B4 with contextual environmental parameters including temperature, humidity, and soil moisture levels. The fusion is mathematically formulated as $F\_total = \alpha F\_image + \beta F\_env$, where optimal weighting coefficients are learned through end-to-end training. We introduce a novel Disease Severity Index (DSI) that quantifies infection progression from 0-100%, enabling precision intervention strategies. Explainable AI techniques including Grad-CAM++ and LIME provide interpretable visual explanations, ensuring farmer trust and adoption. Experimental validation on a dataset of 12,847 images across eight disease classes demonstrates superior performance, achieving 97.3% accuracy, 96.8% precision, 97.1% recall, and an F1-score of 96.9%. The multimodal approach outperforms unimodal image-only baselines by 4.7% in accuracy and shows robust generalization across different geographic regions. Ablation studies confirm the significant contribution of environmental context, improving early-stage detection accuracy by 6.2%. The system demonstrates practical deployment feasibility with edge inference latency of 143ms per sample. This work advances the field by bridging the gap between laboratory research and field-deployable precision agriculture solutions, offering smallholder farmers an accessible, interpretable, and effective disease management tool.

Keywords: Computer Vision; Deep Learning; Explainable Artificial Intelligence; Internet of Things(IoT); Multimodal Learning.

ICFTSEMAP171

Cast Liner for Early Detection and Prevention of Skin Infections

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ABSTRACT

Orthopedic casting is a widely used method for fracture immobilization; however, patients frequently experience complications such as moisture accumulation, skin maceration, irritation, itching, unpleasant odor, and in severe cases, secondary infection. These issues often occur because the skin environment beneath the cast cannot be visually inspected and is rarely monitored continuously. As a result, complications are typically identified only after discomfort worsens, leading to delayed intervention and avoidable cast replacement. This paper presents the design and development of a wearable smart cast liner aimed at real-time monitoring and early warning of cast-related skin complications. The proposed system integrates a compact moisture sensing module embedded into a thin liner placed between the skin and cast. When abnormal moisture levels are detected, the system triggers an alert to notify the patient and caregiver. Additionally, to address itch-induced discomfort and reduce unsafe scratching behavior, the liner incorporates a low-amplitude micro-vibration feedback mechanism to provide non-invasive itch relief without electrical stimulation. The proposed solution emphasizes low-cost implementation, ease of use, and compatibility with conventional plaster and fiberglass casts, making it suitable for routine clinical use and resource-limited healthcare environments. By enabling early detection and preventive response, the smart liner system aims to improve patient comfort, compliance, and safety during cast immobilization.

Keywords: Orthopedic Casting, Micro-Vibration Feedback, Wearable Smart Cast Liner, Preventive Healthcare.

AI Credit Card Fraud Detection

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ABSTRACT

Credit card fraud poses significant threats to financial institutions and consumers worldwide, necessitating the development of advanced techniques for predicting and detecting these incidents. This project explores the application of the Random Forest algorithm to enhance the identification and forecasting of fraudulent transactions by analyzing historical payment data. Random Forest, an ensemble supervised learning model, combines multiple decision trees to uncover patterns and anomalies indicative of potential fraud with robust classification performance. The project focuses on feature extraction from transaction records, spending behavior, and merchant data, creating predictive models that can forecast fraud risks and detect irregular activities in real time. The effectiveness of Random Forest is evaluated using metrics such as accuracy, precision, recall, and F1-score, demonstrating its superior capability to handle imbalanced datasets and identify both established and emerging fraud patterns. Findings confirm that Random Forest significantly improves the accuracy, stability, and generalization of fraud detection compared to single decision tree methods. By integrating Random Forest into fraud prevention frameworks, organizations can more effectively anticipate and mitigate financial risks, ultimately enhancing their security posture and resilience against sophisticated fraud schemes in the digital payment landscape.

Keywords: **Credit Card Fraud, Random Forest, Fraud Detection, Ensemble Learning, Feature Extraction.**

ICFTSEMAP173

An Efficient Machine Learning Approach for Crime Detection in India

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ABSTRACT

As the population is rising, levels of wealth are also reaching various social sections, and urbanization is also taking place in India, so crime is also changing unpredictably, which is making traditional ways of policing less efficient. Waiting for the occurrence of crimes to take place is not a good approach; rather, sensing a change in advance can be very helpful in catching crimes early, and in this case, the use of data technology such as pattern recognition is very effective. Comparing past crime statistics with population and income trends can help identify important relationships between events and their contexts. Rather than implementing a single approach, three distinct approaches are validated - Linear Regression, Random Forest Regressor, and Gradient Boosting Regressor - all of which are optimized by systematically adjusting their respective parameters. When comparing the results, the Gradient Boosting Regressor is able to provide more precise forecasts because it is able to interpret complicated relationships. This is especially true when comparing it to other models with similar characteristics. The new system, developed using Flask, functions within an efficient web tool. Crime predictions are provided to the police without any kind of delay. This allows them to modify their strategies since results are provided instantly. Their decisions become more refined since data is provided instantly.

Keywords: Machine Learning, Crime Prediction, Gradient Boosting, Random Forest, Predictive Policing, Flask, Linear Regression.

ICFTSEMAP174

Decentralized Edge-AI and Blockchain Framework for Intelligent Traffic Management

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ABSTRACT

The advent of new era transportation systems is creating large volumes of real time data which requires smart management. This is possible with the amalgamation of connected vehicles, autonomous-movement, and smart-infrastructure. The current research proposes a framework of decentralized smart traffic management. This incorporates AI, edge-computing, blockchain, multiple agent systems for the purpose of real-time, secure and adaptive traffic coordination. The proposed framework provides a non-central decision making, predictive traffic-analysis, protected information exchange and coordinated mechanism through all the vehicles, signals, and infrastructure. The current research analyse how modern transportation system merges different technologies, that can improve traffic efficiency, safety, scalability and system stability. The proposed approach made available the basis for developing secure and intelligent transportation eco-systems capable of supporting future smart mobility and autonomous transport environments.

Keywords: Decentralized Traffic Management, Intelligent Transportation Systems, Edge Computing, Blockchain, Multi-Agent Systems, Artificial Intelligence, Smart Cities, Traffic Prediction, Secure Vehicular Networks.

A Comprehensive Analysis of Machine Learning Techniques for Brain Tumor Detection and Prediction: Methods, Outcomes, And Comparisons

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ABSTRACT

Brain cancer diagnosis remains a complex task because tumors vary significantly in location, size, shape, and appearance across patients. These variations make precise detection and classification essential for timely treatment and improved clinical outcomes. This study presents a comprehensive review of brain tumor analysis using Magnetic Resonance Imaging (MRI) and compares the performance of multiple machine learning algorithms on a pre-processed dataset containing both tumor and healthy brain scans. The algorithms evaluated include Support Vector Machine (SVM), Random Forest, Decision Tree, K-Nearest Neighbors (KNN), Naive Bayes, Logistic Regression, CatBoost, and Neural Networks. To enhance classification performance, meaningful features were extracted from the MRI images, including texture characteristics, pixel intensity patterns, and shape-based descriptors. These features were used to train and test the models systematically. The performance of each algorithm was measured using standard evaluation metrics such as accuracy, sensitivity, specificity, and precision to ensure a comprehensive assessment of diagnostic effectiveness. The results demonstrate that all models achieved satisfactory performance in distinguishing tumor from non-tumor cases. Among them, CatBoost and Random Forest delivered the highest accuracy and sensitivity, indicating strong predictive capability and reliability. SVM showed consistent efficiency in tumor detection tasks, while Neural Networks were particularly effective in classifying tumor subtypes. Overall, the findings highlight the strengths and limitations of different machine learning approaches and suggest that ensemble techniques offer significant potential for improving the accuracy, robustness, and reliability of automated brain tumor prediction systems.

Keywords: Brain Cancer, Machine Learning Algorithms, Magnetic Resonance Imaging, Accuracy, Texture, Precision.

ICFTSEMAP176

Detection of Deepfake QR Codes in Street Vendor Digital Payment Systems Using Frequency Domain Analysis

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ABSTRACT

QR code-based digital payments are widely used in street vendor transactions, especially in India through UPI systems. These payments are fast and easy, but they also face security risks. One major problem is the use of fake or deepfake QR codes. Fraudsters replace or modify original QR codes and redirect the payment to their own bank accounts. Customers and vendors often do not notice this change, which leads to financial loss. Most existing research focuses on detecting fraud after the transaction is completed using machine learning models. Very few studies focus on detecting manipulated QR codes before the payment happens. This paper proposes a new method to detect deepfake QR codes using frequency domain analysis. Instead of only checking the visible structure of the QR code, the proposed method studies hidden frequency patterns using techniques like Fast Fourier Transform (FFT) and Discrete Cosine Transform (DCT). When a QR code is edited or tampered with, small changes occur in its frequency components. These changes are not clearly visible to the human eye but can be detected through frequency analysis. The proposed system is lightweight and suitable for real-time use in street vendor payment systems. Experimental results show that the method can effectively identify manipulated QR codes with good accuracy and low computational cost. This approach can improve security in QR-based digital payments and help prevent fraud in small-scale business environments.

Keywords: QR Code, Frequency Domain Analysis, Fast Fourier Transform (FFT), Discrete Cosine Transform (DCT), Digital Payment Security.

ICFTSEMAP177

Perceptible Investigation of Established Building to Endorse Repair Approaches

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ABSTRACT

A perceptible investigation of established educational institution buildings—specifically those aged 18 years and above—serves as a critical process for identifying necessary repair approaches and endorsing appropriate rehabilitation strategies. Over nearly two decades of continuous use, institutional structures often encounter material degradation, environmental wear, compliance gaps with updated safety codes, and evolving functional requirements. Perceptible investigation involves a combination of visual assessments, non-destructive testing (NDT), damage diagnostics, and structural health monitoring to evaluate the current condition of key building components, including concrete, steel reinforcements, partitions, and service infrastructures. Findings from these methods pinpoint both immediate hazards and latent deficiencies, such as cracks, spalling, moisture ingress, reinforcement corrosion, and compromised joints. Based on these investigations, targeted repair approaches are recommended—such as crack sealing, cathodic protection, fiber-reinforced polymer (FRP) strengthening, and surface repair. The integration of self-healing concrete, cathodic protection systems, and advanced crack injection techniques further enhances long-term serviceability and resilience of institutional buildings. Effective endorsement of repair methodologies is underpinned by a meticulous understanding of the extent of deterioration and tailored interventions that ensure structural integrity, safety, and operational continuity. Emphasis is placed on proactive maintenance and sustainable rehabilitation, securing the building's role as a safe and functional educational facility. These abstract highlights the necessity for systematic and perceptible investigation as a foundation for implementing innovative and durable repair techniques in aging educational institution buildings.

Keywords: Assessment, Corrosion, Deterioration, Rehabilitation, Repair.

ICFTSEMAP178

UART WITH FIFO for Serial Communication

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ABSTRACT

The Universal Asynchronous Receiver Transmitter (UART) is one of the most fundamental and widely used serial communication protocols in digital systems, enabling reliable data exchange between microprocessors, microcontrollers, and peripheral devices. To overcome these limitations, this project presents the design and RTL implementation of a UART integrated with a First-In First-Out (FIFO) buffer using Verilog HDL. The RTL simulation results validate the correct transmission and reception of data frames with proper start, data, parity, and stop bit patterns. The FIFO's read and write control logic ensures robust data buffering and synchronization. Furthermore, synthesis and timing reports demonstrate optimized area utilization and stable timing performance under standard design constraints. This work provides a modular and reusable UART-FIFO architecture suitable for System-on-Chip (SoC) and embedded communication applications, ensuring high throughput, low latency, and reliable operation even in variable-speed data environments.

Keywords: UART-FIFO, Verilog Design, Peripheral Devices, System On Chip.

ICFTSEMAP179

Comparative Analysis of Global Management Philosophies: A Review of American, Japanese, European, Chinese, Singaporean, and Indian Approaches

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ABSTRACT

This paper offers a thorough comparative examination of management philosophies in various countries, including the United States, Japan, Europe, China, Singapore, and India. The study employs a qualitative design grounded in a thorough examination of secondary sources, which encompasses peer-reviewed journal articles, foundational management literature, cross-cultural studies, policy reports, and modern empirical investigations. A systematic thematic examination is utilised to integrate current academic work and pinpoint fundamental concepts, consistent themes, and contextual differences in national management philosophies. Analytical frameworks like Hofstede's Cultural Dimensions, the GLOBE study, and institutional theory serve to inform comparison and interpretation. The approach consists of structuring the literature according to essential dimensions, including leadership style, decision-making processes, organisational structure, human resource practices, performance orientation, ethical foundations, and long-term strategic outlook. Comparative matrices are crafted to systematically analyse the performance-driven and individualistic orientation of American management; the collectivist, consensus-based, and quality-focused Japanese model; the European social-market and participative approach; the Confucian-influenced Chinese philosophy that emphasises hierarchy, guanxi (relationship networks), and long-term orientation; Singapore's hybrid model characterised by meritocracy, pragmatic governance, multicultural integration, and strategic state guidance; and India's value-driven and relationship-oriented philosophy shaped by cultural diversity, spiritual traditions, jugaad (frugal innovation), family-owned business structures, and an emerging blend of traditional ethos with modern global practices. Significant attention is directed towards understanding how historical, cultural, and institutional contexts influence managerial behaviour and the effectiveness of organisations. The comparative review emphasises the trends of convergence driven by globalisation alongside the persistent cultural uniqueness in managerial practices. The findings indicate that management philosophies are intricately linked to socio-cultural contexts, and their success relies on situational alignment instead of universal applicability. This paper presents a structured framework by synthesising cross-national literature, enhancing the understanding of comparative management philosophies and offering valuable insights for scholars, policymakers, and global business leaders in multicultural settings.

Keywords: Management Philosophies , American management , Indian Philosophies.

ICFTSEMAP180

Confidence-Aware Phishing Website Detection using Evidence Vector Machine (EVM)

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ABSTRACT

Phishing attacks continue to pose a serious threat to online users by mimicking legitimate websites to steal sensitive information. Traditional machine learning classifiers often produce overconfident predictions when encountering unfamiliar or evolving phishing patterns, which reduces their reliability in real-world deployment. To address this limitation, this study proposes a confidence-aware phishing website detection framework based on the Evidence Vector Machine (EVM). The proposed system focuses on URL-based feature extraction to ensure practical applicability without requiring webpage content or third-party services. Relevant lexical and structural features are derived from URLs, and an EVM classifier is trained to distinguish between legitimate and phishing websites while also estimating the confidence of each prediction. Unlike conventional classifiers, EVM incorporates open-set recognition capability, enabling the model to flag suspicious or previously unseen patterns more effectively. The model is trained and evaluated using a benchmark phishing dataset after appropriate preprocessing and feature selection. Experimental results demonstrate that the EVM-based approach improves detection reliability and provides meaningful confidence scores that help reduce false trust in uncertain predictions. The system is further integrated with an interactive interface for real-time URL analysis. This work highlights the potential of confidence-aware learning for strengthening phishing detection systems and offers a scalable solution suitable for deployment in modern cybersecurity environments.

Keywords: Phishing Detection, Evidence Vector Machine, URL Features, Open-Set Recognition, Machine Learning, Cybersecurity.

ICFTSEMAP181

Intelligent Prompt Optimization Tool for Generative AI Models

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ABSTRACT

This abstract explores the concept, application, and future of intelligent prompt optimization tools for generative AI models. It defines these tools as automated systems designed to enhance the effectiveness of user prompts, moving beyond manual trial-and-error. The core function of these tools is to improve the quality of AI-generated output by refining and enriching initial prompts with greater clarity, context, and constraints. The application of such tools spans diverse fields, including content creation, software development, education, and customer service. They serve as a crucial intermediary, translating human intent into the precise language required for optimal AI performance. Key applications include generating high-quality ad copy, creating accurate code, drafting personalized emails, and producing detailed images.

Keywords: Prompt Optimization, Prompt Engineering, Generative AI (GenAI), Large Language Model (LLM), Prompt Refinement, Automated Prompting, Chain-of-Thought (CoT), Retrieval-Augmented Generation (RAG).

ICFTSEMAP182

AN Agent-Driven Framework for Task-Level SDLC Automation

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ABSTRACT

Although most current systems are designed to automate isolated SDLC phases or high-level project planning, recent breakthroughs in agentic AI have made possible the partial automation of software development life cycle (SDLC) processes. This paper introduces a task-specific, agent-based framework for the automation of SDLC by breaking down requirements into systems and tasks with dynamic assignment of priorities, deadlines, and sequence of execution. In contrast to existing solutions, which are designed to automate SDLC phases or projects at a high level of abstraction, this framework allows task-level with parallel execution. The framework has been implemented using a multi-agent architecture based on large language model (LLM) with CrewAI framework and illustrates how autonomous agents can work together to control SDLC activities while keeping cost in mind. The framework is evaluated using software project examples and contrasted with existing project management software such as Jira.

Keywords: **Software Development Life Cycle, Multi-Agent Architecture, CrewAI.**

ICFTSEMAP183

Superhydrophobic Coating on Glass Using Magnetron Sputtering Technique: A Comprehensive Review

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ABSTRACT

Superhydrophobic (SH) surfaces, typically defined by a water contact angle (WCA) greater than 150° and a sliding angle below 10°, have attracted extensive research attention due to their self-cleaning, anti-fogging, anti-icing, and corrosion-resistant properties. Among various substrates, glass holds particular importance because of its widespread use in optical devices, photovoltaic panels, architectural glazing, and automotive windshields. Achieving durable and transparent superhydrophobic coatings on glass remains a critical scientific and technological challenge. Magnetron sputtering has emerged as a promising thin-film deposition technique owing to its excellent film adhesion, uniformity, scalability, and compatibility with industrial glass processing. This review critically surveys recent progress (2020–2025) in the development of superhydrophobic coatings on glass fabricated using magnetron sputtering techniques. The fundamental principles of wetting, including the Wenzel and Cassie–Baxter models, are first discussed to provide a theoretical basis for surface design. Subsequently, the working principles of magnetron sputtering and its suitability for fabricating functional and nanostructured coatings are presented. A detailed literature survey summarizes reported sputtered materials such as metal oxides (SiO_2, TiO_2, ZnO), metals (Al, Cu), and hybrid multilayer systems combined with post-deposition surface functionalization. Key performance metrics, including water contact angle, sliding angle, optical transparency, and mechanical durability, are systematically compared. Despite promising results, challenges related to long-term durability, abrasion resistance, fluorinated chemistry, and large-area scalability remain unresolved. Finally, future research directions focusing on fluorine-free surface modification, nanocomposite architectures, and hybrid sputtering-based approaches are proposed. This review provides a consolidated reference for researchers aiming to develop robust and industrially viable superhydrophobic glass coatings using magnetron sputtering techniques [1–5].

Keywords: Superhydrophobic Surface; Glass Coating; Magnetron Sputtering; Thin Films; Surface Wettability; Functional Coatings.

ICFTSEMAP184

Port Guard AI-Smart Vulnerability & Threat Scanner Using Artificial Intelligence

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ABSTRACT

Modern digital infrastructures, including enterprise and cloud environments, depend heavily on secure network communication. Despite advancements in cybersecurity, misconfigured ports, exposed services, and unpatched vulnerabilities continue to serve as major attack vectors. Traditional vulnerability assessment tools such as Nmap, Nessus, and OpenVAS provide reliable port discovery and vulnerability detection but often produce false positives, require manual interpretation, and lack adaptive intelligence for emerging threats. This paper introduces **Port Guard AI**, an intelligent vulnerability and threat scanning framework that integrates automated port scanning, AI-driven threat classification, dynamic risk scoring, and honeypot-based malicious activity detection. The system scans configurable port ranges, identifies open services, and evaluates their risk levels based on access frequency, anomaly patterns, and behavioral analysis. A key feature is the deployment of user-configured honeypots—intentionally exposed trap ports designed to attract unauthorized access attempts. These honeypots help analyze attacker methodologies, detect repeated intrusion behavior, and enhance threat intelligence. Port Guard AI incorporates a web-based dashboard for real-time visualization of scan results, displaying open ports, risk scores, threat levels, and activity logs through structured tables and interactive bar charts. Color-coded indicators (green for normal/low risk, blue for medium risk, and red for malicious activity) enable rapid security assessment and decision-making. Experimental results indicate improved detection accuracy, reduced false positives, and enhanced response efficiency, making Port Guard AI a scalable and proactive solution for modern network security environments.

Keywords: Port Scanning, Vulnerability Assessment, AI-Based Threat Classification, Honeypot Deployment, Dynamic Risk Scoring, Network Security.

Intelligent RTO Monitoring and Forecasting System

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ABSTRACT

The exponential increase in car ownership has become extremely difficult for Regional Transport Offices (RTOs) to handle licensing, registration, fraud detection, and compliance verification. There is a need for more intelligent automation because traditional manual processes take a lot of time in processing, prone to mistakes, and susceptible to corruption. And so, to improve vehicle monitoring and decision-support systems, this paper examines how artificial intelligence (AI) and machine learning (ML) can transform RTO operations. Machine learning, deep learning, optical character recognition (OCR), and data analytics are some of the technologies that are merged together in the proposed Smart RTO Vehicle Intelligence and Cross-State Management System to automate important RTO tasks like tracking pollution compliance, predicting vehicle resale value, and document verification.

Keywords: Machine Learning, Natural Language Processing, Deep Learning, Fraud Detection.

Sarv Sampoorna Kisan Mitra (AI Based Crop Disease Detection and Management System)

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ABSTRACT

The contemporary challenge of sustainable agriculture necessitates the deployment of robust, data-driven decision support systems. This review paper details the architecture and efficacy of an integrated AI-based system, termed 'Sarv Sampoorna Kisan Mitra,' for comprehensive crop disease identification and management. The system is predicated on two core technological pillars: the use of advanced Deep Learning (DL) models, specifically Convolutional Neural Networks (CNNs), for rapid, visual diagnosis of crop diseases; and the implementation of a Knowledge Graph (KG) for prescriptive management recommendations. The paper explores the full lifecycle, from data acquisition via remote sensing and IoT, through predictive modeling for yield forecasting, and culminating in the generation of actionable, customized advice for fertilizer application and pest control. By transforming raw diagnostic data into contextualized, actionable knowledge, this integrated AI-KG framework offers a scalable solution to enhance precision farming efficiency and minimize resource wastage.

Keywords: Crop Disease Identification; Knowledge Graph (KG); Deep Learning (DL); Smart Agriculture.

ICFTSEMAP187

LEARNSCOPE: A Training-Free and Privacy-Preserving Learning Engagement Analysis System Using MediaPipe-Based Behavioral Signal Fusion

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ABSTRACT

The rapid growth of online and hybrid education has created an urgent need for scalable, ethical, and real time methods to assess student engagement. Most existing solutions rely on supervised deep learning models trained on large scale facial or behavioral datasets, which introduces challenges related to privacy, bias, computational overhead, and limited explainability. This paper presents LEARNSCOPE, a training free and privacy preserving learning engagement analysis system that leverages MediaPipe based edge vision processing and rule based temporal fusion of behavioral signals. The proposed framework extracts eye gaze, blink rate, head pose, facial activity proxies, and presence consistency directly on device, without storing video data, performing face recognition, or requiring labeled datasets. Engagement levels are inferred using an interpretable heuristic scoring mechanism combined with temporal smoothing, enabling stable trend analysis and transparent instructor facing analytics. The system architecture prioritizes low latency, explainability, and ethical compliance, making it suitable for real world educational deployments. Experimental observations demonstrate consistent engagement trend detection with minimal computational overhead, validating the practicality of the proposed approach. By eliminating dataset dependency and opaque inference pipelines, LEARNSCOPE offers a scalable and responsible alternative for learning analytics in modern digital classrooms.

Keywords: Learning Analytics, Student Engagement, MediaPipe, Edge AI, Explainable AI, Privacy-Preserving Systems, Computer Vision.

ICFTSEMAP188

Skin Lesion Analysis Towards Melanoma Detection Using Deep Learning

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ABSTRACT

This project focuses on advancing automated skin lesion analysis for melanoma detection by leveraging deep learning techniques, particularly Convolutional Neural Networks (CNNs). Melanoma is one of the most aggressive forms of skin cancer, and early diagnosis is critical to improving survival rates. Traditional diagnostic approaches often rely on manual interpretation of dermoscopic images, which can be subjective and prone to variability across clinicians. To address these challenges, this project proposes a robust AI-driven solution that utilizes CNNs trained on large-scale dermoscopic image datasets to assist in the early and accurate classification of skin lesions.The methodology encompasses data acquisition, preprocessing, model development, and performance evaluation. Preprocessing techniques such as image normalization, resizing, and augmentation are employed to enhance model generalization and reduce overfitting. The CNN is implemented using deep learning frameworks like TensorFlow or PyTorch, and optimized through hyperparameter tuning and transfer learning strategies. Evaluation metrics including accuracy, precision, recall, F1-score, and ROC-AUC are used to assess the model's diagnostic capability.To improve interpretability and clinical trust, visualization tools such as Grad-CAM are integrated to highlight lesion regions influencing predictions. The final model is designed for deployment within a clinical decision support framework, offering dermatologists a scalable and consistent tool for melanoma screening. This system holds promise for improving diagnostic accuracy, reducing workload, and enabling timely intervention in diverse healthcare environments.

Keywords: Convolutional Neural Network, Dermascopic Image.

ICFTSEMAP189

Software-Based Low-Power Approximate Mac Unit for AI Accelerators

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ABSTRACT

Energy-efficient arithmetic units are critical in AI accelerators, where Multiply–Accumulate (MAC) operations dominate overall computation and power consumption. This work presents a low-power approximate MAC architecture developed using approximate computing techniques to improve hardware efficiency while maintaining acceptable computational accuracy. The proposed design replaces conventional exact adders and multipliers with optimized approximate arithmetic blocks, introducing controlled approximation in the least significant computation stages to reduce switching activity, critical path delay, and overall resource utilization. The architecture is modeled at Register Transfer Level (RTL) using Verilog HDL and implemented entirely in Xilinx Vivado for functional simulation, synthesis, timing analysis, and power estimation. Performance is evaluated in terms of power consumption, propagation delay, area utilization, and accuracy, and compared with a conventional exact MAC implementation. Experimental results obtained from Vivado synthesis reports demonstrate reductions in power and hardware resources with minimal degradation in output accuracy, indicating that the proposed design is suitable for energy-constrained AI accelerators, image processing applications, and edge-based inference systems where power efficiency is a primary design requirement.

Keywords: Approximate Computing, Multiply–Accumulate (MAC), Low Power Design, AI Accelerators, Verilog HDL, RTL Design.

Improved Detection and Adaptive Noise Suppressive Strategy for SEE-OFDM VLC Links

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ABSTRACT

This paper examines the optimization of Spectral and Energy Efficient Orthogonal Frequency Division Multiplexing (SEE-OFDM) for Visible Light Communication (VLC) systems. It focuses on reducing the harmful effects of high Peak-to-Average Power Ratio (PAPR). SEE-OFDM improves power efficiency by only transmitting positive signal components, but it is still very vulnerable to LED nonlinearity. This nonlinearity causes clipping noise and lowers transmission quality. By using effective signal processing methods, such as transform-based precoding, nonlinear companding, and time-domain noise cancellation, the study shows a significant reduction in PAPR and an improvement in Bit Error Rate (BER) performance. The results provide a strong framework for achieving fast, reliable optical data transmission while protecting hardware from nonlinear distortion. This ultimately makes SEE-OFDM more practical for real-world VLC applications.

Keywords: BER; LED Nonlinearity; Noise Cancellation; Nonlinear Companding; PAPR; SEE-OFDM; VLC; WHT Precoding.

ICFTSEMAP191

Symptom Checker and Doctor Recommendation System

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ABSTRACT

Early disease identification is crucial for preventing complications and improving healthcare outcomes. Many individuals rely on unreliable online sources to interpret symptoms, leading to misinformation and delayed treatment. This paper presents an AI-Based Symptom Checker and Doctor Recommendation System that provides preliminary healthcare assistance. Users enter symptoms through a chatbot interface using natural language. Natural Language Processing (NLP) techniques preprocess and structure the input data. A supervised machine learning model predicts probable diseases with confidence scores. Based on the prediction, the system recommends appropriate medical specialists. The application is developed using Python and web technologies to ensure real-time performance. Experimental results show reliable disease prediction and accurate specialist mapping. The system serves as a scalable decision-support tool that promotes timely medical consultation and healthcare awareness.

Keywords: Symptom Checker, Disease Prediction, Machine Learning, Natural Language Processing, Chatbot, Doctor Recommendation, Healthcare Decision Support System.

ICFTSEMAP192

Next-Gen FarmSense: Intelligent Agriculture Monitoring & Automation Network

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ABSTRACT

Agriculture productivity is highly dependent on environmental conditions such as soil moisture, temperature, humidity, pH level, and light intensity. In traditional farming practices, these parameters are monitored manually, leading to inaccurate observations, delayed decision-making, excessive water usage, and reduced crop yield. With increasing climate variability and water scarcity, there is a critical need for intelligent and automated agricultural systems. This project presents Next-Gen FarmSense, an IoT-based intelligent agriculture monitoring and automation network that enables real-time sensing, data analysis, and automated control of farming operations. The system integrates multiple environmental sensors with a microcontroller and wireless communication module to continuously monitor field conditions. Sensor data is transmitted to a cloud platform, where it is stored, visualized, and analyzed for decision-making. An automated irrigation mechanism is implemented based on soil moisture thresholds to ensure optimal water usage. The proposed system minimizes human intervention, improves resource efficiency, and supports precision farming. Experimental evaluation confirms the reliability of real-time monitoring and effective automation, making the system suitable for smart and sustainable agriculture applications.

Keywords: Internet of Things, Smart Agriculture, Automated Irrigation, Precision Farming, Environmental Monitoring.

ICFTSEMAP193

LEARNSCOPE: Federated Intrusion Detection Framework for Heterogeneous Network Traffic Using Multi-Client Deep Learning Aggregation

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ABSTRACT

The rapid expansion of cloud computing, Internet of Things (IoT) ecosystems, and large-scale enterprise networks has intensified the complexity and frequency of cyber attacks. Traditional centralized Intrusion Detection Systems (IDS) require aggregation of raw network traffic from multiple distributed sources, raising significant concerns related to data privacy, regulatory compliance, communication overhead, and scalability. These challenges limit effective collaborative security across heterogeneous network environments. This paper presents a federated intrusion detection framework that enables decentralized collaborative learning without sharing raw network traffic data. A lightweight Multilayer Perceptron (MLP) model is trained across three independent federated clients using the Federated Averaging (FedAvg) algorithm. Each client is assigned a distinct real-world benchmark dataset — CICIDS2017, UNSW-NB15, and NSL-KDD introducing authentic non-IID data heterogeneity in terms of dataset size, feature dimensionality, class imbalance, and attack taxonomy. The framework ensures strict data locality by transmitting only model parameters during aggregation. Experimental results demonstrate that the global federated model achieves 87% accuracy with balanced precision (0.86), recall (0.86), and F1-score (0.85), representing a 20 percentage-point improvement over the weakest local model. Convergence analysis confirms stable performance under non-IID conditions, while communication overhead is reduced by approximately 1000× compared to centralized training. These findings validate federated learning as a scalable, privacy-preserving, and communication-efficient alternative for intrusion detection across diverse network domains.

Keywords: Federated Learning, Intrusion Detection System, Network Security, Privacy Preservation, Non-IID Data, FedAvg, Distributed Learning.

ICFTSEMAP194

IoT-Enabled Stress Prediction System with Multi-Sensor Integration

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ABSTRACT

The rapid proliferation of Internet of Things (IoT)-enabled wearable devices has created unprecedented opportunities for continuous, non-invasive physiological monitoring in real-world settings. This paper presents the design, simulation, and implementation of a multi-modal stress detection system based on an Arduino Uno microcontroller interfaced with an infrared (IR) sensor for eye-blink rate detection, a heart beat sensor for cardiac activity monitoring, a sound detector for respiratory rate estimation, and a thermometric module for body temperature acquisition. Recognised as a major global health concern by the World Health Organization, stress manifests through measurable autonomic nervous system (ANS) responses — including elevated heart rate, irregular breathing patterns, suppressed eye-blink rate, and peripheral temperature changes — which this system captures simultaneously. The proposed architecture was validated through Proteus Professional circuit simulation, successfully demonstrating stable, real-time acquisition of all four physiological parameters with consistent serial output (Temp = 62, Eye Blink = 38, Resp = 44, Heart Beat = 40). The system offers a low-cost, portable, and extensible platform suitable for IoT-based remote stress monitoring and future integration with machine learning classifiers for automated stress-level prediction.

Keywords: Stress Detection, Wearable Sensors, IoT Health Monitoring, Arduino, Physiological Signals, Heart Rate Variability.

ICFTSEMAP195

GAN-Based Framework for Decision Support in Stock Market Forecasting

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ABSTRACT

Stock market forecasting is not easy. Financial markets are constantly changing, highly dynamic, and often unpredictable. Prices move due to many factors economic news, global events, investor psychology, and sudden market reactions. Because of this, investors and traders are always looking for reliable tools that can help them make smarter decisions while reducing financial risk. However, traditional statistical models and many machine learning methods often struggle to handle sudden fluctuations and hidden patterns in stock price movements.To overcome these challenges, this paper introduces a GAN-based framework designed to support better decision-making in stock market forecasting. The system uses Generative Adversarial Networks (GANs) to understand complex market behavior by learning from historical stock data and important technical indicators. In this framework, the generator predicts future stock price movements, while the discriminator checks how realistic those predictions are by comparing them with actual market data. This competitive learning process helps the model continuously improve its accuracy and adaptability. Beyond just predicting prices, the framework also provides practical decision support. It interprets market trends and generates buy, sell, or hold signals, helping investors make more confident and informed investment choices. Overall, the proposed system aims to deliver a reliable, intelligent, and data-driven solution for stock market forecasting.

Keywords: GANs, Robustness, Stock Forecating, Accuracy.

ICFTSEMAP196

Flex Sensor Enabled Smart Sign Language Gloves

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ABSTRACT

Communication barriers continue to be a major problem between the general public and people with speech and hearing impairments. A wearable Smart Sign Language Glove was created to solve this problem by translating preset hand gestures into speech and text outputs in real time. The system incorporates an MPU6050 motion sensor to record hand orientation and movement dynamics, as well as three flex sensors to detect finger bending. Sensor data collection, preprocessing, and wireless transmission to a Flask-based backend via Wi-Fi are all managed by an ESP32 Dev Module. Python-based logic is used to implement gesture classification, and a Streamlit web interface with integrated text-to-speech functionality is used to display the recognized output. Near real-time communication is made possible by experimental results showing an average recognition accuracy of about 90% and a response time of almost one second. The system offers an assistive communication solution that is scalable, portable, and reasonably priced.

Keywords: Sign Language Recognition, Flex Sensor, ESP32, MPU6050, Flask, Assistive Technology.

ICFTSEMAP197

SleepSight: Sleep Disorder Risk Assessment and Recommendations

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ABSTRACT

Sleep disorders such as insomnia, sleep apnea, narcolepsy and restless leg syndrome have been on the rise due to reasons like stress, irregular lifestyle habits and lack of public awareness. Early detection is the cure, but most contemporary solutions track only sleep patterns without providing accurate risk estimates, hence creating a vast chasm in preventive healthcare. SleepSight is a web application based on machine learning technology that estimates the likelihood of different sleep disorders based on demographic and lifestyle details like age, gender, sleep duration, stress and activity. The architecture encompasses data preprocessing, optimized multi-class classification and lean architecture to provide accurate and rapid predictions. A set of algorithms like Random Forest, XGBoost, Logistic Regression and Support Vector Machines will be tested and the most efficient model will be implemented. Users are provided with percentage-based risk scores and personalized improvement tips through an easy-to-use interface. The process facilitates proactive lifestyle intervention, encourages frequent self-checks and on-time medical consultations as and when needed. The target users are individuals, wellness professionals, telemedicine practitioners and corporate wellness programs with a larger goal of sensitizing the public, enabling early intervention and encouraging overall sleep wellness at community level.

Keywords: Logistic Regression, Random Forest, SleepSight, Support Vector Machines, XGBoost.

ICFTSEMAP198

A Review of VirtualGems – Virtual Try-On and Customization

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ABSTRACT

The rapid growth of online jewelry shopping has introduced several challenges, including a lack of user confidence, limited personalization, and increased return rates due to the absence of a "try-before-you-buy" experience. To address these challenges, this paper reviews recent advancements in virtual jewelry try-on and recommendation systems through the development of "VirtualGems," an AI-powered web-based platform that integrates computer vision, machine learning, and intelligent recommendation techniques to enhance the online jewelry shopping experience. The system utilizes real-time facial detection and landmark analysis using MediaPipe and OpenCV to accurately overlay jewelry images onto the user's live camera feed, enabling an interactive virtual try-on experience. Additionally, the platform analyzes the user's facial structure and automatically classifies face types to generate personalized jewelry recommendations using a database-driven recommendation model. Unlike traditional systems that rely on heavy augmented reality frameworks such as ARCore or Unity, VirtualGems employs a lightweight and scalable architecture built with a Django-based backend and a React-based frontend, ensuring flexibility, performance, and ease of deployment. The system also stores user interaction data, try-on history, and customization records to improve recommendation accuracy and user engagement over time. This paper evaluates the architecture, performance, advantages, and limitations of AI-assisted virtual try-on systems like VirtualGems and highlights their potential to improve user confidence, personalization, and satisfaction in online jewelry shopping environments.

Keywords: Virtual Try-On, Facial Landmark Detection, Jewelry Recommendation, Computer Vision.

ICFTSEMAP199

Quantitative Assessment of Groundwater in Chirang District of Assam

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ABSTRACT

Quantitative assessment and analysis of groundwater is equally crucial as that of the qualitative analysis. In order to address the disparity between available water potential and crop water demand, the groundwater of Chirang district was subjected to quantitative assessment in the present study. Water samples were collected on seasonal basis, from 15 numbers of locations through open wells and tube wells in HDPE plastic water bottles of 1 litre capacity. Depths to water table were determined by measuring the water level in open wells below ground level in pre-monsoon, monsoon and post-monsoon seasons for the years 2022 and 2023. The two-year average revealed that Chapaguri had the highest depth to water table during pre-monsoon season and shallowest was in Bagargaon during monsoon season. These values were further analyzed through graphs to understand the trend of fluctuation levels in ground water table throughout the district. The water table fluctuation pattern across all locations shows a decrease in depth from pre-monsoon to monsoon, followed by a gradual increase from monsoon to post-monsoon. The highest positive fluctuation was found in Bagargaon during transition phase of monsoon to post-monsoon and highest negative fluctuation was found in Agrong No. II during transition phase of pre-monsoon to monsoon.

Keywords: Chirang, Quantitative, Groundwater, Fluctuation, Monsoon.

Face-Authenticated Touchless Interface for Gesture-Based Interaction and Personalized File Access using Raspberry pi

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ABSTRACT

This research introduces a multi-modal, touchless interface for the Raspberry Pi 4, designed to enhance Human-Computer Interaction (HCI) through the fusion of facial recognition and hand gesture control. By utilizing lightweight deep learning frameworks, including MediaPipe for gesture tracking and TensorFlow Lite for CNN-based facial embeddings, the system provides secure, identity-based file access with 95% accuracy and sub-100ms latency. Robustness in diverse lighting is achieved through CLAHE preprocessing, while an ergonomic gesture set integrated with pyautogui enables seamless cursor navigation and scrolling. Security is reinforced through identity-specific file management using os/shutil and VeraCrypt-encrypted volumes, which unlock only upon a verified facial match. Experimental results involving 50 users demonstrate a 98% gesture precision rate and a consistent 30fps processing speed with zero false positives in authentication process and This innovative framework offers a high-performance, low-latency solution for post-pandemic computing, shared kiosks, and edge-based security applications for effective performing in traditional single purpose authentication systems through its will integration of contactless navigation and automated the data privacy.

Keywords: Human-Computer Interaction (HCI), Raspberry Pi 4, Touchless Interface, Hand Gesture Recognition, Face Recognition, Mediapipe, TensorFlow Lite, CNN, Real-Time Authentication, CLAHE, Contactless Computing, Access Control, Veracrypt, And Pyautogu.

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BRAINDUM - An Thought Sharing App

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ABSTRACT

In recent years, social media platforms have become a common medium for sharing thoughts and ideas; however, many users hesitate to express themselves freely due to privacy concerns and fear of judgment. To address this issue, this project presents Brain Dump, an anonymous thought sharing web application that provides a safe and open environment for users to express their ideas without revealing their identity. The system allows users to post thoughts anonymously, while other users can view the posts, like them, and add comments to encourage interaction and discussion. The application is developed using web technologies with a Python-based backend and a database for secure storage of posts and interactions. By ensuring user anonymity and providing interactive features, the proposed system promotes honest expression, improves user engagement, and supports open communication. The Brain Dump platform offers a simple, scalable, and privacy-focused solution for sharing thoughts and ideas in the digital environment.

Keywords: Brain Dump, Anonymous Posting, Thought Sharing, Web Application, Like and Comment System, Privacy, Social Interaction.

ICFTSEMAP202

An Automated Ensemble Learning Framework using BERT for Critical Software Bug Reports

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ABSTRACT

In order to prioritize important concerns in large-scale software development and increase the effectiveness of bug triage, automated classification of software bug reports is essential. Though there are still issues with robustness, class imbalance, and adaptability to various software settings, recent research has shown that transformer-based language models like BERT are excellent at capturing the contextual semantics of bug report content. This paper suggests an automated ensemble learning framework using BERT for significant software issue reports, motivated by the future research directions found in earlier BERT-based bug classification studies. In order to improve prediction stability and lessen sensitivity to noisy and unbalanced data, the suggested method uses BERT to create deep contextual representations from bug report titles and descriptions. These representations are then processed by an ensemble of machine learning classifiers. The system enhances recall and reliability in identifying important bug categories including security and performance issues by combining ensemble learning with transformer-based feature extraction. Since the system is completely automated, human involvement in prediction is not necessary. When compared to single-model techniques, the suggested framework provides better and consistent classification performance, according to experimental results on real-world bug report datasets. This study allows additional extensions like multimodal inputs and domain-specific adaption and shows a scalable and extensible foundation for future automated bug triage systems.

Keywords: Bug Severity Classification, BERT, Ensemble Learning, Automated Bug Triage, Software Maintenance.

ICFTSEMAP203

Ayurvedic Health Advisor

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ABSTRACT

The rising prevalence of lifestyle-related health issues has increased the demand for preventive and holistic healthcare solutions. Ayurveda, a traditional system of medicine, emphasizes balance in daily routine, diet, and lifestyle to maintain long-term well-being. However, access to authentic Ayurvedic guidance remains limited for the general population due to lack of awareness, expert availability, and complexity of traditional concepts. This paper presents the design and implementation of an Ayurvedic Health Advisor, a digital wellness advisory system that provides personalized Ayurvedic recommendations based on user-reported symptoms, lifestyle habits, and basic Dosha assessment. The system uses a structured knowledge base and rule-based recommendation engine to deliver diet suggestions, daily routine guidance, lifestyle modifications, and safe home remedies. The platform emphasizes preventive healthcare, explainable recommendations and ethical boundaries by advising professional consultation for serious symptoms. Experimental evaluation using test scenarios and user feedback indicates that the system provides relevant and understandable wellness guidance, demonstrating the feasibility of integrating traditional healthcare knowledge with modern digital platforms.

Keywords: Ayurveda, Ayurvedic Health Advisor, Symptom checker, Health assessment, Preventive Healthcare, Digital Health.

ICFTSEMAP204

The Role of Explainability in Human–AI Co-Decision Making

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ABSTRACT

The rapid deployment of artificial intelligence (AI) in decision-support systems has transformed the way humans interact with computational models. While modern AI systems often achieve high predictive accuracy, their lack of transparency can undermine user trust and limit effective collaboration. Human–AI co-decision making, where both human judgment and AI recommendations jointly influence outcomes, requires explainability as a foundational capability rather than an optional feature. This paper investigates the role of explainable artificial intelligence (XAI) in improving co-decision quality, trust calibration, and accountability. A comprehensive literature review is presented, followed by identification of key research gaps. We propose an Explainable Co-Decision Framework (ECDF) that integrates predictive modeling, explanation generation, and adaptive human feedback. Using a simulated risk-assessment dataset comprising 5,000 instances, the framework is evaluated across multiple conditions. Experimental results demonstrate that structured explanations improve joint decision accuracy by up to 10%, reduce trust calibration error by more than 60%, and enhance human engagement with AI outputs. The findings highlight that explanation quality—not merely availability—plays a decisive role in human–AI teaming. The paper concludes with design recommendations and future research directions for robust explainable co-decision systems.

Keywords: Explainable AI, Human-AI Interaction, Trust Calibration, Decision Support, Co-Decision Making.

ICFTSEMAP205

THREATINSIGHT: A Profile-Aware and Explainable Cyber Threat Intelligence System Using Real-Time Vulnerability Analysis

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ABSTRACT

The rapid increase in cyber attacks and the growing complexity of modern software environments have created an urgent need for scalable, ethical, and real time cyber threat intelligence systems. Most existing vulnerability assessment solutions rely on generic threat feeds, static scanning tools, or opaque machine learning models, which introduce challenges related to contextual relevance, explainability, scalability, and operational usability. This paper presents THREATINSIGHT, a profile aware and explainable cyber threat intelligence system that leverages real time vulnerability intelligence and contextual risk analysis. The proposed framework aggregates publicly available cybersecurity data sources, including CVE, CVSS, EPSS, and known exploited vulnerability intelligence, and processes them through a structured data pipeline. User declared system profiles, such as operating systems and deployed applications, are used to correlate vulnerabilities with real world environments, enabling precise and actionable threat prioritization. Risk levels are inferred using an interpretable scoring mechanism that combines severity metrics, exploit likelihood, exploitation status, and profile relevance, ensuring transparency and reliability in decision making. The system architecture prioritizes explainability, modularity, and low latency analysis, supported by an interactive dashboard, alerting mechanisms, and a retrieval augmented intelligence layer for natural language querying. Experimental evaluation demonstrates effective risk prioritization, accurate profile based threat detection, and meaningful security insights with minimal computational overhead, validating the practicality of the proposed approach. By eliminating system level scanning and opaque inference pipelines, THREATINSIGHT offers a scalable and responsible cyber threat intelligence solution for proactive security management in modern computing environments.

Keywords: Cyber Threat Intelligence, Vulnerability Analysis, CVE, EPSS, Explainable AI, Profile-Aware Security, Risk Assessment, Cybersecurity Analytics.

ICFTSEMAP206

A Review Paper on: "Digital Transformation and Cybersecurity in Healthcare: Balancing Innovation and Safety"

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ABSTRACT

The growing use of digital technologies in healthcare has led to major improvements in patient care, communication, and operational efficiency. However, it has also introduced significant cybersecurity challenges. Healthcare systems are increasingly targeted by threats such as data breaches, ransomware attacks, and vulnerabilities in connected medical devices. This paper reviews the current cybersecurity risks in healthcare and their potential impact on patient safety and data privacy. Notable incidents like the Anthem Blue Cross breach and the WannaCry ransomware attack are examined to highlight the real-world consequences of poor cybersecurity practices. The paper also discusses emerging technologies—such as AI, telehealth, cloud computing, and wearables—and their associated risks and benefits. Best practices for mitigating cybersecurity threats are presented, including regulatory compliance, data encryption, staff training, risk assessment, and incident response plan.

Keywords: Healthcare Cybersecurity, Zero-Trust Model, Medical IOT Risks, Patient Data Protection.

ICFTSEMAP207

RideShield: Bike Theft Analysis

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ABSTRACT

Bike theft remains a recurring urban problem that causes significant financial loss and social inconvenience. Conventional crime monitoring approaches are largely reactive, relying on manual reporting and post-incident investigation, which limits timely preventive action. This paper presents RideShield, a spatial–temporal bike theft analysis and mapping system designed to transform historical crime data into actionable visual intelligence for smart city decision support. The proposed framework integrates a MongoDB database, FastAPI backend, and a React–Tailwind interactive dashboard to process multi-year theft records. Robust data preprocessing techniques, including field normalization, geolocation validation, and temporal parsing, ensure reliable analytical inputs. Spatial aggregation and time-based profiling are performed to identify high-risk zones, peak theft hours, and frequently targeted vehicle categories. The system generates interactive geospatial heatmaps and trend visualizations that highlight theft density and recurring patterns across police jurisdictions. Experimental findings reveal consistent hotspot concentrations in high-traffic urban regions and peak activity during evening hours (18:00–22:00). By converting raw crime records into intuitive visual analytics, RideShield enhances situational awareness and supports data-driven patrol allocation and preventive planning. The framework establishes a scalable foundation for future integration of predictive modeling and real-time crime data streams within smart city infrastructures.

Keywords: Bike Theft Analysis, Spatial–Temporal Analytics, Crime Mapping, Geospatial Visualization, Heatmap Analysis, Smart City Surveillance, Urban Safety, FastAPI, MongoDB, Interactive Dashboard.

ICFTSEMAP208

A Survey on Artificial Intelligence Techniques for Demand Forecasting and Supply Chain Optimization in Grocery Retail

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ABSTRACT

Artificial Intelligence (AI) has emerged as a transformative technology in modern supply chain management, particularly in demand forecasting for grocery retail, where high demand variability, product expiration, and complex consumer behaviour pose significant challenges. Traditional statistical forecasting techniques often fail to capture non-linear patterns, seasonal fluctuations, and the impact of external factors such as promotions, holidays, and disruptions. Recent advances in Machine Learning (ML) and Deep Learning (DL) have enabled data-driven forecasting models capable of learning complex temporal dependencies from large-scale retail data. This survey presents a comprehensive review of AI-powered demand forecasting techniques applied to grocery retail and supply chain optimization. The study systematically analyses ten recent and relevant research works, focusing on deep learning architectures such as LSTM, GRU, CNN, attention mechanisms, ensemble learning, and reinforcement learning-based approaches. The reviewed studies are classified based on forecasting methodology, data characteristics, optimization integration, and performance metrics. The comparative analysis highlights the strengths, limitations and research gaps of existing approaches. This survey aims to help researchers and practitioners understand current trends, identify limitations, and design robust AI-based demand forecasting systems for grocery supply chains.

Keywords: Demand Forecasting, Supply Chain Optimization, LSTM, Self-Attention, XGBoost, Retail Analytics.

ICFTSEMAP209

Emotional Regulation and Marital Satisfaction Among Married Males

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ABSTRACT

This study examines the relationship between emotional regulation and marital satisfaction among married males in south Kerala. The study was conducted Using quantitative descriptive correlational research**.** The Convenience Sampling technique is used to draw 100 sample for the study**. The sample** Participants were categorized on the bases of their age between 40-50 and who is married more than five years. The Revised Marital Satisfaction Scale (MSS) created by Roach, Frazier, and Bowden (198) (1.000) and the Emotion Regulation Questionnaire (ERQ) created by Gross and John (2003) (.142), which measures cognitive reappraisal and expressive suppression, were used to gather data. For better understanding, the questionnaires were given out in Malayalam. Findings of the study revealed that there is **no significant relationship between emotional regulation and marital satisfaction among married males. But there is relationship between cognitive reappraisal and expressive suppression among married males.** These findings show that there can be other factors that affect married men's emotional regulation and marital satisfaction.

Keywords: Emotional Regulation, Marital Satisfaction, Married Males, Age, Married More Than Five Years.

ICFTSEMAP210

AI-Based Smart Attendance and Behaviour Monitoring System

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ABSTRACT

The conventional attendance systems are plagued by time inefficiency, proxy attendance issues, and a lack of engagement data. This paper proposes an AI-powered smart attendance and behaviour tracking system that combines facial recognition and real-time behavioural analysis. The proposed system uses Multi- task Cascaded Convolutional Networks (MTCNN) for accurate face detection, FaceNet for the production of 128-dimensional facial embeddings, and a designed Convolutional Neural Network for the classification of pedagogically significant emotional states. Experimental results show a 96.8% accuracy rate in multi- person attendance recognition in a classroom setting and close to 90% accuracy in engagement-related emotional state detection. The system uses a Flask-based web interface for real-time analysis and in-depth analytical data. The system's real-world implementation in a classroom setting proves that it can recoup 5-10 minutes of lost class time per session, eliminate proxy attendance, and facilitate data-driven pedagogical interventions, establishing the system's feasibility in an academic setting.

Keywords: Face Recognition, Automated Attendance Systems, Emotion Recognition, Deep Learning Techniques, MTCNN, Behavioral Analytics.

ICFTSEMAP211

Detection and Classification of Skin Disease Using Transfer Learing in Deep Neural Network

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ABSTRACT

Every year, countless people around the world struggle to get a proper skin disease diagnosis not because treatments don't exist, but because of the specialist doctors simply aren't available in their area. Conditions like melanoma can become fatal when caught too late, making early and accessible detection critical. To address this, we developed a deep learning system that can identify seven types of skin conditions from images. We used DenseNet201 and EfficientNet as our base models, adapting them through transfer learning to work with dermatological data from the HAM10000 dataset. The models were trained on over 10,000 dermoscopic images after careful preprocessing and augmentation. DenseNet201 delivered the best results, reaching 85.8% accuracy while remaining light enough for mobile or web deployment. This system is not meant to replace doctors, it is designed to work alongside them, offering a quick second opinion that could make a real difference in places where dermatologists are hard to reach.

Keywords: Skin Disease Classification, Transfer Learning, Deep Learning, DenseNet201, Dermatoscopic Image Analysis.

ICFTSEMAP212

Campus Surveillance and Safety using Computer Vision

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ABSTRACT

Surveillance systems have become essential components of modern institutional security infrastructure. This research introduces an intelligent surveillance framework that leverages advanced deep learning techniques to enhance campus safety through automated event detection. The system incorporates YOLOv8, a state-of-the-art object detection architecture, alongside ByteTrack, a robust multi-object tracking solution, to process video streams in real-time. The framework is engineered to identify four categories of security-relevant events through sophisticated spatio-temporal analysis: the detection of loitering behavior via occupancy duration metrics, identification of violent interactions through kinematic pattern recognition, unauthorized presence in restricted zones through geospatial boundary monitoring, and early fire/smoke detection leveraging visual signature analysis with temporal consistency filtering. The architecture generates immediate notifications upon event confirmation and maintains comprehensive incident logs for post-event analysis. Its modular design facilitates seamless integration with existing camera infrastructure while maintaining computational efficiency suitable for continuous operation.

Keywords: Computer Vision, YOLOv8, Intelligent Surveillance, Campus Safety.

Lightweight Transfer-Learner Framework for Multistage Alzheimer’s Detection in MRI Scans

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ABSTRACT

Millions of people around the world experience memory loss and cognitive decline due to Alzheimer's disease (AD), which is a progressive neural disorder. Detecting it early with MRI imaging can significantly improve diagnosis and treatment planning. However, interpreting MRI scans by hand is subjective and takes a lot of time. This paper introduces an automated, lightweight deep learning method for classifying the stages of Alzheimer's using transfer learning with MobileNetV3. The system prepares MRI images by resizing, normalizing, and enhancing them to improve generalization. Images are slightly rotated, flipped and adjusted in brightness to help the network learn from multiple variations allowing model to become more reliable and less overfitted. MobileNetV3 has frozen feature extraction layers along with custom dense layers. It divides MRI scans into four categories: Non-Demented, Very Mild Dementia, Mild Dementia, and Moderate Demented. The evaluation covers how well classification performs considering model’s accuracy, precision, recall and F1-score. A web interface built on Streamlit allows real-time image uploads and instant predictions, making it useful for clinical and research applications. This study connects AI-based image analysis to healthcare by providing an efficient and clear way to detect Alzheimer’s. Additionally, it demonstrates high efficiency in computation making it suitable to deploy on low-resource devices and environments.

Keywords: Alzheimer’s Disease; Anomaly Detection; Deep Learning; MRI Image Classification; Transfer Learning, Overfitting.

ICFTSEMAP214

Smart Inhaler Monitoring System

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ABSTRACT

Asthma and Chronic Obstructive Pulmonary Disease (COPD) are long-term respiratory conditions. Managing these conditions requires regular and correct use of inhalers. Unfortunately, many people struggle with sticking to their medication and using their inhalers properly. This often leads to worse symptoms and a higher chance of hospital visits. This project suggests a Smart Inhaler Monitoring System that uses sensors, a microcontroller unit, and wireless communication to check inhaler usage in real time. The system automatically notices each time the inhaler is used, records the data, and sends it to a mobile or web app for ongoing monitoring. It gives timely reminders for taking medication and creates daily and weekly reports to help users see how well they are sticking to their routine. Another feature allows users to set reminders based on their past usage, making the alerts more personalized and effective. The system is designed to be affordable and easy to use in both home and clinical settings. By providing accurate tracking, timely reminders, and insights from data, the Smart Inhaler Monitoring System aims to improve medication adherence, lower health risks, and enhance management of respiratory diseases.

Keywords: Smart Inhaler, Asthma and COPD, Medication Adherence, Real-Time Monitoring, IoT in Healthcare, History-Based Reminder.

Oral Cancer Detection Using Convolutional Neural Network

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ABSTRACT

Oral Squamous Cell Carcinoma (OSCC) is a common and potentially fatal form of oral cancer, where timely and correct diagnosis can greatly enhance the chances of survival. The traditional histopathological analysis requires manual analysis of slides, which is a tedious and subjective process. To overcome this drawback, this paper proposes an automated deep learning solution for the classification of normal and cancerous oral tissues using a Convolutional Neural Network (CNN) with transfer learning. The histopathological images were standardized into a common dataset to eliminate bias, and various preprocessing methods like resizing, normalization, and augmentation were used to improve generalization. The EfficientNetB3 architecture was used and fine-tuned with batch normalization, regularized dense layers, and dropout to reduce the risk of overfitting and improve feature representation. The proposed model performed well in terms of accuracy, precision, recall, and F1-score, and the confusion matrix analysis validated the effectiveness of the model in classifying unknown samples. The trained model can be used for inference and real-time applications in the future and validates the potential of transfer learning-based CNN models as fast, objective, and scalable diagnostic tools.

Keywords: Oral Squamous Cell Carcinoma, Histopathology, Convolutional Neural Network, Transfer Learning, EfficientNetB3, Deep Learning, Medical Image Classification.

ICFTSEMAP216

A Web-Based Automated Programming Practice and Assessment Platform

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ABSTRACT

Developing strong programming skills requires consistent practice and immediate feedback. However, in many traditional classroom settings, evaluation is often delayed and manually intensive, slowing the learning process. This paper presents the design and implementation of a web-based programming practice platform that combines a structured Multiple Choice Question (MCQ) management system with an interactive code playground. The platform enables instructors to organize questions by subject domain and difficulty level, while allowing students to practice coding in a flexible, self-paced environment. The system incorporates secure remote code execution with multi-language support, including Python, C, and Java. A test case–driven automated grading mechanism evaluates submissions using output normalization and partial scoring to provide immediate and accurate feedback. Experimental evaluation demonstrates a measurable reduction in manual grading effort, faster feedback cycles, and increased learner engagement, ultimately enhancing programming proficiency through continuous, practice-oriented learning.

Keywords: **Web-Based Learning, Automated Code Evaluation, Programming Practice System, MCQ Management, Remote Code Execution, Educational Technology.**

ICFTSEMAP217

Tribological Performance Evaluation of Graphene-Coated Grey Cast Iron Using Pin-On-Disc Testing

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ABSTRACT

Friction and wear in grey cast iron components significantly affect the efficiency and lifespan of mechanical systems. Surface modification using graphene offers promising tribological improvements due to its exceptional mechanical strength and self-lubricating properties. In this study, graphene coatings were deposited on grey cast iron pins and evaluated using a pin-on-disc tribometer under dry sliding conditions. Coefficient of friction (COF), wear rate, and surface roughness were measured and compared with uncoated samples. Surface morphology and elemental composition were analyzed using Scanning Electron Microscopy (SEM) coupled with Energy Dispersive Spectroscopy (EDS), while phase identification was performed using X-ray Diffraction (XRD). The graphene-coated samples demonstrated significant reduction in friction and wear compared to uncoated specimens, confirming the effectiveness of graphene as a solid lubricant coating.

Keywords: Graphene Coating, Grey Cast Iron, Tribology, Wear Rate, Pin-On-Disc, SEM-EDS, XRD.

ICFTSEMAP218

Clinical Video Intelligence in Healthcare: A Dual-Perspective Review of Healthcare Applications and AI Methodologies

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ABSTRACT

Clinical video analysis is increasingly being explored as a means to improve patient monitoring, procedural understanding, and workflow efficiency in modern healthcare. However, existing studies are often limited to specific applications such as pain assessment or surgical analysis, resulting in a fragmented understanding of the field. This review aims to provide an integrated perspective that connects healthcare needs with advances in AI-driven video intelligence. A PRISMA-guided literature review was conducted, resulting in a curated set of fifty-five research articles. The selected studies were analyzed from both healthcare and computing perspectives, with attention to application focus, methodological trends, strengths, and practical limitations. The review shows a clear shift from early CNN-based approaches toward temporal, transformer-based, and multimodal models, alongside an evolution from patient-level perception tasks to workflow-aware clinical intelligence. Key outcomes include the identification of four major healthcare directions: patient-state monitoring, procedure intelligence, team workflow analytics, and safety-oriented quality assessment. Overall, the field is moving toward integrated clinical intelligence systems rather than isolated recognition models. Future work should emphasize robust multimodal learning, generalization across clinical settings, and human-AI collaboration, while addressing limitations related to data heterogeneity and real-world deployment challenges.

Keywords: Clinical Video Intelligence, Healthcare Video Analytics, Medical Computer Vision, Multimodal Learning in Healthcare, Patient State Monitoring.

ICFTSEMAP219

Employee Perspectives on Organizational Assistance in Achieving Work-Life Harmony

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ABSTRACT

Achieving work-life harmony has become a critical focus in contemporary workplaces, driven by increasing employee demands for balance and organizational needs for productivity. This study examines employees' perspectives on the role of organizational support in promoting work-life balance. Using a mixed-methods approach, through surveys and in-depth interviews, we gathered information from workers in various industries. The findings highlight key factors, including flexible work policies, leadership support, and workplace culture, that significantly influence employees' ability to balance professional and personal responsibilities. Additionally, the research identifies barriers such as inconsistent policy implementation and a lack of managerial empathy. The study underscores the importance of tailored organizational initiatives that address diverse employee needs, emphasizing the interplay between perceived support and employee well-being. Practical recommendations for organizations include fostering open communication, enhancing managerial training, and adopting inclusive work-life policies. This study adds to the expanding corpus of research on work-life balance by providing firms looking to establish peaceful and sustainable work environments with practical insights.

Keywords: Work–Life Harmony; Perceived Organizational Support; Flexible Work Arrangements; Leadership Support; Workplace Culture; Employee Well-Being; Work–Family Interface; Managerial Empathy; Policy Implementation; Employee Engagement; Organizational Climate; Sustainable Workplaces.

IoT Based Real-Time Road Hazard Detection and Alert System Using Smart Streetlights and Smart Road Studs

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ABSTRACT

Road safety is a major problem in modern transportation systems. Many accidents happen because drivers do not get information about road problems in time. These problems include potholes, uneven roads, sudden speed breakers, accident-prone areas, and damaged road surfaces. Traditional methods like road signs and manual inspection are not enough because they do not change based on real road conditions. Also, road inspections take time and cannot cover all areas frequently. This paper presents an IoT-based system that can detect road hazards in real time and warn drivers quickly. The system uses sensors placed in vehicles to collect data such as vibration and location using accelerometers, ultrasonic sensor and GSM Modem. When a vehicle passes over a bad road surface, the sensor data changes. This data is sent to a cloud server through mobile networks, where it is analysed to confirm if there is a real road problem. Once a hazard is confirmed, the system sends information to nearby smart streetlights or roadside units. These units then activate smart road studs using wireless communication. The road studs light up and give a clear visual warning to drivers, especially at night or in bad weather. This helps drivers slow down and avoid accidents. The system was tested in different road conditions. The results show that it can detect road problems quickly and accurately. The system is reliable and can work well in busy city roads. This smart road safety system is useful for smart cities. It reduces the need for manual road inspection and improves safety using modern technology. By combining vehicles, cloud computing, and smart roadside devices, this system makes roads safer and helps save lives.

Keywords: IoT, Intelligent Transportation Systems, Road Safety, Pothole Detection, Smart Streetlights, Cloud Computing.

ICFTSEMAP221

Public Sector Undertakings and Women Employees in Bengaluru

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ABSTRACT

Karnataka Antibiotics and Pharmaceuticals Private Ltd (Kapl) a central public sector undertaking with the branch in Karnataka l which was started in 1981 in the capital city of Bengaluru. The main objectives was to study the sales of the company, to analyze the performance of the company, to know about working traditions of the employees, to study the facilities given by the company for well-being of the employee, and limitations of the study They manufactured antibiotics medicines and other essential medical antibiotics.Its a important sector as it is concerned with very important healthcare sector .Though the public sector undertaking has been running under losses it has pulled itself up and since 2021 it has pulled itself out of the pandemic phase with very less profit but with running capital intact .The main aim here is to study the women employees and their situation they work environment the welfare activity the welfare facilities transportation facilities healthcare facilities live facilities and workers and over time aspects of women employees in public sector undertakings. The sample size is 150 women from the age group of 21 - 30 years old in various working in various capacities as office assistant, steno, computer operators, supervisors and managers. Sampling technique stratified sampling and the research gap spoke referred to the women employees being ready to work overtime with minimum pay as they were ready to work more even past normal working hours indicating the secure work atmosphere. The research methodology is concerned with both the primary data as well as the secondary data. Primary data was the main research methodological tool used for analysis purposes. Primary data was collected by from the respondents by designing the destination add interviewee personally the various women employees The second data was collected from the many government reports and from other previously published articles. Data analysis was done by using stratified random sampling. Simple percentage method and descriptive analysis was was used for analyzing the selected data The findings said that the women employees in public sector undertaking of KAPL where a very satisfied lot though they suffered while commuting for work during the pandemic season. Post pandemic slowly life limped back to normal and they were all getting their salary and wages. The conclusion tells us that the public sector undertakings were a godsend as jobs were very protected by many labor laws and regulations as most of the women employees are satisfied with settlement of their grievances.

Keywords women employees public sector undertakings salaries antibiotics.

ICFTSEMAP222

AgriConnect: AI-Powered Digital Marketplace and Advisory Platform for Farmers

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ABSTRACT

The increasing digitalization of the global industrial sector has yet to offer a holistic solution to the systemic problems presented by small-scale farmers, including market fragmentation, the lack of formal credit, and generic advisory platforms. Conventional agricultural practices are inefficient in managing markets characterized by middlemen and non-actionable weather information, resulting in a 15-25% loss of potential profit. This paper presents AgriConnect, an AI-powered digital marketplace and advisory platform that seeks to address these systemic problems using a hybrid intelligence framework. The proposed platform integrates an Agentic AI Chatbot powered by the Google Gemini API and grounded in ICAR (Indian Council of Agricultural Research) data sets with hyperlocal weather intelligence to provide scientifically validated, crop-type advisories. Data preprocessing and high-performance design using Next.js 15 and Redis caching are incorporated to ensure 90%+ efficiency in external API calls and rapid execution in the rural environment. A weighted fusion approach is used to merge scientific advisory and environmental information in such a way that the chances of errors occurring during decision-making are minimized, which is market and climate change sensitive. The experimental outcome of the result shows a 40% improvement in the estimated output and a 10-20% improvement in the farmer's price realization. The approach increases economic security, efficiency, and sustainable resource management in the present agricultural context by enabling real-time market entry and scientific advisory.

Keywords: Agentic AI; Digital Marketplace; Explainable AI (XAI); Google Gemini; Precision Agriculture; ICAR, Sustainable Farming.

ICFTSEMAP223

Influence of Fly ash and Titanium Dioxide on Al7075 in Three Body Tribological Behaviour

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ABSTRACT

The growing need for materials that are both light and strong, alongside being wear-resistant, particularly within aerospace, automotive, and maritime sectors, has spurred significant advancements in Aluminum Metal Matrix Composites (AMMCs). Al7075, a widely used aluminum alloy, is prized for its excellent mechanical strength, its ability to withstand fatigue, and its resistance to corrosion. However, its wear performance under harsh adhesive and abrasive conditions needs improvement to satisfy the tough demands of high-stress applications. This research focuses on creating and evaluating Al7075 hybrid composites, incorporating Titanium Dioxide (TiO_2) and fly ash through a stir casting process. Reinforcement particles, specifically 3%, 6%, and 9% TiO_2 by weight, were carefully dispersed, and the resulting samples were subjected to detailed mechanical and tribological analysis. Hardness tests, following ASTM E10 standards, showed a considerable increase from 19.62 BHN for the unreinforced Al7075 to 68.76 BHN with the inclusion of 9% TiO_2 reinforcement. Abrasive wear behavior was examined using the ASTM G65 dry sand rubber wheel test, with loads of 11.77 N, 53.2 N, and 102.4 N. The results showed a notable decrease in volume loss as the reinforcement level increased, with an approximate 66% wear reduction at 102.4 N for the 9% TiO_2 composite compared to the base alloy. Scanning Electron Microscopy (SEM) analysis revealed that micro-cutting and micro-ploughing were the dominant mechanisms, with limited adhesive wear. X-ray Diffraction (XRD) confirmed the existence of the aluminum matrix and the anatase/rutile phases of TiO_2, with no unwanted phase formation detected. The study's findings reveal a clear relationship between increased hardness and a reduced wear rate, thus highlighting the effectiveness of TiO_2 and fly ash reinforcements in improving the tribological performance of Al7075 composites. This, in turn, opens up possibilities for advanced engineering applications.

Keywords— Al7075; Hybrid metal matrix composites; TiO_2 reinforcement; Fly ash particles; Abrasive wear resistance; Adhesive wear; Tribology; Microstructural characterization.

END

Thank You